West Bengal State Council of Technical & Vocational Education and Skill Development
(Technical Education Division)



Syllabus
of

Diploma in Civil Engineering [CE]

Part-II (3rd Semester)

Revised 2022

CURRICULAR STRUCTURE FOR PART – II FIRST SEMESTER (THIRD SEMESTER) OF THE FULL-TIME DIPLOMA COURSE IN CIVIL ENGINEERING

Sl. No.	Subject Code	Subject of Study	HOURS PER WEEK			CREDITS
		Theoretical Papers	Lecture	TUTORIAL	PRACTICAL	
1	CEPC301	Construction Materials	2	0	0	2
2	CEPC302	Basic Surveying	3	0	0	3
3	CEPC303	Mechanics of Materials	3	0	0	3
4	CEPC304	Building Construction	2	0	0	2
5	CEPC305	Concrete Technology	2	0	0	2
6	CEPC306	Civil Engineering Planning and Drawing	1	0	0	1
7	CEPC307	Transportation Engineering	2	0	0	2
		LABORATORY/SESSIONAL PAPERS				
8	CEPC308S	Civil Engineering Planning and Drawing Practices	0	0	4	2
	CEPC309S : Civil Engineering Lab- I (consisting of following four Modules with 8 Practical Classes and 4 Credits)					
9	CEPC309S/I	Module-I: Construction Materials Lab	0	0	2	1
10	CEPC309S/II	Module-II: Mechanics of Materials Lab	0	0	2	1
11	CEPC309S/III	Module-III: Concrete Technology Lab	0	0	2	1
12	CEPC309S/IV	Module-IV: Transportation Engineering Lab	0	0	2	1
		INTERNSHIP				
13	CEI310S	Internship-I after IInd Sem	0	0	0	1
		TOTAL	15	0	12	22

Name of the Course	**Diploma in Civil Engineering**	Course duration	6 semester
Course Title	**Construction Materials**	Course Code	**CEPC301**
Subject offered in Semester	Third	Number of Credits	2(L:2, T:0, P: 0)
Prerequisites	NIL	Course Category	**PC**
Question distribution		Marks distribution	

Course Objectives: Following are the objectives of this course:

- To learn about various construction materials, and understand their relevant characteristics.
- To be able to identify suitability of various materials for different construction purposes.
- To know about natural, artificial, and processed materials available for various purposes of construction activities.

Course Content:

Module	Distribution of unit
Module A	Unit 1 and 2
Module B	Unit 3 and 4
Module C	Unit 5

Unit – I: Overview of Construction Materials

- Scope of construction materials in Building Construction, Transportation Engineering, environmental Engineering, Irrigation Engineering (applications only).
- Selection of materials for different civil engineering structures on the basis of strength,durability, Eco friendly and economy.
- Broad classification of materials – Natural, Artificial, special, finishing and recycled

Unit – II: Natural Construction Materials

- Requirements of good building stone; general characteristics of stone; quarrying and dressing methods and tools for stone.
- Structure of timber, general properties and uses of good timber, different methods of seasoning for preservation of timber, defects in timber, use of bamboo in construction.
- Asphalt, bitumen and tar used in construction, properties and uses.
- Properties of lime, its types and uses.
- Properties of sand and uses
- Classification of coarse aggregate according to size and its use

Unit- III: Artificial Construction Materials

- Constituents of brick earth, Conventional / Traditional bricks, Modular and Standard bricks, Special bricks –fly ash bricks, Characteristics of good brick, Field tests on Bricks, Classification of burnt clay bricks and their suitability, Manufacturing process of burnt clay brick, fly ash bricks, Aerated concrete blocks.
- Flooring tiles – Types, uses
- Manufacturing process of Cement - dry and wet (only flow chart), types of cement and its uses. Field tests on cement.
- Precast concrete blocks- hollow, solid, pavement blocks, and their use.
- Plywood, particle board, Veneers, laminated board and their uses.
- Types of glass: soda lime glass, lead glass and borosilicate glass and their uses
- Ferrous and non-ferrous metals and their uses.

Unit– IV: Special Construction Materials

- Types of material and suitability in construction works of following materials: Waterproofing, Termite proofing; Thermal and sound insulating materials.
- Fibers – Types –Jute, Glass, Plastic Asbestos Fibers (only uses).
- Geopolymer cement: Geo-cement: properties, uses.

Unit– V: Processed Construction Materials

- Constituents and uses of POP (Plaster of Paris), POP finishing boards, sizes and uses.
- Paints- whitewash, cement paint, Distemper, Oil Paints and Varnishes with their uses. (Situations where used).
- Industrial waste materials- Fly ash, Blast furnace slag, Granite and marble polishing waste and their uses.
- Agro waste materials - Rice husk, Bagasse, coir fibres and their uses.
- Special processed construction materials; Geosynthetic, Ferro Crete, Artificial timber, Artificial sand and their uses.

References:

1. Ghose, D. N., Construction Materials, Tata McGraw Hill, New Delhi.

2. S.K. Sharma, Civil Engineering Construction Materials, Khanna Publishing House, Delhi

3. Varghese, P.C. , Building Materials, PHI learning, New Delhi.

4. Rangwala, S.C., Engineering Materials, Charator publisher, Ahemdabad.

5. Somayaji, Shan, Civil Engineering Materials, Pearson education, New Delhi.

6. Rajput, R.K, Engineering Materials, S. Chand and Co., New Delhi.

7. Sood H., Laboratory Manual on Testing of Engineering Materials, New Age Publishers, New Delhi.

8. Sharma C. P., Engineering Materials, PHI Learning, New Delhi.

9. Duggal, S. K, Building Materials, New International, New Delhi.

10. Singh, Parveen, Civil Engineering Materials, S.K. Kataria & sons

11. Soni, S.K., Building Materials and Constructions, S.K. Kataria & sons

12. Singh, Gurucharan, Building Construction & Materials, Standard Book House, New Delhi.

13. Gambhir, M.L, Jamwal Neha, Lab Manual Building & Construction Materials, Tata McGraw Hill, New Delhi.

14. Subramanian, N., Building Materials, Oxford University Press.

Course outcomes: After completing this course, student will be able to:

- Identify relevant construction materials.
- Identify relevant natural construction materials.
- Select relevant artificial construction materials.
- Select relevant special type of construction materials.
- Identify and use of processed construction materials.

Name of the Course	**Diploma in Civil Engineering**	Course duration	6 semester
Course Title	**Basic Surveying**	Course Code	**CEPC302**
Subject offered in Semester	Third	Number of Credits	3 (L:3, T: 0, P: 0)
Prerequisites	NIL	Course Category	**PC**
Question distribution		Marks distribution	

Objective:- Following are the objectives of this course:

- To understand types of surveying works required.
- To know methods to be used for different surveys.
- To know the types, use and operational details of various Surveying equipments.

Contents:

Module	Distribution of unit
Module A	Unit 1 and 2
Module B	Unit 3 and 4
Module C	Unit 5

Unit – I Overview and Classification of Survey

- Survey-Definition, Purpose and Use.
- Types of surveying- Primary and Secondary, Classification: Primary-Plane & Geodetic, Secondary Classification-Based on Instrument, Based on Method, Based on Purpose, Based on Nature of Field.
- Principles of Surveying.
- Scales: Engineer's scale, Representative Fraction (RF), Plain, Diagonal, Vernier and Chord Scale.

Unit– II Chain Surveying

- Instruments used in chain survey: Chain, Tapes, Arrow, Ranging rod, Line ranger, Pegs, Offset rod, cross staff, Optical square, Plumb Bob.
- Methods of Chaining, obstacles in chaining.
- Principles of chain survey.
- Well and ill Conditioned Triangle.
- Errors in length: Instrumental error, personal error, error due to natural cause, random error.
- Calculation of Chain and Tape Correction.
- Types of offsets: Perpendicular and Oblique.
- Chain survey Station, Base line, Check line, Tie line, Tie station.
- Ranging: Direct and Indirect Ranging.
- Conventional Signs, Recording of measurements in a field book.
- area calculation from field book entry
- Simple numerical problems.

Unit– III Compass Traverse Survey

- Compass Traversing- open and closed.
- Technical Terms: Meridians and Bearings of a Line-True, Magnetic, Arbitrary
- Whole Circle Bearing system and Reduced Bearing system and examples on conversion of given bearing to another bearing (from one form to another), Fore Bearing and Back Bearing, Calculation of internal and external angles from bearings at a station and Vice-versa, Dip of Magnetic needle, Magnetic Declination and It's variation, Isogonic and Agonic line.
- Principles of Compass Traversing.
- Components of Prismatic Compass and their Functions, Difference between Prismatic and Surveyor compass. Methods of using Prismatic Compass - Temporary adjustments and observing bearings.
- Local attraction, different methods of correction of observed bearings - Correction at station and correction to included angles including cases when all stations in a traverse are affected with local attraction
- Methods of plotting a traverse and closing error, Graphical adjustment of closing error.
- Errors in Compass Surveying-Instrumental, Personal and Natural.
- Simple numerical problems.

Unit– IV Levelling and Contouring

- Levelling-Definition, Object and Uses.
- Basic terminologies: Level, Horizontal and vertical surfaces & Lines, Datum surface or Line, Bench Marks- GTS, Permanent, Arbitrary and Temporary, Reduced Level, Rise, Fall, Line of collimation, Axis of the Telescope, Axis of Bubble Tube, Station,

Back sight, Fore sight, Intermediate sight, Change point, Height of instruments, Focussing and Parallax.

- Types of levels: Dumpy, Tilting, Auto level, Digital level, Components of Dumpy Level and its fundamental axes, Temporary adjustments of Level.
- Types of Levelling Staff: Self-reading staff and Target staff.
- Curvature and Refraction correction.
- Reduction of level by Line of collimation or Height of Instrument and Rise and Fall Method.
- Levelling Types: Simple, Differential, Fly, Profile, Cross sectional, Check and Reciprocal Levelling.
- Entry in level book and arithmetical check and accuracy check of level work
- Errors in Levelling-Instrumental, Personal and Natural, Permissible limits of Error in Levelling.
- Contour, contour intervals, horizontal equivalent. Contour gradient.
- Uses of contour maps, Characteristics of contours, Methods of Contouring: Direct and indirect.
- Simple numerical problems

Unit – V Plane Table Surveying

- Principles of plane table survey.
- Accessories of plane table survey and their uses,
- Setting of plane table- Leveling, Centering, Orientation.
- Orientation of plane table - Back sighting and Magnetic needle method.
- Methods of Plane table survey- Radiation, Intersection, Traversing, and Resection (No Two-Point and Three-Point Problem).
- Merits and demerits of plane table survey
- Errors in Plane table Surveying- Instrumental, Personal and Plotting.
- No numerical problems

Suggested learning resources :

1. Punmia, B.C, Jain, Ashok Kumar; Jain, Arun Kumar, Surveying I, Laxmi Publications, New Delhi.
2. Basak, N. N., Surveying and Levelling, McGraw Hill Education, New Delhi.
3. Kanetkar, T. P., Kulkarni, S. V., Surveying and Levelling volume I, Pune Vidyarthi Gruh Prakashan.
4. Duggal, S. K., Survey I, McGraw Hill Education, New Delhi.
5. Saikia, M D, Das. B.M., Das. M.M., Surveying, PHI Learning, New Delhi.
6. Subramanian, R., Fundamentals of Surveying and Levelling, Oxford University Press. New Delhi.

7. Rao, P. Venugopala Akella, Vijayalakshmi, Textbook of Surveying, PHI Learning New Delhi.
8. Bhavikatti, S. S., Surveying and Levelling, Volume 1, I. K. International, New Delhi.
9. Arora K R , Surveying Vol. I, Standard Book House.
10. C.L. Kochher, Surveying, Dhanpat Rai Publication, New Delhi.
11. Dr. N.R. Chandak and Prof. H.R. Kumavat, Surveying, SK Kataria & Sons, New Delhi
12. Saurabh Kumar Soni, Surveying-I, SK Kataria & Sons, New Delhi
13. Agor, R., A Text Book of Surveying & Levelling, Khanna Publishers.
14. Chandra A.M., Plane Surveying, New Age International Publishers.

Course outcomes: After completing this course, student will be able to:

- Measure linear,angular, horizontal & vertical distances in field , adopting precautions and standard data recording methods.
- Conduct traversing in the field using chain and compass.
- Use levelling instruments to determine reduced level for preparation of contour maps
- Prepare plans using Plane Table Surveys.
- Select the type of survey required for given situation.

Name of the Course	**Diploma in Civil Engineering**	Course duration	6 semester
Course Title	**Mechanics of Materials**	Course Code	**CEPC303**
Subject offered in Semester	Third	Number of Credits	3 (L:3, T: 0, P: 0)
Prerequisites	NIL	Course Category	**PC**
Question distribution		Marks distribution	

Objective:- Following are the objectives of this course:

- To learn properties of structural sections and structural materials.
- To understand the concept of stress and strain.
- To calculate shear force, bending moment for different structural elements and corresponding stresses.
- To understand different methods of finding axial forces in trusses.

Contents:

Module	Distribution of unit
Module A	Unit 1 and 2
Module B	Unit 3 and 4
Module C	Unit 5

Unit – I Moment of Inertia

- Moment of inertia (M.I.): Definition, M.I. of plane lamina, Radius of gyration, section modulus, Parallel and Perpendicular axes theorems, M.I. of rectangle, square, circle, semi-circle, quarter circle and triangle section

- M.I. of symmetrical and unsymmetrical sections: I-section, Channel section, T-section, Angle section, Hollow sections and built up sections about centroidal axes and any other reference axis.

- Polar Moment of Inertia of solid circular sections.
- Simple numerical problems.

Unit– II Simple Stresses and Strains

- Definition of rigid, elastic and plastic bodies, deformation of elastic body under various forces, Definition of stress, strain, elasticity, Hooke's law, Elastic limit, Modulus of elasticity.
- Types of Stresses- Tensile and Compressive stresses, Shear and Bending stresses.
- Standard stress strain curve for Mild steel and HYSD bar under tension: Yield stress, Proof stress, Ultimate stress, Strain at various critical points, strain hardening, Percentage elongation and Factor of safety.
- Deformation of body due to axial force, forces applied at intermediate sections, maximum and minimum stress induced, Composite section under axial loading.
- Concept of temperature stresses and strain, Stress and strain developed due to temperature variation in homogeneous simple bar (no composite section, without yielding).
- Longitudinal and lateral strain, Modulus of Rigidity, Poisson's ratio, Biaxial and tri-axial stresses [concept only], volumetric strain, change in volume, Bulk modulus (Introduction only).
- Relation between modulus of elasticity, modulus of rigidity and bulk modulus
- Simple numerical problems.

Unit– III Shear Force and Bending Moment

- Types of supports, beams and loads.
- Concept and definition of shear force and bending moment, Relation between load, shear force and bending moment.
- Shear force and bending moment diagram for cantilever, simply supported and over hang on one or both side beams subjected to point loads, uniformly distributed loads or combination of these two types of loading (for udl - over full span or partial span), point of contra flexure.
- Simple numerical problems.

Unit– IV Bending and Shear Stresses in beams

- Concept and theory of pure bending, assumptions, flexural equation, bending stresses and their nature, concept of neutral plane and neutral axis, bending stress distribution diagram for square, rectangular, circle, I-section, T- section.

- Concept of moment of resistance and simple numerical problems using flexural equation

- Shear stress equation (without derivation), relation between maximum and averages hear stress for rectangular and circular section, shear stress distribution diagram for square, rectangular, circle, I-section, T- section. Simple numerical problems based on shear equation.

- Simple numerical problems.

Unit– V Simple Trusses

- Types of trusses (Simple, Fink, compound fink, French truss, Pratt truss, Howe truss, North light truss, King post and Queen post truss)
- Calculate support reactions for trusses subjected to point loads at joints
- Calculate forces in members of truss using Method of joints and Method of sections.

Suggested learning resources:

1. Bedi D.S. , Strength of Materials, Khanna Publishing House, Delhi, Ed. 2018
2. Timoshenko, S., Strength of Materials, Vol. I, CBS, New Delhi.
3. Khurmi, R.S., Strength of Materials, S Chand and Co. Ltd. New Delhi.
4. Ramamurtham, S, Strength of Materials, Dhanpat Rai and sons, New Delhi.
5. Punmia B C, Strength of Materials, Laxmi Publications (p) Ltd. New Delhi.
6. Rattan S.S., Strength of Materials, McGraw Hill Education; New Delhi.
7. Bansal R K, Strength of Materials, Laxmi Publications.
8. Subramaniam R, Strength of Materials, Oxford University Press.
9. Ramamrutham.S, Theory of structures, Dhanpatrai & Sons.
10. Khurmi, R. S. , Theory of Structures S. Chand and Co., New Delhi.
11. Bhavikatti, S S , Structural Analysis Vol-1, Vikas Publishing House Pvt Ltd. New Delhi.
12. Bhattachariya, B., Engineering Mechanics, Oxford University Press.
13. Chakraborty, M., Strength of Materials, Kolkata.
14. Adarsh, Swaroop, Mechanics of Materials, New Age International Publishers.

Course outcomes: After completing this course, student will be able to:

- Articulate practical applications of moment of inertia of symmetrical and unsymmetrical structural sections.
- Analyse structural behaviour of materials under various loading conditions.
- Interpret shear force and bending moment diagrams for various types of beams and loading conditions.

- Determine the bending and shear stresses in beams under different loading conditions.
- Evaluate axial forces in the members of simple truss.

Name of the Course	**Diploma in Civil Engineering**	Course duration	6 semester
Course Title	**Building Construction**	Course Code	**CEPC304**
Subject offered in Semester	Third	Number of Credits	2 (L:2, T: 0, P: 0)
Prerequisites	NIL	Course Category	**PC**
Question distribution		Marks distribution	

Course Objectives: Following are the objectives of this course:

- To identify different components of building.
- To understand different types of foundation and their significance.
- To know different types of masonry and their construction.
- To highlight the importance of communications in building.

Course Content:

Module	Distribution of unit
Module A	Unit 1 and 2
Module B	Unit 3 and 4
Module C	Unit 5

Unit – I: Overview of Building Components

- Classification of Buildings as per National Building Code Group A to I, As per Types of constructions- Load Bearing Structure, Framed Structure, Composite Structure.
- Building Components - Functions of Building Components, Substructure – Foundation, Plinth.

- Superstructure – Walls, Partition wall, Cavity wall, Sill, Lintel, Stair, Doors and Windows, Floor, Mezzanine floor, Roof, Columns, Beams, Parapet.

Unit – II: Construction of Substructure

- Job Layout: Site Clearance, Layout for Load Bearing Structure and Framed Structure by Center Line and Face Line Method, Precautions.
- Earthwork: Excavation for Foundation, Timbering and Strutting, Earthwork for embankment, Material for plinth Filling, Tools and plants used for earthwork.
- Foundation: Functions of foundation, Types of foundation – Shallow Foundation: Stepped Footing, Wall Footing, Column Footing, Isolated and Combined Column Footing, Raft Foundation, Grillage Foundation. Deep Foundation: Pile Foundation, Well foundation and Caissons, Pumping Methods of Dewatering, Deep wells, Well points, Cofferdams (Introduction only).

Unit- III: Construction of Superstructure

- **Stone Masonry:** Terms used in stone masonry- facing, backing, hearting, Through stone, corner stone, cornice. Types of stone masonry: Rubble masonry, Ashlar Masonry and their types. Joints in stone masonry and their purpose. Selection of Stone Masonry, Precautions to be taken in Stone Masonry Construction.
- **Brick masonry:** Terms used in brick masonry- header, stretcher, closer, quoins, course, face, back, hearting, bat, bonding, joints, lap, frog, level and plumb and other related important terms, Bonds in brick masonry- header bond, stretcher bond, English bond and Flemish bond. Requirements of good brick masonry. Junctions in brick masonry and their purpose and procedure. Precautions to be observed in Brick Masonry Construction. Comparison between stone and Brick Masonry. Tools and plants required for construction of stone and brick masonry. Hollow concrete block masonry and composite masonry.
- **Scaffolding, Shoring and formwork:** Purpose, Types of Scaffolding, Process of Erection and Dismantling. Purpose and Types of Shoring, purpose and types of Underpinning. Formwork: Definition of Formwork, Requirements of Formwork, Materials used in Formwork, Types of Formwork, Removal of formwork as per BIS.

Unit– IV: Building Communication and Ventilation

• **Horizontal Communication: Doors** –Components of Doors, Full Panelled Doors, Partly panelled and Glazed Doors, Flush Doors, Collapsible Doors, Rolling Shutters, Revolving Doors, Fully glazed Doors. Sizes of Door recommended by BIS.

• **Windows:** Component of windows, Types of Windows - Full Panelled, Partly Panelled and Glazed, wooden, Steel, Aluminum windows, Sliding Windows, Louvered Window, Bay window, Corner window, clear-storey window, Gable and Dormer window, Skylight. Size of Windows recommended by BIS. Ventilators. Fixtures and fastenings for doors and windows- material used; Functions of Window Sill , Lintels, Shed / Chajja.

• **Vertical Communication:** Means of Vertical Communication- Stair Case, Ramps, Lift, Elevators and Escalators. Terms used in staircase - steps, tread, riser, nosing, soffit, waist slab, baluster, balustrade, scotia, hand rails, newel post, landing, headroom, winder, stringer beam, going, rising. Types of staircase (On the basis of shape): Straight, dog-legged, open well, Spiral, quarter turn, bifurcated, Three quarter turn and Half turn, (On the basis of Material): Stone, Brick, R.C.C., wooden and Metal.

Unit– V: Building Finishes

- **Floors and Roofs**: Types of Floor Finishes and its suitability- Kota, Marble, Granite, Ceramic Tiles, Vitrified, Chequered Tiles, Paver Blocks, Concrete Floors, wooden Flooring, Skirting and Dado. Process of Laying and Construction, Finishing and Polishing of Floors, Roofing Materials- RCC, Mangalore Tiles, AC Sheets, G.I. sheets, Corrugated G.I. Sheets, Plastic and Fibre Sheets. Types of Roof: Flat roof, Pitched Roof-King Post truss, queen Post Truss, terms used in roofs.
- **Wall Finishes:** Plastering – Necessity of Plastering, Procedure of Plastering, Single Coat Plaster, Double Coat Plaster, Rough finish, Neeru Finishing and Plaster of Paris (POP). Special Plasters- Stucco plaster, sponge finish, pebble finish. Plaster Board and Wall Claddings. Precautions to be taken in plastering, defects in plastering. Pointing – Necessity, Types of pointing and procedure of Pointing. Painting – Necessity, Surface Preparation for painting, Methods of Application, Whitewashing and colour washing, distempering.

Suggested learning resources:

1. S. P. Arora and Bindra., Building Construction, Dhanpat Rai Publication, Delhi.

2. Sushil Kumar., Building Construction, Standard Publication.

3. Rangawala, S. C., Building Construction, Charotar Publication, Anand.

4. Punmia B. C., and Jain A. K., Building Construction ,Firewall Media.

5. Sharma S. K., Building Construction, S. Chand and Co. Pvt. Ltd., New Delhi.

6. Janardan Zha , Building Construction, Khanna Publication.

7. Bhavikatti S. S., Building Construction, Vikas Publication House Pvt. Ltd., Delhi.

8. Mantri S., A to Z Building Construction, Satya Prakashan, New Delhi.

9. Neelam Sharma, Building Construction, S.K.Kataria and Sons

10. Singh, Gurucharan, Building Construction & Materials, Standard Book House, New Delhi.

11. Gambhir, M.L. and JAMWAL, NEHA, Lab Manual Building & Construction Materials, Tata McGraw Hill, New Delhi.

Course outcomes: After completing this course, student will be able to:

- Identify components of building structures.
- Propose suitable type of foundation for building structures.
- Select suitable type of masonry for building structures.
- Propose relevant means of communications for different types of buildings.
- Select relevant material for finishing works.

Name of the Course	**Diploma in Civil Engineering**	Course duration	6 semester
Course Title	**Concrete Technology**	Course Code	**CEPC305**
Subject offered in Semester	Third	Number of Credits	2 (L:2, T: 0, P: 0)
Prerequisites	NIL	Course Category	**PC**
Question distribution		Marks distribution	

Course Objectives: Following are the objectives of this course:

- To know properties of cement, aggregate and water used in concrete.
- To understand different characteristics and properties of concrete.
- To learn about role of admixtures in concrete.

Course Content:

Module	Distribution of unit
Module A	Unit 1 and 2
Module B	Unit 3 and 4
Module C	Unit 5

Unit – I Cement, Aggregates, Water and Admixture

- Physical properties of OPC and PPC: fineness, standard consistency, setting time, soundness, compressive strength. Different grades of OPC and relevant BIS codes
- Testing of cement: Field test and Laboratory tests -fineness, standard consistency, setting time, soundness, compressive strength. Storage of cement and effect of storage on properties of cement.
- BIS Specifications and field applications of different types of cements: Rapid hardening, Low heat, Portland pozzolana, Sulphate resisting, Blast furnace slag, High Alumina and White cement.
- Aggregates: Requirements of good aggregate, Classification according to size and shape.

- Fine aggregates: Properties, size, specific gravity, bulk density, water absorption and bulking, fineness modulus and grading zone of sand, silt content and their specification as per IS 383. Concept of crushed Sand.
- Coarse aggregates: Properties, size, shape, surface texture, water absorption, soundness, specific gravity and bulk density, fineness modulus of coarse aggregate, grading of coarse aggregates, crushing value, impact value and abrasion value of coarse aggregates with specifications.
- Water: Quality of water, impurities in mixing water and permissible limits for solids as per IS: 456.
- Admixtures in concrete: Purpose, properties and application for different types of admixture such as accelerating admixtures, retarding admixtures, water reducing admixtures, air entraining admixtures and super plasticizers.

Unit– **II Concrete**

- Concrete: Different grades of concrete, provisions of IS 456.
- Duff Abraham water cement (w/c) ratio law, significance of w/c ratio, selection of w/c ratio for different grades, maximum w/c ratio for different grades of concrete for different exposure conditions as per IS 456.
- Properties of fresh concrete: Workability: Factors affecting workability of concrete. Determination of workability of concrete by slump cone, compaction factor, Vee-Bee Consistometer. Value of workability requirement for different types of concrete works. Segregation, bleeding and preventive measures.
- Properties of Hardened concrete: Strength, Durability, Impermeability.

Unit– **III Concrete Mix Design and Testing of Concrete**

- Concrete mix design: Objectives, ordinary concrete and controlled concrete, methods of mix design, study of mix design as per IS 10262 (only procedural steps), factors affecting concrete mix design
- Testing of concrete, determination of compressive strength of concrete cubes and cylinder at different ages, interpretation and co-relation of test results.
- Non-destructive testing of concrete: Rebound hammer test, working principle of rebound hammer and factor affecting the rebound index, Ultrasonic pulse velocity test as per IS13311 (part 1 and 2), Importance of NDT tests.

Unit– **IV Quality Control of Concrete**

- Concreting Operations: Batching, Mixing, Transportation, Placing, Compaction, Curing and Finishing of concrete.
- Forms for concreting: Different types of form works for different structural members, requirement of good form work. Stripping time for removal of form works per IS 456.
- Waterproofing: Importance and need of waterproofing, methods of waterproofing and materials used for waterproofing.
- Joints in concrete construction: Types of joints, methods for joining old and new concrete, materials used for filling joints.

***Unit–* V Special Concrete and Extreme Weather concreting**

- Special Concrete: Properties, advantages and limitation of following types of Special concrete: Ready mix Concrete, Fiber Reinforced Concrete, High performance Concrete Self-compacting concrete and light weight concrete.
- Cold weather concreting: effect of cold weather on concrete, precautions to be taken while concreting in cold weather condition.
- Hot weather concreting: effect of hot weather on concrete, precautions to be taken while concreting in hot weather condition.

Suggested learning resources:

1. Gambhir, M.L., Concrete Technology, Tata McGraw Hill Publishing Co. Ltd., Delhi.
2. Shetty, M.S., Concrete Technology, S. Chand and Co. Pvt. Ltd., Ram Nagar, Delhi.
3. Santhakumar, A. R., Concrete Technology, Oxford University Press, New Delhi.
4. Neville, A. M. and Brooks, J.J., Concrete Technology, Pearson Education Pvt. Ltd.
5. Neville, A. M., Concrete Technology, Pearson Education Pvt. Ltd., New Delhi.
6. Sood, H., Kulkarni P. D., Mittal L. N., Laboratory Manual in Concrete Technology, CBS Publishers, New Delhi.
7. IS 456 : 2000
8. IS 10262: 2009
9. SP 23: 1982
10. Kulkarni, P. D., Ghosh, R. K. & Phull V. R., New Age International Publishers.
11. Vazirani, V. N. & CHANDOLA, S.P., Concrete Technology, Khanna Publishers.
12. Laskar, Aminul Islam, Concrete Technology, Laxmi Publication Pvt. Ltd.

Course outcomes: ***After completing this course, student will be able to:***

- Use of different types of ingredient in concrete: cement, aggregates, water, admixture
- Prepare concrete of desired compressive strength.
- Prepare concrete of required specification.
- Maintain quality of concrete under different conditions.
- Apply relevant admixtures for concreting.

Name of the Course	**Diploma in Civil Engineering**	Course duration	6 semester
Course Title	**Civil Engineering Planning and Drawing**	Course Code	**CEPC306**
Subject offered in Semester	Third	Number of Credits	1 (L:1, T: 0, P: 0)
Prerequisites	NIL	Course Category	**PC**
Question distribution		Marks distribution	

Course Objectives:

Following are the objectives of this course:

- To learn basic principles Civil Engineering drawing.
- To know graphical representation of various components of Civil Engineering structure mainly building
- To draw complete plan and elevation of a building.
- To learn basics of Computer Aided Drawings.

Course Content:

Module	Distribution of unit
Module A	Unit 1 and 2
Module B	Unit 3 and 4
Module C	Unit 5

Unit – I Conventions and Symbols

- Conventions as per IS 962, symbols for different materials such as earthwork, brickwork, stonework, concrete, woodwork and glass.
- Graphical symbols for doors and windows, Abbreviations, symbols for sanitary and electrical installations.

- Types of lines-visible lines, centre line, hidden line, section line, dimension line, extension line, pointers, arrow head or dots. Appropriate size of lettering and numerals for titles, sub-titles, notes and dimensions.
- Types of scale- Monumental, Intimate, criteria for Proper Selection of scale for various types of drawing.
- Sizes of various standard papers/sheets.
- Reading and interpreting readymade Architectural building drawing (To be procured from Architect, Planning Consultants, Planning Engineer).

Unit– II Planning of Building

- Principles of planning for Residential and Public building- Aspect, Prospect, Orientation, Grouping, Privacy, Elegance, Flexibility, Circulation, Furniture requirements, Sanitation, Economy.
- Space requirement and norms for minimum dimension of different units in the residential and public buildings as per IS 962.
- Rules and bye-laws of sanctioning authorities for construction work.
- Plot area, built up area, super built up area, plinth area, carpet area, floor area and FAR (Floor Area Ratio).
- Line plans for residential building of minimum three rooms including water closet (WC), bath, kitchen and staircase as per principles of planning.
- Line plans for public building, school building & primary health centre,

Unit– III Drawing of Load Bearing Structure

- Drawing of Single storey Load Bearing residential building (2 BHK) with staircase.
- Data drawing –plan, elevation, section, site plan, schedule of openings, construction notes with specifications, area statement, Planning and design of staircase- Rise and Tread for residential and public building.
- Working drawing – developed plan, elevation, sections passing through staircase, and/or WC and bath.
- Foundation plan of Load bearing structure.
- Drawing with CAD - Draw commands, modify commands, layer commands.
- Area statement of building.

Unit – IV Culverts

- Introduction to culvert & its different components and types and specific use along with demonstration of a model
- Half sectional plan and half sectional elevation and side view of a single span slab culvert
- Half sectional elevation of a single span two hume-pipe culvert

Unit – V Steel connections & Steel Roof truss

- Plan elevation and side view of stanchion connected with base plate with gusset plate in concrete foundation
- Connection of main beam with secondary beam in one side
- Connection of beam with column
- Unequal column splicing
- Equal beam splicing
- Column bracket
- Introduction to truss – wooden (king post and queen post), steel (with angles and tubular truss) along with demonstration of model
- Details of a fink truss with welded/riveted joints and details of column connection
- drawing of a wooden king post and queen post truss of the building drawing in sheet

Suggested learning resources:

1. Shah. M.G. Kale, CM, Patki, S.Y., Building Drawing, Mcgraw Hill Publishing company Ltd. New Delhi.

2. Malik and Mayo, Civil Engineering Drawing, Computech Publication Ltd New Asian Publish- ers, New Delhi.

3. M. G. Shah and C. M. Kale, Principles of Perspective Drawing, Mcgraw Hill Publishing compa- ny Ltd. New Delhi.

4. Swamy, Kumara; Rao, N, Kameshwara, A ., Building Planning and Drawing, Charotar Publica- tion, Anand.

5. Bhavikatti, S. S., Building Construction, Vikas Publication House Pvt. Ltd., New Delhi.

6. Mantri, Sandip, A to Z Building Construction, Satya Prakashan, New Delhi.

7. Singh, Ajit, Working with Auto CAD 2000, Mcgraw Hill Publishing company Ltd. New Delhi.

8. Sane, Y.S., Planning and design of Building, Allied Publishers, New Delhi.

9. Venugopal, K. & Prabhu, Raja V., Engineering Drawing & AUTO-CAD, New Age International Publishers.

10. Jeyapoovan, T., Engineering Graphics using AUTO-CAD, Vikas Publication House Pvt. Ltd., New Delhi.

11. Jolhe, Engineering Drawing with an introduction to CAD, Tata McGraw Hill, New Delhi.

Course outcomes: After completing this course, student will be able to:

1. Interpret the symbols, signs and conventions from the given drawing.

2. Prepare line plans of residential and public buildings using principles of planning.

3. Prepare submission and working drawing for the given requirement of civil engineering structure (CAD may be used in addition to conventional drawing).

Assignment to be produced in Sketch Book [compulsory]

1	Draw various types of lines, graphical symbols for materials, doors and windows, symbols for sanitary, water supply and electrical installations and write abbreviations as per IS 962.
2	Study of a working drawing on Two BHK accommodation obtained from professional persons (Group activity) and make a report on this
3	a) Measure the units of existing building (Load Bearing / Frame structure/his or her own dwelling unit: home or house or flat).
	b) Draw line plan of measured existing building at **serial no 3a** to the suitable scale.
4	Draw line plan to suitable scale (Minimum 2BHK, flat roof, staircase, 2 nos WC and Bathroom(one attached), venrandah) [the same may be used in the drawing practical classes also]
	a) Residential single storied building (two plans)
	b) Apartment (two plans)
5	Draw line plans to suitable scale for any **three** Public Buildings from the following (School Building, Primary Health Centre, Bank, Post Office, Hostel, Restaurant, Community Hall and Library).
6	Draw three principal views of a channel section, I section, H section and angle section

Name of the Course	**Diploma in Civil Engineering**	Course duration	6 semester
Course Title	**Transportation Engineering**	Course Code	**CEPC307**
Subject offered in Semester	Third	Number of Credits	2 (L:2, T: 0, P: 0)
Prerequisites	NIL	Course Category	**PC**
Question distribution		Marks distribution	

Course Objectives: Following are the objectives of this course:

- To identify the types of roads as per IRC recommendations.
- To understand the geometrical design features of different highways.
- To perform different tests on road materials.
- To identify the components of railway tracks

Course Content:

Module	Distribution of unit
Module A	Unit 1 and 2
Module B	Unit 3 and 4
Module C	Unit 5

Unit – I Overview of Highway Engineering

- Role of transportation in the development of nation, Scope and Importance of roads in India and its' Characteristics.
- Different modes of transportation – land way, waterway, airway. Merits and demerits of roadway and railway;
- General classification of roads.
- Selection and factors affecting road alignment.

Unit– II Geometric Design of Highway

- Design speed, design vehicle, PCU, volume of traffic, terrain classification
- Camber: Definition, purpose, types as per IRC – recommendations.
- Kerbs, Road margin, road formation, right of way as per IRC – recommendations..
- Design speed and various factors affecting design speed as per IRC – recommendations.
- Gradient: Definition, types as per IRC – Recommendations, grade compensation at curve
- Sight distance: Definition, types IRC – recommendations, simple numerical.
- Curves: Necessity, types: Horizontal including transition curve, vertical curves [no design, no problems on curve design]
- Extra widening of roads: numerical examples.
- Super elevation: Definition, formula for calculating minimum and maximum Super elevation and method of providing super-elevation.
- Standards cross-sections of national highway in plain, embankment and cutting.

Unit– III Construction of Road Pavements

- **T**ypes of road materials and their Tests – Test on aggregates-Flakiness and Elongation Index tests, Angularity Number test, test on Bitumen- penetration, Ductility, Flash and Fire point test and Softening point test.
- Pavement – Definition, Types, Structural Components of pavement and their functions
- Construction of WBM road. Merits and demerits of WBM & WMM road.
- Construction of Flexible pavement / Bituminous Road, Types of Bitumen and its properties, emulsion, Cutback, Tar, Terms used in Bituminous Road - prime coat, tack coat, seal coat, premix carpet., penetration macadam, mastic asphalt, merits and Demerits of Bituminous Road.
- Cement concrete road - methods of construction, Alternate and Continuous Bay Method, joints in concrete pavement, filler and sealers, merits and demerits of concrete roads.

Unit– IV Basics of Railway Engineering

- Classification of Indian Railways, zones of Indian Railways
- Permanent way: Ideal requirement, Components; Rail Gauge, types, factors affecting selection of a gauge
- Rail, Rail Joints - requirements, types.
- Creep of rail: causes and prevention, wear in rail.
- Sleepers - functions and Requirement, types and sleeper density [numerical problems]
- Ballast - function and types, suitability.

• Rail fixtures and fastenings – fish plate, spikes, bolts, keys, bearing plates, chairs-types of anchors and anti-creepers.

Unit– V Track geometrics, Construction and Maintenance

• Alignment- Factors governing rail alignment.

• Track Cross sections – standard cross section of single and double line in cutting and embankment. Important terms - formation width, side drains, etc

- Railway Track Geometrics: Gradient, curves - types and factors affecting, grade compensation, super elevation, limits of Super elevation on curves, cant deficiency, negative cant, coning of wheel, tilting of rail.
- Branching of Tracks, Points and crossings, Turn out- types, components, functions and inspection. Track junctions: crossovers, scissor cross over, diamond crossing, track triangle.
- Station - Purpose, requirement of railway station, important technical terms, types of railway station, factors affecting site selection for railway station.
- Station yard: Classification- Passenger, goods, locomotive and marshalling yards. Function & drawbacks of marshalling yards.
- Track Maintenance- Necessity, Classification, Tools required for track maintenance with their functions, Organisation of track maintenance, Duties of permanent way inspector, gangmate and key man.

Suggested learning resources:

1. L.R. Kadiyali, Transportation Engineering, Khanna Book Publishing Co., Delhi (ISBN: 978-93- 82609-858) Edition 2018

2. Khanna S.K., Justo, C E G and Veeraragavan, A., Highway Engineering, Nem Chand and Brothers, Roorkee.

3. Arora, N. L., Transportation Engineering, Khanna Publishers, Delhi.

4. Saxena S C and Arora S P, A Textbook of Railway Engineering, Dhanpat Rai Publication.

5. Birdi, Ahuja, Road, Railways, Bridge and Tunnel Engg , Standard Book House, New Delhi.

6. Sharma, S.K., Principles, Practice and Design of Highway Engineering,, S. Chand Publication, New Delhi.

7. Duggal, Ajay K. and Puri, V. P., Laboratory Manual in Highway Engineering, New Age international (P) Limited, Publishers, New Delhi.

8. Subramanian, K.P., Highway, Railway, Airport and Harbour Engineering, Scitech Publications, Hyderabad.

9. Chandola, S.P., A Text Book of Transportation Engineering, S. Chand Publication.

10. Vazirani, V. N. & Chandola, S.P., Transportation Engineering Vol.-I, Khanna Publishers.

11. Upadhyaya, A. K., Transportation Engineering, S. K. Kataria & Son, New Delhi.

Course outcomes: After completing this course, student will be able to:

- Identify the types of roads as per IRC recommendations.
- Implement the geometrical design features of different highways.
- knowledge on different tests on road materials.
- Preliminary knowledge on different type of highway construction
- Identify the components of railway tracks.
- Identify the defects in railway tracks.

Name of the Course	**Diploma in Civil Engineering**	Course duration	6 semester
Course Title	**Civil Engineering Planning and Drawing Practices**	Course Code	**CEPC308S**
Subject offered in Semester	Third	Number of Credits	2 (L:0, T: 0, P: 4)
Prerequisites	NIL	Course Category	**PC**
Question distribution		Marks distribution	

Course Objectives:

Following are the objectives of this course:

- To learn the basic principles of Civil Engineering Drawing.
- To make graphical representation of various components of Civil Engineering structure.
- To draw complete plan and elevation of a building.
- To learn basics of Computer Aided Drawings.

List of Practicals/Drawings to be completed:

Drawing to be produced in Full Imperial Size Sheet (A1)	
1	Draw submission drawing to the scale 1:100 of a single storey load bearing residential building (2BHK) with flat Roof, staircase, 2 nos WC and Bathroom(min. one attached), one Varandah showing
	a)Developed plan, Front elevation, one side elevation
	b) Section passing through Stair **,** W.C. and Bath
	c) Foundation plan and schedule of openings.
	d) Site plan (1:200), area statement, construction notes.
2	Draw working drawing for above mentioned drawing at serial number 1 showing:
	a) excavation trench plan

	b) Foundation plan to the scale 1:50
	c) Detailed enlarged section of Lintel and Chajjas including standard reinforcement.
	d) Detailed enlarged section of RCC staircase and landing slab including standard reinforcement
	e) Plan of Roof slab showing ridge line, drainage, position of rain water pipe, etc.
3	Culverts (problems to be supplied by the subject teacher(s)
	a) Introduction to culvert & its different components and types and specific use along with demonstration of a model
	b)Half sectional plan and half sectional elevation and side view of a single span slab culvert
	c)Half sectional elevation of a single span two hume-pipe culvert
4	Steel connections [bolted or welded] (problems to be supplied by the subject teacher(s)
	a. Plan elevation and side view of stanchion connected with base plate with gusset plate in concrete foundation
	b. Connection of main beam with secondary beam in one side
	c. Connection of beam with column
	d. Unequal column splicing
	e. Equal beam splicing
	f. Column bracket
5	Steel Roof truss (problems to be supplied by the subject teacher(s) Study of different trusses used in construction and draw a welded fink truss [angle or tubular] having standard dimension and minute detailing

NOTE: <u>In addition to conventional approach to Civil Engineering Drawing, student may also take the help of CAD in preparing their sessional works if he/she desires so.</u>

Suggested learning resources:

1. Shah. M.G. Kale, CM, Patki, S.Y., Building Drawing, Mcgraw Hill Publishing
2. Malik and Mayo, Civil Engineering Drawing, Computech Publication Ltd
3. M. G. Shah and C. M. Kale, Principles of Perspective Drawing, Mcgraw Hill
4. Swamy, Kumara; Rao, N, Kameshwara, A ., Building Planning and Drawing, Charotar Publica- tion, Anand.
5. Bhavikatti, S. S., Building Construction, Vikas Publication House Pvt. Ltd., Delhi.
6. Mantri, Sandip, A to Z Building Construction, Satya Prakashan, New Delhi.
7. Singh, Ajit, Working with Auto CAD 2000, Mcgraw Hill Publishing company Ltd.

8. Sane, Y.S., Planning and design of Building, Allied Publishers, New Delhi.
9. Venugopal, K. & Prabhu, RAJA V., Engineering Drawing & AUTO-CAD, New Age International Publishers.
10. Jeyapoovan, T., Engineering Graphics using AUTO-CAD, Vikas Publication House Pvt. Ltd., New Delhi.
11. Jolhe, Engineering Drawing with an introduction to CAD, Tata McGraw Hill, New Delhi.

Course outcomes: After completing this course, student will be able to:

- Interpret the symbols, signs and conventions from the given drawing.
- Prepare line plans of residential and public buildings using principles of planning.
- Prepare working drawing for the given requirement of Civil engineering structure

Name of the Course	**Diploma in Civil Engineering**	Course duration	6 semester
Course Title	**Civil Engineering Laboratory I** **Module-I: Construction Material Lab**	Course Code	**CEPC309S/I**
Subject offered in Semester	Third	Number of Credits	1 (L:0, T: 0, P: 2)
Prerequisites	NIL	Course Category	**PC**
Question distribution		Marks distribution	

Course Objectives: Following are the objectives of this course:

- To learn about various construction materials, and understand their relevant characteristics.
- To be able to identify suitability of various materials for different construction purposes.
- To know about natural, artificial, and processed materials available for various purposes of construction activities.

List of practical to be performed:

1. Identify various sizes of available coarse aggregates from sample of 10 kg in laboratory and prepare report (60, 40, 20, 10 mm)
2. Identify the available construction materials in the laboratory on the basis of their sources.
3. Identify the grain distribution pattern in given sample of teak wood in the laboratory and draw the various patterns. (along and perpendicular to the grains)
4. Prepare the lime putty by mixing lime (1 kg) with water in appropriate proportion and prepare report on slaking of lime.
5. Identify various layers and types of soil in foundation pit by visiting a construction sites and prepare report consisting photographs and samples.
6. Select first class, second class and third-class bricks from the stack of bricks and prepare report on the basis of its properties.

7. Measure dimensions of 10 bricks and find average dimension and weight. Perform field tests - dropping, striking and scratching by nail and correlate the results obtained.
8. Identify different types of flooring tiles such as vitrified tiles, ceramic tiles, glazed tiles, mosaic tiles, anti-skid tiles, chequered tiles, paving blocks and prepare report about the specifications.
9. Apply the relevant termite chemical on given damaged sample of timber.
10. Identify the type of glasses from the given samples.
11. Apply two or more coats of selected paint on the prepared base of a given wall surface for the area of 1m x 1m using suitable brush/rollers adopting safe practices.
12. Prepare the cement mortar of proportion 1:3 or 1:6 using sand and apply on 1 m X 1 m surface as plastering
13. Prepare mortar using cement and Fly ash or Granite/marble polishing waste in the proportion 1:6 or 1:3 and apply on 1 m X 1 m surface as plastering

Suggested learning resources:

1. Ghose, D. N., Construction Materials, Tata McGraw Hill, New Delhi.

2. S.K. Sharma, Civil Engineering Construction Materials, Khanna Publishing House, Delhi

3. Varghese, P.C. , Building Materials, PHI learning, New Delhi.

4. Rangwala, S.C., Engineering Materials, Charator publisher, Ahemdabad.

5. Somayaji, Shan, Civil Engineering Materials, Pearson education, New Delhi.

6. Rajput, R.K, Engineering Materials, S. Chand and Co., New Delhi.

7. Sood H., Laboratory Manual on Testing of Engineering Materials, New Age Publishers, New Delhi.

8. Sharma C. P., Engineering Materials, PHI Learning, New Delhi.

9. Duggal, S. K, Building Materials, New International, New Delhi.

10. Singh, Parveen, Civil Engineering Materials, S.K. Kataria & sons

11. Soni, S.K., Building Materials and Constructions, S.K. Kataria & sons

12. Singh, Gurucharan, Building Construction & Materials, Standard Book House, New Delhi.

13. Gambhir, M.L. and Jamwal, NEHA, Lab Manual Building & Construction Materials, Tata McGraw Hill, New Delhi.

14. Subramanian, N., Building Materials, Oxford University Press.

Course outcomes: After completing this course, student will be able to:

1) Identify relevant construction materials.

2) Identify relevant natural construction materials.

3) Select relevant artificial construction materials.

4) Select relevant special type of construction materials.

5) Identify and use of processed construction materials.

Name of the Course	**Diploma in Civil Engineering**	Course duration	6 semester
Course Title	**Civil Engineering Laboratory I** **Module-II: Mechanics of Materials Lab**	Course Code	**CEPC309S/II**
Subject offered in Semester	Third	Number of Credits	1 (L:0, T: 0, P: 2)
Prerequisites	NIL	Course Category	**PC**
Question distribution		Marks distribution	

Objective:- Following are the objectives of this course:

- To know the procedure for the conduct of tensile and compressive strength.
- To understand the concept of stress and strain through testing of different materials.
- To calculate shear force, bending moment and their corresponding stresses.

List of Practicals to be performed:

- Study and understand the use and components of Universal Testing Machine (UTM).
- Perform Tension test on mild steel as per IS:432(1).
- Perform tension test on Tor steel as per IS:1608, IS:1139.
- Conduct compression test on sample test piece using Compression Testing Machine.
- Plot Shear force and Bending Moment diagrams for cantilever, simply supported beams
- Plot Shear force and Bending Moment diagrams for overhanging beams for different types of loads including moment loading.

Suggested learning resources:

1. Bedi D.S., Strength of Materials, Khanna Publishing House, New Delhi (Edition 2018)
2. Timoshenko, S., Strength of Materials, Vol. I, CBS, New Delhi.
3. Khurmi, R.S., Strength of Materials, S Chand and Co. Ltd. New Delhi.
4. Ramamurtham, S, Strength of Materials, Dhanpat Rai and sons, New Delhi.

5. Punmia B C, Strength of Materials, Laxmi Publications (p) Ltd. New Delhi.
6. Rattan S.S., Strength of Materials, McGraw Hill Education; New Delhi.
7. Bansal R K, Strength of Materials, Laxmi Publications.
8. Subramaniam R, Strength of Materials, Oxford University Press.
9. Ramamrutham.S, Theory of structures, Dhanpatrai & Sons.
10. Khurmi, R. S. , Theory of Structures S. Chand and Co., New Delhi.
11. Bhavikatti, S S , Structural Analysis Vol-1, Vikas Publishing House Pvt Ltd. New Delh
12. Bhattachariya, B., Engineering Mechanics, Oxford University Press.
13. Chakraborty, M., Strength of Materials, Kolkata.
14. Adarsh, Swaroop, Mechanics of Materials, New Age International Publishers.

Course outcomes: After completing this course, student will be able to:

- Test different Civil engineering materials on Universal Testing Machine.
- Analyse structural behaviour of materials under various loading conditions.
- Interpret shear force and bending moment diagrams for various types of beam sections and different loading conditions.

Name of the Course	**Diploma in Civil Engineering**	Course duration	6 semester
Course Title	**Civil Engineering Laboratory I** **Module-III: Concrete Technology Lab**	Course Code	**CEPC309S/III**
Subject offered in Semester	Third	Number of Credits	1 (L:0, T: 0, P: 2)
Prerequisites	NIL	Course Category	**PC**
Question distribution		Marks distribution	

Course Objectives: Following are the objectives of this course:

- To know properties of cement, aggregate and water used in concrete.
- To understand different characteristics of concrete.
- To learn about role of admixtures in concrete.

List of Practical to be performed: ***(At least 12 experiments should be performed)***

1	Determine fineness of cement by Blaine's air permeability apparatus Or by sieving.
2	Determine standard consistency of cement.
3	Determine initial and final setting times of cement.
4	Determine compressive strength of cement.
5	Determine silt content in sand.
6	Determine bulking of sand.
7	Determine bulk density of fine and coarse aggregates.
8	Determine water absorption of fine and coarse aggregates.
9	Determine surface moisture of fine and coarse aggregates.
10	Determine grading of fine aggregate and coarse aggregate by sieve analysis.

11	Determine workability of concrete by slump cone test.
12	Determine workability of concrete by compaction factor test.
13	To prepare concrete mix of a particular grade [nominal mix may be allowed] and determine compressive strength of concrete for 7 and 28 days.
14	Demonstration of NDT equipment .

Suggested learning resources:

1. Gambhir, M.L., Concrete Technology, Tata McGraw Hill Publishing Co. Ltd., Delhi.
2. Shetty, M.S., Concrete Technology, S. Chand and Co. Pvt. Ltd., Ram Nagar, Delhi.
3. Santhakumar, A. R., Concrete Technology, Oxford University Press, New Delhi.
4. Neville, A. M. and Brooks, J.J., Concrete Technology, Pearson Education Pvt. Ltd.
5. Neville, A. M., Concrete Technology, Pearson Education Pvt. Ltd., New Delhi.
6. Sood, H., Kulkarni P. D., Mittal L. N., Laboratory Manual in Concrete Technology, CBS Publish- ers, New Delhi.
7. Kulkarni, P. D., Ghosh, R. K. & Phull, V. R., New Age International Publishers.
8. Vazirani, V. N. & CHANDOLA, S. P., Concrete Technology, Khanna Publishers.
9. Laskar, Aminul Islam, Concrete Technology, Laxmi Publication Pvt. Lltd.
10. IS 456 : 2000
11. IS 10262: 2009
12. **SP 23: 1982**

Course outcomes: After completing this course, student will be able to:

- **determine the engineering properties of cement and its suitability in using in different site condition.**
- **determine the engineering properties of coarse aggregate and fine aggregate and their suitability in using in preparing design mix of concrete**
- **determine the engineering properties of fresh and hardened concrete**
- **perform non-destructive testings of concrete**

Name of the Course	**Diploma in Civil Engineering**	Course duration	6 semester
Course Title	**Civil Engineering Laboratory I** **Module-IV: Transportation Engineering Lab**	Course Code	**CEPC309S/IV**
Subject offered in Semester	Third	Number of Credits	1 (L:0, T: 0, P: 2)
Prerequisites	NIL	Course Category	**PC**
Question distribution		Marks distribution	

Course Objectives: Following are the objectives of this course:

- To identify the types of roads as per IRC recommendations.
- To understand the geometrical design features of different highways.
- To perform different tests on road materials.
- To identify the components of railway tracks.

List of Practicals to be performed [at least ten]:

1	Draw the sketches showing standard cross sections of Expressways, Freeways, NH/SH, MDR/ODR
2	Flakiness and Elongation Index of aggregates.
3	Aggregate impact value test
4	Los Angeles Abrasion test
5	Aggregate crushing value test
6	Softening point test of bitumen.

7	Penetration test of bitumen.
8	Flash and Fire Point test of bitumen.
9	Ductility test of Bitumen.
10	Visit the constructed road for visual inspection to identify defects and suggest remedial measures and Prepare the photographic report containing details
11	Visit the hill road constructed site to understand its components and prepare the photographic report containing details
12	Visit the road of any one type (flexible or rigid) to know the drainage condition and prepare a photographic report containing details.
13	Prepare the photographic report suggesting possible repairs and maintenance for a. flexible pavement and b. rigid pavement
14	Visit to railway track for visual inspection of fixtures, fasteners and yards and prepare a photographic report containing details.

Suggested learning resources:

1. L.R. Kadiyali, Transportation Engineering, Khanna Book Publishing Co., Delhi (ISBN: 978-93- 82609-858) Edition 2018

2. Khanna S.K., Justo, C E G and Veeraragavan, A., Highway Engineering, Nem Chand and Brothers, Roorkee.

3. Arora, N. L., Transportation Engineering, Khanna Publishers, Delhi.

4. Saxena S C and Arora S P, A Textbook of Railway Engineering, Dhanpat Rai Publication.

5. Birdi, Ahuja, Road, Railways, Bridge and Tunnel Engg , Standard Book House, New Delhi.

6. Sharma, S.K., Principles, Practice and Design of Highway Engineering,, S. Chand Publication, New Delhi.

7. Duggal, Ajay K. and Puri, V. P., Laboratory Manual in Highway Engineering, New Age international (P) Limited, Publishers, New Delhi.

8. Subramanian, K.P., Highway, Railway, Airport and Harbour Engineering, Scitech Publications, Hyderabad.

9. Chandola, S.P., A Text Book of Transportation Engineering, S. Chand Publication.

10. Vazirani, V. N. & Chandola, S. P.,Transportation Engineering Vol.-I, Khanna Publishers.

11. Upadhyaya. K., Transportation Engineering, S. K. Kataria & Son, New Delhi.

Course outcomes: After completing this course, student will be able to:

- Identify the types of roads as per IRC recommendations.
- Implement the geometrical design features of different highways.
- Perform different tests on road materials.
- Identify the components of railway tracks.
- Identify the defects in railway tracks

Name of the Course	**Diploma in Civil Engineering**	Course duration	6 semester
Course Title	**Internship-I after II nd Sem**	Course Code	**CEI310S**
Subject offered in Semester	**Third**	Number of Credits	**1 (L:0, T: 0, P: 0)**
Prerequisites	NIL	Course Category	**I**
Question distribution		Marks distribution	

Course Objectives: Following are the objectives of this course:

- To understand industrial environment and Civil Engineering activities.
- To get exposure to field level works.
- To get brief idea on drawings.
- To be accustomed with online courses.

NOTE: A**ctivity(s) from the following has to be performed and prepare a report on the activity together with presentation before the subject teacher.**

- **Activities may be arranged by the West Bengal State Council of Technical and Vocational Education & Skill Development.**
- **Board of Practical Studies, MSME or Department of Small Scale Industries or other engineering department of State Government may be involved. Initiative from the Department of Technical Education, Training and Skill Development is highly solicited.**
- **Activities centering Private organization in the arena of civil engineering construction/ planning and design/ supervision/marketing etc may also be considered.**
- **It may be arranged in-campus or off-campus; online or offline mode or blended mode.**
- **Activities may be conducted continuously for stipulated period of time or may be arranged in a staggered fashion – in the later case Saturday and Sunday may be utilized for the Internship Program and accordingly class schedule will have be to arranged.**

- **Activities may be performed by a group of students or may be done by individually under the guidance of subject teacher(s)**

After the 2nd Semester students are required to be involved in Inter/ Intra Institutional Activities viz: training and simulation program with different institute like workshop of ITI , other polytechnics, other technical institute, institutions; soft skill training organized by the Training and Placement Cell of the respective institute; contribution at innovation/entrepreneurship cell of the institute, participation in the workshop/competition etc; learning at departmental laboratory/institutional workshop. It may also cover subjects taught in 1st, 2nd and 3rd semester (on any one or combination of the following subjects) a. science and humanities b. engineering mechanics c. computer application d. engineering graphics e. environmental science f. departmental subjects g. any other current topic on latest trend in civil engineering.

Activity(s) may include -

- Training / Skill Development from any institute [private/govt/govt aided] or from any individual expert on topics related to civil engineering field;
- Soft skill training organized by Training and Placement Cell of the respective institutions;
- Review study and preparation of report based on traditional library or digital library or internet ;
- Learning at Departmental Lab/ Institutional workshop;
- Working for consultancy job/project work within the institutes or outside the institute;
- Any training on open online learning (like internshala, Moocs etc) - **NOT Compulsory TO PRODUCE ANY CERTIFICATE from the training organisation or trainer;**
- Visit to construction site and prepare a report along with labeled photograph or drawing ;
- An industrial visit to a testing laboratory of civil engineering materials and prepare a report on visit
- Industrial visit to a manufacturing unit related to the civil engineering construction or any other engineering materials;
- Participation in a socio-economic survey on any current topics or like and know the methodology
- Conducting a market survey on determination/variation on rate of any civil

engineering materials and preparation a report on the same

If a subject teacher or any other teachers from civil engineering department or any other department of respective polytechnic desires to involve students to study/review/visit/etc etc on any subject or topic not covered in the list produced above as per his/her convenience then he/she should be given authority to proceed with the same.

After completion of each internship, the student should prepare a comprehensive report to indicate what he/she has observed and learnt in the training period. The student may contact Industrial supervisor/Faculty member/TPO for assigning topics and problems and should prepare the final report on assigned topics. The training report should be signed by the Industrial supervisor/Internship Faculty member, TPO and HOD.

The internship report will be evaluated on the basis of following criteria (as applicable):

Sl no	Criteria for evaluation of Internship Report
1.	Origianlity
2.	Adequacy and purposeful writeup
3	Orgainizaton, format, drawing,sketches style language
4	Practical applications and relationship with basic theory
5	Concepts taught in the course outcomes
6	Attendance record, daily diary,quality of the internship report

Seminars must be arranged for the students based on his/her training report, before an internal committee constituted by the concerned department of the institute. The evaluation will be based on the following criteria:

Sl no	Criteria for evaluation of Internship Report
1	Quality of content presented
2	Proper planning for presentation
3	Effectiveness of presentation
4	Depth of knowledge and skills
5	Viva voce

Course outcomes: After completing this course, student will be able to:

- Understand the overall brief knowledge of industrial activities.
- Relate the industrial activities with his/her different courses of Diploma in Civil Engineering.

West Bengal State Council of Technical & Vocational Education and Skill Development
(Technical Education Division)



Syllabus
of

Diploma in Civil Engineering [CE]

Part-II (4th Semester)

Revised 2022

CURRICULAR STRUCTURE FOR PART – II SECOND SEMESTER (FOURTH SEMESTER) OF THE FULL-TIME DIPLOMA COURSE IN CIVIL ENGINEERING

SL. NO.	SUBJECT CODE	SUBJECT OF STUDY	HOURS PER WEEK			CREDITS
		THEORETICAL PAPERS	LECTURE	TUTORIAL	PRACTICAL	
1	CEPC401	Hydraulics	2	0	0	2
2	CEPC402	Advanced Surveying	3	0	0	3
3	CEPC403	Theory of Structure	3	0	0	3
4	CEPC404	Geotechnical Engineering	3	0	0	3
5	CEPC405	Design of RCC and Steel Structure	3	0	0	3
		LABORATORY/SESSIONAL PAPERS				
6	CEPC406S	Basic Surveying Field Practices	0	0	2	1
	CEPC407S : Civil Engineering Lab- II (consisting of following two Modules with 4 Practical Classes and 2 Credits)					
7	CEPC407S/I	Module-V: Hydraulics Lab	0	0	2	1
8	CEPC407S/II	Module-VI: Geotechnical Engineering Lab	0	0	2	1
		ELECTIVE & PROJECT				
9	CEPE408	Elective - I:one subject out of Precast and Prestressed Concrete (subject code: CEPE 408/I)/Rural Construction Technology (subject code: CEPE 408/II)	3	0	0	3
10	CEPR409S	Minor Project	0	0	4	2
		TOTAL	17	0	10	22

NOTE: -

1. All subjects (theoretical as well as sessional/practical) are full paper with 100 marks in aggregate as per AICTE and WBSCT&VE&SD
2. Basic Surveying Field Practices may also be conducted in 2-3 weeks field work continously instead of having 2 practical class per week for the entire semester.

Name of the Course	**Diploma in Civil Engineering**	Course duration	6 semester
Course Title	**Hydraulics**	Course Code	**CEPC401**
Subject offered in Semester	**Fourth**	Number of Credits	2 (L:2, T: 0, P: 0)
Prerequisites	NIL	Course Category	**PC**
Question distribution	As per standing norms of WBSCT&VE&SD	Marks distribution	As per standing norms of WBSCT&VE&SD

Course Objectives: Following are the objectives of this course:

- To understand parameters associated with fluid flow and hydrostatic pressure.
- To know head loss and water hammer in fluid flowing through pipes.
- To learn different types of pumps and their uses.

Module /Group [as per directives from WBSCT&VE&SD in framing questions of end semester]	Distribution of unit
Module A/Group A	Unit I and II
Module B/Group B	Unit III
Module C /Group C	Unit IV and V

Course Content:

Unit – I Pressure measurement and Hydrostatic pressure

- Technical terms used in Hydraulics–fluid, fluid mechanics, hydraulics, hydrostatic and hydrodynamics - ideal and real fluid, application of hydraulics.

- Physical properties of fluid – density-specific volume, specific gravity, surface tension, capillarity, viscosity - Newton's law of viscosity.
- Various types of pressure – Atmospheric Pressure, Gauge Pressure, Absolute Pressure, Vacuum Pressure. Concept of Pressure head and its unit, Pascal's law of fluid pressure and its uses. [simple numerical problems]
- Measurement of pressure at a point and differential Pressure between two points by different methods. [simple numerical problems]
- Variation of pressure with depth, Pressure diagram, hydrostatic pressure and center of pressure on immersed surfaces inclined at any angle and on tank walls. Determination of total pressure and center of pressure on sides and bottom of water tanks, sides and bottom of tanks containing two liquids, vertical surface in contact with liquid on either side. [simple numerical problems]

Unit– II Fluid Flow Parameters

- Types of flow – Gravity and pressure flow, Uniform, Non-uniform, Steady, Unsteady flow. Reynolds number, Laminar, Turbulent.
- Discharge and its unit, continuity equation of flow.
- Energy of flowing liquid: potential, kinetic and pressure energy.
- Bernoulli's theorem: statement, assumptions, equation [no deduction]
- simple numerical problems

Unit– III Flow through Pipes

- Major head loss in pipe: Frictional loss and its computation by Darcy's Weisbach equation [no deduction, simple numerical problems]
- Minor losses in pipe: loss at entrance, exit, sudden contraction, sudden enlargement and fittings. [no deduction, simple numerical problems]
- Flow through pipes in series, pipes in parallel and Dupuit's equation for equivalent pipe. Hydraulic gradient line and total energy line. [no deduction, simple numerical problems]
- Water hammer in pipes: concept only
- Discharge measuring device for pipe flow: Venturimeter - construction and working, orifice meter- construction and working [no deduction, simple numerical problems]
- Discharge measurement using Orifice, Hydraulic Coefficients of Orifice. [no deduction, simple numerical problems]

Unit– IV Flow through Open Channel

- Geometrical properties of channel section: Wetted area, wetted perimeter, hydraulic radius for rectangular and trapezoidal channel section.

- Determination of discharge by Chezy's equation and Manning's equation. [no deduction, simple numerical problems]
- Conditions for most economical rectangular and trapezoidal channel section. [no deduction, simple numerical problems]
- Discharge measuring devices: Triangular and rectangular Notches [no deduction, simple numerical problems]
- Velocity measurement devices: current meter, floats and Pitot's tube. [no deduction, simple numerical problems]
- Specific energy diagram, Froudes' Number, critical flow, super critical flow and subcritical flow, critical depth of flow, critical velocity of flow, minimum specific energy, concept of sequent depth. [numerical problems only]
- Hydraulic jump, condition of its occurrence, finding alternate depth of flow [simple numerical problems]

Unit– V Hydraulic Pumps

- Concept of pump, Types of pump - centrifugal, reciprocating, submersible.
- Centrifugal pump: components and working
- Reciprocating pump: single acting and double acting, components and working.
- Suction head, delivery head, static head, Manometric head Power of centrifugal pump
- Selection and choice of pump.

Suggested learning resources:

1. Modi, P. N.and Seth, S.M., Hydraulics and Fluid Mechanics, Standard book house, Delhi.

2. S.S. Rattan, Fluid Mechanics & Hydraulic Machines, Khanna Book Publishing Co., New Delhi

3. Ramamrutham, and Narayan, R., Hydraulics, Fluid Mechanics and Fluid Machines, Dhanpat Rai Publishing Company, New Delhi.

4. Khurmi R S, Hydraulics, Fluid Mechanics, Hydraulic machines, S. Chand Publishers

5. Rajput, R K, Fluid Mechanics, S Chand, New Delhi.

6. Ojha, C S P, Berndtsson, R, and Chandramoulli P. N., Fluid Mechanics and Machinery, Oxford University Press, New Delhi.

7. Upadhyay A.K., Fluid Mechanics(Hydraulics), S.K.Kataria & Sons.

8. Bansal R k, A Text Book of Fluid Mechanics, Laxmi Publication (P) Ltd.

Course outcomes: After completing this course, student will be able to:

- Measure pressure and determine total hydrostatic pressure for different conditions.
- Understand various parameters associated with fluid flow.
- Determine head loss of fluid flow through pipes.
- Find the fluid flow parameters in open channels and calculate discharge.
- Select relevant hydraulic pumps for different applications.

Name of the Course	**Diploma in Civil Engineering**	Course duration	6 semester
Course Title	**Advanced Surveying**	Course Code	**CEPC402**
Subject offered in Semester	**Fourth**	Number of Credits	3 (L:3, T: 0, P: 0)
Prerequisites	NIL	Course Category	**PC**
Question distribution	As per standing norms of WBSCT&VE&SD	Marks distribution	As per standing norms of WBSCT&VE&SD

Course Objective:- Following are the objectives of this course:

- To know methods of Theodolite Surveying and its use.
- To learn Tacheometric Surveying and Curve Setting.
- To understand the Principles of Electronic Distance Measurement equipment and Total station and their use.
- To know the concept of Remote Sensing, GPS and GIS.

Module /Group [as per directives from WBSCT&VE&SD in framing questions of end semester]	Distribution of unit
Module A/Group A	Unit I , IV and V
Module B/Group B	Unit II
Module C /Group C	Unit III

Contents:

Unit–I Measurement of Area and Volume

- Components and use of Digital Planimeter.

- Calculation of cross section area - i) From field book entry, ii) From plotted Plan -Mid ordinate rule, The average ordinate rule, The trapezoidal rule, Simpson's rule.
- Computation of volume - i) The trapezoidal rule (Av. End area rule), ii) prismoidal formula.
- Measurement of volume of reservoir from contour map, volume of earth work involved in highway and canal construction.
- Simple numerical problems.

Unit– II Theodolite Surveying

- Definition, Types and uses of Theodolite, Components of transit Theodolite and their functions, Reading the Vernier of transit Theodolite.
- Technical terms- Swinging the Telescope, Centering, Transiting, Face left, Face right, Changing Face.
- Fundamental Lines of transit Theodolite and their relationship
- Temporary adjustment of transit Theodolite.
- Measurement of horizontal angle- Direct, Repetition and Reiteration method
- Measurement of magnetic bearing of a line, Prolonging and ranging a line
- Measurement of deflection angle.
- Measurement of vertical Angle.
- Theodolite traversing by Included angle method, Deflection angle method and Magnetic bearing Method.
- Checks for open and closed traverse.
- Calculations of Magnetic bearing from angles.
- Traverse computation-Latitude, Departure, Consecutive coordinates, Independent coordinates, balancing the traverse by Bowditch's rule Transit rule and Third Rule, Gale's Traverse table computation.
- Errors in Theodolite Surveying- Instrumental, Personal and Natural.
- Simple numerical problems

Unit– III Tacheometric Surveying and Curve Setting

- Principles of Tacheometry, Instruments used in Tacheometry, Characteristics of a Tacheometer, Anallatic lens, uses of a Tacheometer.
- Tacheometric formula for horizontal distance with telescope horizontal and staff vertical.

- Field method for determining constants of tacheometer, Determining horizontal and vertical distances with tacheometer by fixed hair method and staff held vertical and nortmal to the line of collimation (both angle of elevation and depression)
- Limitations of Tacheometry Survey.
- Types of curves used in roads and railway alignments. Designation of curves, Properties of Simple Circular Curve.
- Setting simple circular curve by offsets from long chord and Rankine's method of deflection angle.
- Simple numerical problems.

Unit– IV Advanced Surveying Equipments

- Principle of Electronic Distance Meter (EDM), Types, its component parts and their Functions, use of EDM.
- Use of micro optic Theodolite and Electronic Digital Theodolite.
- Total Station: Features, Basic components, Functional operation, Use, Description and use of Function keys.
- Measurements of Horizontal angles, vertical angles, distances and coordinates using Total Station, Traversing, Profile Survey and Contouring with Total Station.
- No numerical problems.

Unit– V Remote Sensing, GPS and GIS

- Remote Sensing: Overview, Principles, Components, Process, Types of Remote sensing system, sensors used in practice mainly in India, Applications of remote sensing in Civil engineering: land use / Land cover, mapping, disaster management, .
- Global Positioning System (G.P.S.): Definition, Overview, Function, Principle, Application.
- Introduction to Differential Global Positioning System(D.G.P.S)
- Geographic Information System (GIS): Definition, Overview, Components, Applications, Software for GIS.
- Introduction to Drone Surveying.
- No numerical problems.

Suggested learning resources :

1. Kanetkar, T. P.; Kulkarni, S. V., Surveying and Levelling Part I and II, Pune Vidyarthi Gruh Prakashan, Pune.

2. Basak, N. N.,Surveying and Levelling, McGraw Hill Education (India) Pvt. Ltd.,Noida.
3. Duggal, S. K., Survey I and Survey II, Tata McGraw Hill Education Pvt. Ltd., Noida.
4. Saikia, M D.; Das. B.M.; Das. M.M., Surveying PHI Learning Pvt. Ltd., New Delhi.
5. Subramanian, R., Surveying and Levelling, Oxford University Press. New Delhi.
6. Punmia, B.C, Jain, Ashok Kumar; Jain, Arun Kumar, Surveying Vol. I and Surveying Vol. II, Laxmi Publications Pvt. Ltd., New Delhi.
7. Rao, P. Venugopala Akella, Vijayalakshmi, Textbook of Surveying, PHI Learning Pvt. Ltd., New Delhi.
8. Venkatramaiah, C, Textbook of Surveying, Universities Press, Hyderabad.
9. Anderson, James M and Mikhail, Edward M, Surveying theory and practice, Mc Graw Hill Education, Noida.
10. Aloke De, Plane Surveying, S.Chand Publications, New Delhi.
11. C.L. Kochher, Surveying, Dhanpat Rai Publication, New Delhi.
12. Dr. N.R. Chandak and Prof. H.R. Kumavat,Surveying, SK Kataria & Sons, New Delhi
13. Saurabh Kumar Soni, Surveying-II, SK Kataria & Sons, New Delhi

Course outcomes: After completing this course, student will be able to:

- Use digital Planimeter to calculate the areas
- Prepare plans using Theodolite surveys.
- Find distances and elevations using Tacheometer.
- Prepare plans using Total Station instrument.
- Locate coordinates of stations using GPS.

Name of the Course	**Diploma in Civil Engineering**	Course duration	6 semester
Course Title	**Theory of Structure**	Course Code	**CEPC403**
Subject offered in Semester	Fourth	Number of Credits	3 (L:3, T: 0, P: 0)
Prerequisites	NIL	Course Category	**PC**
Question distribution	As per standing norms of WBSCT&VE&SD	Marks distribution	As per standing norms of WBSCT&VE&SD

Objective:- Following are the objectives of this course:

- To understand the concept of buckling loads for short and long columns.
- To learn concept of eccentric loading and stresses in vertical members like column, chimneys, dam.
- To analyze beams using various methods like slope deflection, three moment, and moment distribution.

Module /Group [as per directives from WBSCT&VE&SD in framing questions of end semester]	Distribution of unit
Module A/Group A	Unit I and II
Module B/Group B	Unit III and V
Module C/Group C	Unit IV

Contents:

Unit– I Column

Concept of compression member, short and long column, Effective length, Radius of gyration, Slenderness ratio, Types of end condition for columns, Buckling of axially loaded columns.

- Euler's theory, assumptions made in Euler's theory and its limitations, Application of Euler's equation to calculate buckling load.
- Rankine's formula and its application to calculate crippling load.
- Concept of working load/safe load, design load and factor of safety.
- Simple numerical problems.

Unit – II Direct and Bending Stresses in vertical members

- Introduction to axial and eccentric loads, concept of combined stress, eccentricity about one principal axis only, nature of stresses, Maximum and minimum stresses, resultant stresses distribution diagram.
- Condition for no tension or zero stress at extreme fiber, Limit of eccentricity, core of section for rectangular and circular cross sections, Middle third rule (without derivation).
- Chimneys of circular cross section subjected to wind pressure, Maximum and minimum stresses, resultant stresses distribution diagram at base.
- Analysis of dams subjected to horizontal water pressure, conditions of stability, Maximum and minimum stresses (Principal stresses excluded), resultant stresses distribution diagram at base.
- Simple numerical problems.

Unit – III Slope and Deflection

- Concept of slope and deflection, stiffness of beams, Relation among bending moment, slope, deflection and radius of curvature, (no derivation).
- Moment Area method to find slope and deflection of cantilever and simply supported beams subjected to concentrated load and uniformly distributed load on entire span/ partial span.
- Simple numerical problems.

Unit- IV Fixed and Continuous Beam

- Concept of fixity, effect of fixity, advantages and disadvantages of fixed beam over simply supported beam.
- Principle of superposition, application of standard formulae in finding end moments, end reactions and drawing S.F. and B.M. diagrams for a fixed beam subjected to point loads, uniformly distributed loads on entire span.
- Continuous Beam: Effect of continuity, nature of moments induced due to continuity, concept of deflected shape, practical examples.
- Clapeyron's theorem of three moment (no derivation), Application of Clapeyron's theorem maximum up to three spans and two unknown support moment only [no end is fixed in continuous beam], Support at same level [no yield] spans having same

and uniform moment of inertia subjected to concentrated loads and uniformly distributed loads over entire span, Drawing SF diagrams showing point of contraflexure, shear and BM diagrams showing net BM and point of contraflexure for continuous beams.

- Simple numerical problems.

Unit–V Moment distribution method

- Introduction to moment distribution method, sign convention, Carry over factor, stiffness factor, distribution factor.
- Application of moment distribution method to continuous beams subjected to concentrated loads and uniformly distributed load over entire span having same moment of inertia, supports at same level [no yield], up to two spans.
- Introduction to portal frames (No derivation, no numerical problem).
- Simple numerical problems.

Suggested learning resources:

1. Ramamrutham.S, Theory of structures, Dhanpatrai & Sons.
2. Khurmi, R. S. , Theory of Structures S. Chand and Co., New Delhi.
3. Bhavikatti, S S , Structural Analysis Vol-1, Vikas Publishing House Pvt Ltd.New Delhi.
4. Junnarkar, S. B., Mechanics of structures, Volume-I and II Charotar Publishing House, Anand.
5. Pandit, G.S. and Gupta, S.P., Theory of Structures Vol I & II, Tata McGraw Hill, New Delhi.
6. Agor R, Structural Analysis, Khanna Publishing House, Delhi.
7. Vazirani and Ratwani, Part I & Part II, Khanna Publishing House, Delhi.
8. Dr. Punmia, B.C, Jain A. K., Theory of Structure, Laxmi Publication

Course outcomes: After completing this course, student will be able to:

- Analyze the column for various loading and end conditions.
- Analyze stresses induced in vertical member subjected to direct and eccentric loads/transverse loads.
- Analyze slope and Deflection in fixed and continuous beams.
- Analyze continuous beam under different loading conditions using the principles of Three Moments.
- Analyze continuous beam using Moment Distribution Method under different loading conditions.

Name of the Course	**Diploma in Civil Engineering**	Course duration	6 semester
Course Title	**Geotechnical Engineering**	Course Code	**CEPC404**
Subject offered in Semester	Fourth	Number of Credits	3 (L:3, T: 0, P: 0)
Prerequisites	NIL	Course Category	**PC**
Question distribution	As per standing norms of WBSCT&VE&SD	Marks distribution	As per standing norms of WBSCT&VE&SD

Course Objectives: Following are the objectives of this course:

- To understand and determine physical and index properties and classification of soil
- To estimate permeability and shear strength of soil
- To know the load bearing capacity of soil
- To learn various soil stabilization and compaction, consolidation, soil exploration methods

Module /Group [as per directives from WBSCT&VE&SD in framing questions of end semester]	Distribution of unit
Module A/Group A	Unit I and II
Module B/Group B	Unit III and IV
Module C /Group C	Unit V

Course Content:

Unit – I: Overview of Geology and Geotechnical Engineering

- Introduction, Definition of a rock: Classification based on their genesis (mode of origin), formation. Classification and engineering uses of igneous, sedimentary and metamorphic rocks.
- Importance of soil as construction material and as supporting medium for structures.
- Field application of geotechnical engineering: for foundation design, pavement design, design of earth retaining structures, design of earthen dam.

Unit– II: Physical and Index Properties of Soil

- Soil as a three phase system, water content, purpose and determination of water content by oven drying method as per BIS code, void ratio, porosity and degree of saturation, density index, Unit weight of soil mass – bulk unit weight, dry unit weight, unit weight of solids, saturated unit weight, submerged unit weight and their interrelationship, purpose and determination of bulk unit weight and dry unit weight by core cutter and sand replacement method, Determination of specific gravity by pycnometer.
- Consistency of soil, Atterberg limits of consistency: Liquid limit, plastic limit and shrinkage limit. Plasticity index, liquidity index, consistency index and toughness index and their significance.
- Particle size distribution test and plotting of curve, uniformity coefficient and coefficient of curvature, Determination of effective diameter of soil, well graded and uniformly graded soils
- Importance of soil classification system, BIS classification of soil.
- Simple numerical problems

Unit– III Permeability and Shear Strength of Soil

- Definition of permeability, Darcy's law of permeability, coefficient of permeability, factors affecting permeability, determination of coefficient of permeability by constant head and falling head tests, simple problems to determine coefficient of permeability.
- Seepage through earthen structures, seepage velocity, seepage pressure, phreatic line, flow lines, equipotential lines, characteristics and application of flow net **(No numerical problems**).
- Shear failure of soil, concept of shear strength of soil. Components of shearing resistance of soil – cohesion, internal friction. Mohr-Coulomb failure theory, Strength envelope, strength equation for purely cohesive and cohesion less soils. laboratory methods on determination of shear strength parameters -Direct shear, triaxial, unconfined compression test and vane shear test [conception only] – their application and significance, unconfined compressive strength

- Simple numerical problems.

Unit– IV Bearing Capacity of Soil and Earth Pressure

- Bearing capacity of shallow foundation, Concept of bearing capacity, ultimate bearing capacity, safe bearing capacity and allowable bearing pressure. Introduction to Terzaghi's analysis and assumptions, effect of water table on bearing capacity, determination of bearing capacity as per BIS.
- Field methods for determination of bearing capacity – Plate load Test and Standard Penetration Test. Test procedures as per IS:1888 & IS:2131.
- Introduction to deep foundation – pile foundation, caisson and well foundation. Determination of bearing capacity of pile foundation – dynamic formula (Engineering News formula and modified Hilley's formula) and static formula – only formula(no numerical problem based on static formula), simple numerical problems based on dynamic formula only, negative skin friction on piles, group action of piles.
- Definition of earth pressure, Active and Passive earth pressure for horizontal backfill (no inclination) including surcharge on moist soil only, coefficient of earth pressure, Rankine's theory and assumptions, extension of earth pressure theory to cohesive soil only for horizontal backfill including surcharge on moist soil [ground water table is well below the foundation level]
- Simple numerical problems.

Unit– V Compaction, consolidation, stabilization and exploration of soil

- Concept of compaction and its importance, Standard and Modified proctor test as per IS code, Plotting of Compaction curve for determining: Optimum moisture content (OMC), maximum dry density(MDD), Zero air voids line. Factors affecting compaction, field methods of compaction – rolling, ramming and vibration. Suitability of various compaction equipments- smooth wheel roller, sheep foot roller, pneumatic tyred roller, Rammer and Vibrator, Difference between compaction and consolidation.
- Preliminary concept on one dimensional consolidation and consolidation settlement (simple numerical problems on compaction but no numerical problems on consolidation settlement).
- Concept of soil stabilization, necessity of soil stabilization, different methods of soil stabilization.
- Necessity of site investigation and soil exploration: Types of exploration, criteria for deciding the location and number of test pits and bores, sample soil exploration report, Field identification of soil – dry strength test, dilatancy test and toughness test.

Suggested learning resources:

1. Punmia, B.C., Soil Mechanics and Foundation Engineering, Laxmi Publication, Delhi.
2. Murthy, V.N.S., A text book of soil mechanics and foundation Engineering, CBS Publishers & Distributors Pvt. Ltd., New Delhi.
3. Moitra, Debashis, Geotechnical Engineering, University Press
4. Ramamurthy, T.N. & Sitharam,T.G., Geotechnical Engineering(Soil Mechanics), S Chand and Company LTD., New Delhi.
5. Raj, P. Purushothama, Soil Mechanics and Foundation Engineering, Pearson India, New Delhi.
6. Kasamalkar, B. J., Geotechnical Engineering, Pune Vidyarthi Griha Prakashan, Pune.
7. Arora K R, Soil Mechanics and Foundation Engineering, Standard Publisher.
8. A.K.Upadhyay, Soil and Foundation Engineering, S.K.Kataria & Sons

Course outcomes: After completing this course, student will be able to:

- Identify types of rocks and sub soil strata of earth.
- Interpret the physical properties of soil related to given construction activities.
- Use the results of permeability and shear strength test for foundation analysis.
- Interpret soil bearing capacity and earth pressure results.
- Compute optimum values for moisture content for maximum dry density of soil through various tests and settlement due to one dimensional consolidation

Name of the Course	**Diploma in Civil Engineering**	Course duration	6 semester
Course Title	**Design of RCC and Steel Structure**	Course Code	**CEPC405**
Subject offered in Semester	Fourth	Number of Credits	3 (L:3, T: 0, P: 0)
Prerequisites	Basic concept of Mechanics of Materials and Theory of Structure	Course Category	**PC**
Question distribution	As per standing norms of WBSCT&VE&SD	Marks distribution	As per standing norms of WBSCT&VE&SD

Course Objectives: Following are the objectives of this course:

- To understand design of RCC elements - RCC beams (rectangular and T beam), lintel
- To learn connection design of steel structure
- To learn the concept of limit state design of steel beams

Module /Group [as per directives from WBSCT&VE&SD in framing questions of end semester]	Distribution of unit
Module A/Group A	Unit I and II
Module B/Group B	Unit III
Module C /Group C	Unit IV and V

Course Content:

Unit I: Introduction

- Aim of design, Concept of R.C.C., Necessity of steel as reinforcement and its position in a Simply Supported and continuous member.
- Design- Functional Design & Structural Design.
- Scope of a Structural Designer- Structural Planning, Calculation of loads, Analysis, Design & Detailing, Advantages of symmetrical planning over un-symmetrical planning
- Loads: Dead load, Imposed load, Wind loads & other loads as per IS 875(Part-I to Part III)
- Earthquake forces. – IS 1893- (Part -1), Seismic zones, Combination of loads.
    - Introduction of IS 456 and SP-16
    - Detailing- introduction of SP-34 & IS 13920.
    - Necessity of IS 13920
- Definition, types of limit states, partial safety factors for materials strength, characteristic strength ,characteristic load, design load. Loading on structure as per I.S 875.

Unit II: Analysis and Design of Singly and Doubly Reinforced Sections by LSM and Shear, Bond and Development Length (LSM)

- Limit State of collapse (Flexure), Assumptions, stress-Strain relationship for concrete and steel, neutral axis, Stress block diagram and Strain diagram for singly reinforced section.
- Concept of under- reinforced, over-reinforced and balanced section, neutral axis co-efficient, limiting value of moment of resistance and limiting percentage of steel required for balanced singly R.C. Section.
- Simple numerical problems on determining design constants, moment of resistance and area of steel.
- General features, necessity of providing doubly reinforced Section, reinforcement limitations.
- Analysis of doubly reinforced section, strain diagram, stress diagram, depth of neutral axis, moment of resistance of the section.
- Simple numerical problems on finding moment of resistance and design of beam sections.
- Nominal Shear stress in R.C. Section, design shear strength of concrete, Maximum shear stress, Design of shear reinforcement, Minimum shear reinforcement, forms of shear reinforcement.
- Bond and types of bond, Bond Stress, check for bond stress, Development length in tension and compression, anchorage value for hooks 90° bend and 45° bend Standard Lapping of bars, check for development length.
- Simple numerical problems on deciding whether shear reinforcement is required or not, check for adequacy of the section in shear. Design of shear reinforcement;

Minimum shear reinforcement in beams; Determination of Development length required for tension reinforcement of cantilevers beam and slab, check for development length.

Unit III: **Analysis and Design of T-Beam and Lintel by LSM**

- General features, advantages, effective width of flange as per IS:456 code provisions.
- Analysis of singly reinforced T-Beam, strain diagram & stress diagram, depth of neutral axis, moment of resistance of T-beam Section
- Designing a T-beam for moment and shear for Neutral axis within or up to flange bottom [Simple numerical problems on deciding effective flange width, on finding moment of resistance, design a T beam for the case of neutral axis lies within the flange only]
- Design of lintel for flexure and shear (consider all possible cases of loading]) - [Numerical problems on design of lintel for flexure and shear – for triangular or rectangular loading only]

Unit IV: Introduction and Design of Steel Connections and Detailing by LSM

- General - Advantages and disadvantages of steel as construction material. Types of sections, Grades of steel(IS 2062) and strength characteristics; Use of steel table (SP6-Part1).
- Geometrical properties of gross and effective cross sections– Classification of Cross Sections as per IS:800-2007– Internal, external (outstands) and tapered elements of sections– Maximum Effective Slenderness ratio.
- General- Types of connections – Bolted, Riveted and Welded connections– Rigid and Flexible connections– Components of connections– Basic requirements of connections- Clearance for holes– Minimum and Maximum spacing of fasteners– Minimum edge/ end distances– Requirements of Tacking fasteners.
- Bolted Connection– Types of bolts– Bearing type Bolts– Nominal and Design shear strengths of bolts– Reduction factors for Long joints, Large grip lengths, Thick packing plates– Nominal and Design bearing strengths of bolts– Reduction factors for over sized and slotted holes– Nominal and Design tensile strengths (tension capacity) of bolts-Simple problems.
- Welded Connection- Types of welds– Fillet welds– Minimum and maximum sizes– Effective length of weld- Fillet welds on inclined faces–-Design strengths of shop/site welds– Butt welds– Effective throat thickness and effective length of butt weld-Simple problems.

Unit- V: Design Of Steel Beams For BM and SF by L.S.M

- General-Formation of Plastic hinges in Flexural members– Plastic Moment of Resistance and Plastic Modulus of Sections– Shape Factors of rectangular / circular/ I / T-Sections– Collapse load.

- Effective span of Beams, Design strength of bending,(Flexure), Limiting deflection of beams –Design of laterally supported Simple beams for Bending moment and Shear force using single / double rolled steel sections (symmetrical cross sections only) – Problems.

Suggested learning resources:

1. Shah, V. L., and Gore, V., Limit State Design of Steel Structures, Structures Publications, Pune.
2. Dr. N.R.Chadak, Design of Steel Structures, S.K.Kataria & Sons.
3. M.R.Shiyekar, Limit State Design in Structural Steel, PHI
4. Subramanian N., Steel Structures: Design and Practice, Oxford University Press.
5. Sairam, K.S., Design of Steel Structures, Pearson Publication, Chennai, Delhi.
6. Shah, V. L., and Karve, S.R., Limit State Theory and Design of Reinforced Concrete Structures, Structures Publications, Pune,2014.
7. S.K.Duggal, limit State Design of Steel Structures, Tata-McgrawHill Publication
8. Krishna Raju, and N. Pranesh, R.N., Reinforced Concrete Design Principles and Practice, New Age International, Mumbai.
9. Pillai, S.U., and Menon, Devdas, Reinforced concrete Design, McGraw Hill Publications, New Delhi.
10. Varghese, P. C., Limit State Design of Reinforced Concrete, Prentice Hall India Learning Pri- vate Limited, Delhi.
11. Neelam Sharma, RCC Design and Drawing, S.K.Kataria & Sons.

Course outcomes: After completing this course, student will be able to :

Design of RCC beam, lintel - for flexure, shear and development length by LSM

Design of steel connections

Design of steel beams

Name of the Course	**Diploma in Civil Engineering**	Course duration	6 semester
Course Title	**Basic Surveying Field Practices**	Course Code	**CEPC406S**
Subject offered in Semester	Fourth	Number of Credits	1(L:0, T: 0, P: 2) Note: **Basic Surveying Field Practices may also be conducted in 2-3 weeks field work continously instead of having 2 practical class per week for the entire semester.**
Prerequisites	Basic Knowledge of surveying and levelling	Course Category	**PC**
Question distribution	As per standing norms of WBSCT&VE&SD	Marks distribution	As per standing norms of WBSCT&VE&SD

Course Objectives: Following are the objectives of this course:

- To understand types of surveying works required
- To know the type of method and equipment to be used for different surveys, namely, chain, compass, plane table surveys and leveling.
- To know the use and operational details of various surveying equipment.
- To understand the basic method to be followed for indirect contouring
- to give layout of a foundation trench of a building

List of Practicals to be performed

- **Chain & compass traverse survey –** a simple closed traverse of minimum five sides enclosing a building or any suitable site: Reconnaissance, preparation of index map, selection of survey station and finalisation of station, taking FB & BB of lines, ranging a line, chaining a line, taking offset with chain and tape, setting out right angles, calculation of included angles, angular error adjustment, closing error and its balancing by Bowditch method, locating details and plotting them on a A1 size imperial drawing Sheet.

- **Profile and Cross-section Levelling –** running a longitudinal section for a length not less than 250 m for a road using dumpy /auto level and levelling staff. At least four cross sections shall be taken suitably. Undertake fly leveling to check the precision of levelling. Plotting alignment (at least one deviation), L- section including profile leveling and Cross section on a A1 size imperial sheet. all the data will be booked in standard level book. Arithmetical check will have to be performed each and every pages.

- **Block contouring –** a block of suitable area with spot levels at suitable interval will be chosen for plotting the contours by indirect method on A-1 size imperial drawing sheet with a contour interval suitable for the site.

- Measure area of irregular figure using formula (Simpson/ trapezoidal), Graph paper and Digital planimeter.

- **Plane table surveying –** demonstration of accessories and their function; centering, levelling and orientation of plane table; traversing by plane table; plane table surveying around a building or a small area of closed traverse of minimum four sides suitably including filling in details by radiation, intersection and traversing method on A1 size imperial drawing sheet.

- **Layout of a building:** Layout of a building for a given problem (building of any type consisting of a frame structure or load bearing walls system or composite structure). Plotting the Layout Plan of the building on A-1 size imperial drawing sheet.

Suggested learning resources:

1. Punmia, B.C, Jain, Ashok Kumar; Jain, Arun Kumar, Surveying I, Laxmi Publications, New Delhi.
2. Basak, N. N., Surveying and Levelling, McGraw Hill Education, New Delhi.
3. Kanetkar, T. P., Kulkarni, S. V., Surveying and Levelling volume I, Pune Vidyarthi Gruh Prakashan.
4. Duggal, S. K., Survey I, McGraw Hill Education, New Delhi.
5. Saikia, M D, Das. B.M., Das. M.M., Surveying, PHI Learning, New Delhi.

6. Subramanian, R., Fundamentals of Surveying and Levelling, Oxford University Press. New Delhi.
7. Rao, P. Venugopala Akella, Vijayalakshmi, Textbook of Surveying, PHI Learning New Delhi.
8. Bhavikatti, S. S., Surveying and Levelling, Volume 1, I. K. International, New Delhi.
9. Arora K R , Surveying Vol. I, Standard Book House.
10. C.L. Kochher, Surveying, Dhanpat Rai Publication, New Delhi.
11. Dr. N.R. Chandak and Prof. H.R. Kumavat, Surveying, SK Kataria & Sons, New Delhi
12. Saurabh Kumar Soni, Surveying-I, SK Kataria & Sons, New Delhi
13. Agor, R., A Text Book of Surveying & Levelling, Khanna Publishers.
14. Chandra A.M., Plane Surveying, New Age International Publishers

Course outcomes: ***After completing this course, student will be able to:***

- Select the type of survey required for given situation.
- Conduct traversing in the field using chain & compass and plane table.
- Use levelling instruments to determine reduced level to prepare contour maps and profile levelling and cross section details
- Use digital planimeter to calculate the areas.
- Perform layout of a foundation trench

Name of the Course	**Diploma in Civil Engineering**	Course duration	6 semester
Course Title	**Civil Engineering Lab- II Module-V: Hydraulics Lab**	Course Code	**CEPC407S/I**
Subject offered in Semester	Fourth	Number of Credits	1(L:0, T: 0, P:2)
Prerequisites	Basic knowledge in Hydraulics	Course Category	**PC**
Question distribution	As per standing norms of WBSCT&VE&SD	Marks distribution	As per standing norms of WBSCT&VE&SD

Course Objectives: Following are the objectives of this course:

- To understand parameters associated with fluid flow and hydrostatic pressure.
- To know head loss and water hammer in fluid flowing through pipes.
- To learn different types of pumps and their uses.

List of Practical to be performed: [Number of experiments should be performed as much as practicable in commensurate with available resources and facilities – at least six experiments]

1	Use of Piezometer to measure pressure at a given point.
2	Use of Bourdon's Gauge to measure pressure at a given point.
3	Use of U tube differential manometer to measure pressure difference between two given points.
4	Use Bernoulli's apparatus to apply Bernoulli's theorem to get total energy line for a flow in a closed conduit of varying cross sections.

5	Use Friction factor Apparatus to determine friction factor for a given pipe.
6	Calibrate Venturimeter to find out the discharge in a pipe.
7	Calibrate the Orifice to find out the discharge through a tank
8	Use Pitot tube to measure the velocity of flow of water in open channel.
9	Use triangular notch to measure the discharge through open channel.
10	Use Rectangular notch to measure the discharge through open channel.

Suggested learning resources:

1. S.K.Likhi, Hydraulics Laboratory Manual, Newage International Publishers
2. Asawa G.L., Laboratory works in Hydraulic Engineering, Newage International Publishers

Course outcomes: After completing this course, student will be able to:

- Measure pressure and determine total hydrostatic pressure for different conditions.
- Understand various parameters associated with fluid flow.
- Determine head loss of fluid flow through pipes.
- Find the fluid flow parameters in open channels.
- Calibrate in order to find out discharge

Name of the Course	**Diploma in Civil Engineering**	Course duration	6 semester
Course Title	Civil Engineering Lab- II Module-VI: **Geotechnical Engineering Lab**	Course Code	**CEPC407S/II**
Subject offered in Semester	Fourth	Number of Credits	1(L:0, T: 0, P: 2)
Prerequisites	Basic Knowledge in Geotechnical Engineering	Course Category	**PC**
Question distribution	As per standing norms of WBSCT&VE&SD	Marks distribution	As per standing norms of WBSCT&VE&SD

Course Objectives: Following are the objectives of this course:

- To understand and determine physical and index properties of soil.
- To estimate the permeability and shear strength of soil.
- To know the procedure for performing C.B.R test.
- To learn various compaction methods for soil stabilization.

List of Practical to be performed: [Number of experiments should be performed as much as practicable in commensurate with available resources and facilities – at least six experiments]

1. Identification of rocks from the given specimen.
2. Determine water content of given soil sample by oven drying method as per IS: 2720 (Part- II).
3. Determine specific gravity of soil by Pycnometer method as per IS 2720 (Part- III).
4. Determine dry unit weight of soil in field by core cutter method as per IS 2720 (Part- XXIX).
5. Determine dry unit weight of soil in field by sand replacement method as per IS 2720 (Part- XXVIII).
6. Determine Plastic and Liquid Limit along with Plasticity Index of given soil sample as per IS 2720 (Part- V).

7. Use different types of soil to identify and classify soil by conducting field tests- Through Visual inspection, Dry strength test, Dilatancy test and Toughness test.
8. Determine coefficient of permeability by constant head test/ by falling head test as per IS 2720 (Part- XVII).
9. Determine shear strength of soil by direct shear test as per IS 2720 (Part-XIII).
10. Determine MDD and OMC by standard Proctor test of given soil sample as per IS 2720 (Part- VII).
11. Determination of CBR value on the field as per IS2720 (Part - XVI).

Suggested learning resources:

1. Reddy E Saibaba & Sastry K.Rama – Mesurement of Engineering Propertiesof soil, Newage International Publishers
2. K V S Apparoa & V C S Rao, Soil Testing, Laxmi Publication Pvt Ltd
3. S. Mittal & J P Shukla, Soil testing for Engineers, Khanna Publishers
4. Dr. D.K.Maharaj, Laboratory Manual for soil testing, S.K. Kataria & Sons.

Course outcomes: ***After completing this course, student will be able to:***

- Identify types of rocks and sub soil strata of earth.
- Interpret the physical properties of soil related to given construction activities.
- Use the results of permeability and shear strength test for foundation analysis.
- Interpret the soil bearing capacity results.
- Compute optimum moisture content values for maximum dry density of soil through various tests.

Name of the Course	**Diploma in Civil Engineering**	Course duration	6 semester
Course Title	Program Elective - I: **Precast and Prestressed Concrete**	Course Code	CEPE408/ I
Subject offered in Semester	Fourth	Number of Credits	**3(L:3, T: 0, P: 0)**
Prerequisites	NIL	Course Category	**PE**
Question distribution	As per standing norms of WBSCT&VE&SD	Marks distribution	As per standing norms of WBSCT&VE&SD

Course Objectives: Following are the objectives of this course:

- To introduce various types of precast and prefabricated concrete elements.
- To know advantages and disadvantages of precast and prefabricated concrete elements.
- To understand prestressing methods, systems for Reinforced Concrete members.
- To learn issues involved in design of prestressing system and loss of prestressing.

Module /Group [as per directives from WBSCT&VE&SD in framing questions of end semester]	Distribution of unit
Module A/Group A	Unit I and II
Module B/Group B	Unit III and IV
Module C /Group C	Unit V

Course Content:

Unit – I Precast concrete Elements

- Advantages and disadvantages of precast concrete members
- Non-structural Precast elements- Paver blocks, Fencing Poles, Transmission Poles,

Manhole Covers, Hollow and Solid Blocks, kerb stones as per relevant BIS specifications (Overall Idea excluding detail dimensions)

- Structural Precast elements – tunnel linings, Canal lining, Box culvert, bridge panels, foundation, sheet piles (Overview without detail design)
- Testing of Precast components as per BIS standards

Unit– II Prefabricated building

- Precast Structural Building components such as slab panels, beams, columns, footings, walls, lintels and chajjas, stair case elements,
- Prefabricated building using precast load bearing and non load bearing wall panels, floor systems - Material characteristics, Plans & Standard specifications
- Modular co-ordination, modular grid, and finishes
- Prefab systems and structural schemes and their classification including design considerations
- Joints – requirements of structural joints and their design considerations
- Testing of Precast components as per BIS standards
- Manufacturing, storage, curing, transportation and erection of above elements, equipment needed

Unit– III Introduction to Prestressed Concrete

- Principles of pre-stressed concrete and basic terminology.
- Applications, advantages and disadvantages of pre stressed concrete
- Materials used and their properties, Necessity of high-grade materials
- Types of Pre-stressing steel -Wire, Cable, tendon, Merits-demerits and applications

Unit– IV Methods and systems of prestressing

- Methods of prestressing – Internal and External pre-stressing, Pre and Post tensioning applications
- Systems for pre tensioning – process, applications, merits and demerits – Hoyer system
- Systems for post-tensioning - process, applications, merits and demerits - Freyssinet sys- tem, Magnel Blaton system, Gifford Udall system.
- Prestressing force in Cable, Loss of prestress during the tensioning process - loss due to friction, length effect, wobbling effect and curvature effect, (Simple Numerical problems to determine loss of prestress), Loss of prestress at the anchoring stage.
- Loss of pre-stress occurring subsequently: losses due to shrinkage of concrete, creep of concrete, elastic shortening, and creep in steel, (Simple Numerical problems to determine loss of prestress).
- BIS recommendations for losses in case of Pre and Post tensioning.

Unit– V Analysis and design of Prestressed rectangular beam section

- Basic assumptions in analysis of pre-stressed concrete beams.

- Cable Profile in simply supported rectangular beam section – concentric, eccentric-straight and parabolic
- Effect of cable profile on maximum stresses at mid span and at support.
- Numerical problems on determination of maximum stresses at mid spans with linear (con- centric and eccentric) cable profiles only.
- Simple steps involved in Design of simply supported rectangular beam section (No numerical problems)

Suggested learning resources

- Krishna Raju, N., Pre-stressed Concrete, Tata McGraw Hill, NewDelhi.
- Nagarajan, Praveen., Pre-stressed Concrete Structures, Pearson EducationIndia
- Shrikant B. Vanakudre, Prestressed Concrete, Khanna Publishing House, NewDelhi
- Nor Ashikin, Marzuki, PreCast and PreStress Technology: Process, Method and Future Technology, Createspace Independent Publication.
- Lin, T.Y., Design of Pre-Stressed Concrete Structures, John Wiley and Sons, New York
- BIS, New Delhi. IS 1343 Prestressed Concrete – Code of Practice, BIS, New Delhi
- Indian Concrete Institute., Handbook on Precast Concrete buildings.
- Elliott, Kim S., Precast Concrete Structures, CRC Press, NewYork.
- BIS, New Delhi. IS 12592 Precast Concrete Manhole Cover and Frame, BIS, New Delhi
- BIS, New Delhi. IS 15658 Precast concrete blocks for paving- Code of Practice, BIS, New Delhi
- BIS, New Delhi. IS 15916 Building Design and Erection Using Prefabricated Concrete - Code of Practice, BIS, New Delhi
- BIS, New Delhi. IS 15917 Building Design and Erection Using Mixed/Composite Construction
- - Code of Practice, BIS, New Delhi
- BIS, New Delhi. IS 458 Precast Concrete Pipes (with and without reinforcement)— Specification, BIS, New Delhi
- Guide for Precast Concrete Tunnel Segments – ACI 533.5R-20 – American Concrete Institute
- Guidelines for Lining of Irrigation Canals – Govt. of Odisha – Department of Water Resources
- Y.R.M Rao, Prestress Concrete analysis and design, S.K.Kataria & Sons

Course outcomes: ***After completing this course, student will be able to:***

- Describe different types of precast concrete elements
- classify different components of pre-fabricated buildings
- Describe the basics of prestressed concrete
- Describe the methods and systems of prestressing
- Analyse and design of prestressed rectangular beam

Name of the Course	**Diploma in Civil Engineering**	Course duration	6 semester
Course Title	Program Elective - I: **Rural Construction Technology**	Course Code	**CEPE 408 / II**
Subject offered in Semester	**Fourth**	Number of Credits	**3(L:3, T: 0, P: 0)**
Prerequisites	NIL	Course Category	**PE**
Question distribution	As per standing norms of WBSCT&VE&SD	Marks distribution	As per standing norms of WBSCT&VE&SD

Course Objectives: Following are the objectives of this course:

- To learn development and planning of low cost housing infrastructure.
- To know about different government schemes for rural development.
- To understand techniques for rural road construction as per IRC stipulations.
- To learn rural irrigation techniques and watershed management.

Module /Group [as per directives from WBSCT&VE&SD in framing questions of end semester]	Distribution of unit
Module A/Group A	Unit I and II
Module B/Group B	Unit III
Module C /Group C	Unit IV and V

Course Contents:

Unit I - Rural Development and Planning

• Scope; development plans; various approaches to rural development planning.

• Significance of rural development.

• Rural development programme/projects.

Unit II -Rural Housing

• Low cost construction material for housing

• Composite material- ferro-cement & fly ash, autoclaved calcium silicate bricks and soil-stabilized un-burnt brick; Plinth protection of mud walls.

• Water-proof and fire-retardant roof treatment for thatch roofs. Pre-cast stone masonry, rattrap bond for walls; Panels for roof, ferro-cement flooring/roofing units.

• Biomass - types of fuels such as firewood, agricultural residues, dung cakes.

• Renewable energy and integrated rural energy program - Objectives, Key elements, Implementation, Financial provisions, sources of renewable energy.

• Working of gobar gas and bio gas plants.

Unit III Water Supply and Sanitation for Rural Areas

• Sources of water: BIS & WHO water standards.

• Quality, Storage and distribution for rural water supply works.

• Hand pumps-types, installation, operation, and maintenance of hand pumps.

• Conservation of water - rainwater harvesting, drainage in rural areas.

• Construction of low cost latrines: Two pit pour flush water seal, septic tank etc.

• Low cost community and individual Garbage disposal systems, Ferro-cement storage tanks

Unit IV - Low Cost Rural Roads

• Broad categories of Pavement Layers, types of Granular Sub-Bases and Bases.

• Guidelines for Surfacing of Rural Road as per relevant IRC codes.

• Pradhan Mantri Gram Sadak Yojna (PMGSY)- Highlights of Scheme.

Unit V - Low Cost Irrigation

• Design consideration and construction of tube-well, drip & sprinkler irrigation systems.

• Watershed and catchment area development –problems and features of watershed management.

• Watershed management structures - Gabian Structure, Cement Plug, Contour, Bunding, Farm pond, Bandhara system.

Suggested learning resources:

1. Madhov Rao A G, and Ramachandra Murthy, D S, Appropriate Technologies for low cost Housing Oxford and IBH Publishing Co. Pvt. Ltd.

2. CBRI, Roorkee, Advances in Building Materials and Constriction.

3. Desai,Vasant , Rural Development in India: Past, Present and Future : a Challenge in the Crisis, Himalaya Publishing House, Delhi.

4. Rastogi, A.K. Rural Development Strategy, Wide Vision, Jaipur.

5. Singh, Katar, Rural Development Principles, Policies and Management, Sage Publications India Pvt Ltd.

6. Gaur, Keshav Dev, Dynamics of Rural Development, Mittal Publications, Delhi.

7. Document Published by Ministry of Rural development, Govt. of India, Ministry of Rural development.

Course outcomes: After completing this course, student will be able to:

- Plan low cost housing using rural materials.
- Make use of relevant government schemes for construction of roads and housing.
- Use guidelines for rural road construction.
- Implement different irrigation systems for rural areas.
- Identify the need of watershed management in rural areas.

Name of the Course	**Diploma in Civil Engineering**	Course duration	6 semester
Course Title	Minor Project	Course Code	**CEPR409S**
Subject offered in Semester	**Fourth**	Number of Credits	**2 (L:0, T: 0, P: 4)**
Prerequisites	Students should have knowledge of Surveying, Building planning drawing and CAD	Course Category	**PR**
Question distribution	As per standing norms of WBSCT&VE&SD	Marks distribution	As per standing norms of WBSCT&VE&SD

Objective:- Following are the objectives of this course:

- To collect the information for a given project.
- To apply principles, theorems and bye-laws in the project planning.
- To interpret and analyze the data.
- To develop professional abilities such as persuasion, confidence, and perseverance and communication skill.
- To develop presentation skill.
- To enhance creative thinking.

[The project report shall be in the following format:
(The project shall be undertaken by a group of students as per convenience)
Topic and objectives
Collection of data, required survey work,
Required drawing set
Utility to society if any
Conclusion]

Contents:-

Title of the Project:-Planning of (G+2) Residential Complex for Middle Income Group(as per SP-7and rules and regulation of local bodies) for sanction before Project approval committee.

NOTE: Same Planning, Drawings and detailings of the problem given in this semester will have to be used in Major Project I and Major Project II in semester 5 and 6.

The details of the Project are given below:-
Each building (RCC framed structure) shall comprise of two symmetrical flat per floor each containing two rooms, bath, WC, kitchen, front verandah with a provision of common staircase and mumty for utilization of roof space and overhead water tank (on 10000 sq m. of total land area). Ground floor to be used for parking spaces.

The following provisions are to be considered during the project planning:- a) Security room(Single room with WC, Load bearing wall structure), b) Central Park, c) Play Ground, d) Hume Pipe Culvert in between the complex and the 12m wide main road, e) Boundary Wall with main gate, f) Submersible Pump, g) Pump House (Load bearing wall structure), h) Surface Drainage System, i) Bituminous road over WBM inside the complex etc.

1)The project report shall include Topographical surveying map, Architecture planning, preparation of drawing sheet –Key Plan, Site plan, a typical floor plan, roof plan with provision of drainage, sectional elevation including staircase, front view, preparation of Rough cost estimate of the project, calculation of FAR, Specification of different item of works and Cost of Land.

2) The project report shall also comprise of drawing sheets of the following-
Floor plan and front elevation of Security Room & Pump house, Plan and sectional elevation of Hume pipe culvert, Plan and front elevation of Boundary Wall with main gate, Sectional elevation of Bituminous road over WBM.

[**NOTE: In addition to conventional approach to Civil Engineering Drawing, student may also take the help of CAD in preparing their sessional works if he/she desires so.**]

Course outcomes: After completing this course, student will be able to:

- List the different data for the project after surveying the plot.
- Apply the principles, rules, regulations & by-laws of project planning.
- Utilise the collected data for arranging different building units in a vacant land.
- Prepare the drawing and detailing of the project by using various drafting softwares.
- Solve the problem by working in a group.
- Improve creative thinking about planning of a housing complex.

West Bengal State Council of Technical & Vocational Education and Skill Development
(Technical Education Division)



Syllabus
of

Diploma in Civil Engineering [CE]

Part-III (5th Semester)

Revised 2022

CURRICULAR STRUCTURE FOR PART – IIIFIRST SEMESTER (FIFTH SEMESTER) OF THE FULL-TIME DIPLOMA COURSE IN CIVIL ENGINEERING

SL. NO.	SUBJECT CODE	SUBJECT OF STUDY	HOURS PER WEEK			CREDITS	Marks	Page No.
		THEORETICAL PAPERS	LECTURE	TUTORIAL	PRACTICAL			
1	CEPC501	Water Resource Engineering	2	0	0	2	100	
2	CEPC502	Estimating, Costing and Valuation	3	0	0	3	100	
		LABORATORY/SESSIONAL PAPERS						
3	CEPC503S	Design of RCC and Steel Structure Practices	0	0	2	1	100	
4	CEPC504S	Estimating, Costing and Valuation Practices	0	0	2	1	100	
5	CEPC505S	Water Resource Engineering Practices	0	0	2	1	100	
		ELECTIVE, PROJECT AND INTERNSHIP						
6	CEPE506	Elective - II: one subject out of Advanced Design of Structures (CEPE506/I) / Traffic Engineering (CEPE506/II)	3	0	0	3	100	
7	CEPE507	Elective - III: one subject out of Building Services and Maintenance (CEPE507/I) /Repair and Maintenance of of Structures (CEPE507/II)	3	0	0	3	100	
8	CEI508	Internship-II after fourth Semester	0	0	0	1	100	
9	CEPR509S	Major Project I	0	0	4	2	100	
10	CEPC510	Safety Engineering & Management in the Construction Sector	2	0	0	2	100	
		TOTAL	13	0	10	19	1000	

Name of the Course	**Diploma in Civil Engineering**	Course duration	6 semester
Course Title	**Water Resource Engineering**	Course Code	**CEPC501**
Subject offered in Semester	Fifth	Number of Credits	2 (L:2, T: 0, P: 0)
Prerequisites	NIL	Course Category	**PC**
Question distribution	As per standing norms of WBSCT&VE&SD	Marks distribution	As per standing norms of WBSCT&VE&SD

Course Objectives: Following are the objectives of this course:

- To learn estimation of hydrological parameters.
- To understand water demand of crops and provisions to meet the same.
- To know planning of reservoirs and dams
- To design irrigation projects, canals and other diversion works.

Module /Group [as per directives from WBSCT&VE&SD in framing questions of end semester]	**Distribution of unit**
Module A/Group A	Unit I and II
Module B/Group B	Unit III
Module C /Group C	Unit IV and V

Course Content:

Unit – I Introduction to Hydrology

- Hydrology: Definition and Hydrological cycle
- Rain Gauge: Symons rain gauge, automatic rain gauge
- Methods of calculating average rainfall: Arithmetic mean, Isohyetal, and Theissen polygon method.

- Runoff, concept of hydrograph, Factors affecting Run off, Computation of run–off [using runoff coefficient method]
- Maximum Flood Discharge measurement: Rational methods, Simple numerical problems.
- Simple numerical problems.

Unit– II Crop water requirement and Reservoir Planning

- Irrigation and its classification
- Crop Water requirement: Cropping seasons, Crop period, base period, Duty, Delta, CCA, GCA, intensity of irrigation, factors affecting duty, methods for improving duties, Problems on water requirement and capacity of canal, Kor irrigation, kor depth of water, kor period, paleo irrigation, capacity factor, outlet factor, time factor, cumec day, overlap allowance
- Methods of application of irrigation water, their merits and demerits, suitability
- Silting of reservoir, Rate of silting, factors affecting silting and control measures.
- Control levels and storage in reservoir - definition of different related technical terms explained with a neat sketch, Area capacity curve

Unit– III Dams and Spillways

- Dams and its classification: Earthen dams and Gravity dams (masonry and concrete)
- Earthen Dams – Components with function, typical cross section, seepage through embankment and foundation and its control.
- Methods of construction of earthen dam, types of failure of earthen dam and preventive measures.
- Gravity Dams – Forces acting on dam, Theoretical and practical profile, typical cross section, inspection gallery, drainage gallery, joints in gravity dam, concept of high dam and low dam.
- Spillways - Definition, function, location, types, spillway gates - types and function, Energy dissipaters and types.

Unit– IV Minor and Micro Irrigation

- Bandhara irrigation: Layout, components, construction and working, solid and open bandhara.
- Percolation Tanks – Need, selection of site.
- Lift irrigation Scheme - Components and their functions
- Drip and Sprinkler Irrigation- Need, components
- Well irrigation: types and yield of wells, advantages and disadvantages of well irrigation

Unit– V Diversion Head Works & Canals

Brief discourses on

- Weirs – components, parts, types only
- Diversion head works – Layout, components and their function.
- Barrages – components and their functions. Difference between weir and Barrage.
- Canals – Classification according to alignment and position in the canal network, Cross section of canal in embankment and cutting, partial embankment and cutting, balancing depth, Design of most economical canal section.
- Canal lining - Purpose, material used and its properties, advantages.
- Cross Drainage works - Aqueduct, siphon aqueduct, super passage, level crossing.
- Canal regulators - Head regulator, Cross regulator, Escape, Falls and Outlets

Suggested learning resources:

1. Punmia, B.C., Pande, B, Lal, Irrigation and Water Power Engineering, Laxmi Publications

2. Subramanayan, Engineering Hydrology, McGraw Hill.

3. Mutreja K N, Applied Hydrology, McGraw Hill

4. Sharma, R.K. and Sharma, T.K., Irrigation Engineering, S.Chand

5. Basak, N.N., Irrigation Engineering, McGraw Hill Education

6. Asawa, G.L., Irrigation and water resource Engineering, New Age

7. Dahigaonkar, J.G., Irrigation Engineering, Asian Book Pvt. Ltd., New Delhi.

8. Garg, S K, Irrigation and Hydraulic Structures, Khanna Publishers, Delhi.

9. Priyani V.B., Irrigation Engineering, Charotar Book Stall, Anand.

10. Relevant BIS codes

Course outcomes: After completing this course, student will be able to:

- ✓ Estimate hydrological parameters.
- ✓ Estimate crop water requirements of a command area and capacity of canals.
- ✓ Execute Minor and Micro Irrigation Schemes.
- ✓ Select the relevant Cross Drainage works for the specific site conditions.
- ✓ Design, construct and maintain simple irrigation regulatory structures

Name of the Course	**Diploma in Civil Engineering**	Course duration	6 semester
Course Title	**Estimating, Costing and Valuation**	Course Code	**CEPC502**
Subject offered in Semester	Fifth	Number of Credits	3 (L:3, T: 0, P: 0)
Prerequisites	NIL	Course Category	**PC**
Question distribution	As per standing norms of WBSCT&VE&SD	Mark distribution	As per standing norms of WBSCT&VE&SD

Objective:- Following are the objectives of this course:

- To learn the procedure for estimating and costing of Civil Engineering works.
- To perform rate analysis for different items associated with construction projects.
- To acquire knowledge on simple terms and concepts related to valuation

Module /Group [as per directives from WBSCT&VE&SD in framing questions of end semester]	**Distribution of unit**
Module A/Group A	Unit I and II
Module B/Group B	Unit III
Module C /Group C	Unit IV and V

Contents:

Unit – I Fundamentals of Estimating and Costing

- Estimating and Costing – Meaning, purpose, Administrative approval, Technical Sanction
- Types of estimates – Approximate and Detailed estimate.
- Types and Uses of Estimates: Revised estimate, Supplementary estimate, Repair and maintenance estimate, renovation estimate.

- Checklist of items in load bearing and framed structure.
- Standard formats of Measurement sheet, Abstract sheet, Face sheet.
- Modes of measurement and desired accuracy in measurements for different items of work as per IS:1200.
- Rules for deduction in different category of work as per IS:1200.
- Description / specification of items of building work as per PWD
- Approximateestimateforroads,culvert

Unit– II Approximate Estimates

- Approximate estimate- Definition, Purpose.
- Methods of approximate estimate - Service unit method, Plinth area rate method, Cubical content method, Approximate quantity method (with simple numerical)

Unit– III Detailed Estimate

- Detailed Estimate- Definition and Purpose, Data required for detailed estimate - Civil cost, GST, Contingencies, Supervision charges, Agency charges, Procedure for preparation of detailed estimate- Taking out quantities and Abstracting.
- Long wall and Short wall method, Centre line method – for building
- Bar bending schedule for footing, column, beam, Lintel, chajja and slab elements
- Provisions in detailed estimate: contingencies, work charged establishment, percentage charges, water supply and sanitary Charges and electrification charges etc.
- Prime cost, Provisional sum, Provisional quantities, Bill of quantities, Spot items or Site items.

Unit– IV Estimate for other Civil Engineering Works

- Earthwork - Quantities for roads, Embankment and canal by – Mid sectional area method, mean sectional area method, Prismoidal and trapezoidal formula method.
- Detailed estimate for septic tank, Community well.
- Case study – steel structure - industrial shed, chimney, transmission tower and like structure, frame and shutter of door window (wooden, steel)

Unit– V Rate Analysis and Valuation

- Rate Analysis: Definition, purpose and importance.
- Lead (Standard and Extra), lift, overhead charges, water charges and contractors' profit,
- Procedure for rate analysis.
- Task work- Definition, types. Task work of different skilled labour for different items.

- Categories of labours, their daily wages, types and number of labours for different items of work.
- Transportation charges of materials - Lead and Lift, Hire charges of machineries and equipments.
- Preparing rate analysis of different items of work pertaining to buildings and roads.
- Definition, purpose and importance of valuation.
- Factors affecting the value of a property. Simple definitions and concepts on scrap value, salvage value, market value, booked value, assessed value, sinking fund, depreciation,obsolescence.
- Different methods of valuation.

Suggested learning resources:

1. Chakraborti,M., Estimating and costing, specification and valuation in civil engineering, Monojit Chakraborti, Kolkata.
2. Datta, B.N., Estimating and Costing in Civil engineering, UBS Publishers Distributors Pvt. Ltd. New Delhi.
3. PWD Schedule of Rates.
4. Ministry of Road Transport and Highways (MORT&H) Specifications and Analysis of Schedule of Rates.
5. Peurifoy, Robert L. Oberlender, Garold, Estimating construction cost (fifth edition), McGraw Hill Education, New Delhi.
6. Rangwala,S.C., Estimating and Costing, Charotar Publishing House PVT. LTD., Anand.
7. Birdie,G.S., Estimating and Costing, Dhanpat Rai Publishing Company(P) Ltd. New Delhi.
8. Patil,B.S., Civil Engineering Contracts and Estimates, Orient Longman, Mumbai.
9. Manual of Specifications and Standards for DBFOT projects, EPC works.

Course outcomes:

After completing this course, student will be able to:

- ✓ Select modes of measurements for different items of works.
- ✓ Prepare approximate estimate of a civil engineering works.
- ✓ Prepare detailed estimate of a civil engineering works.
- ✓ Justify rate for given items of work using rate analysis techniques.
- ✓ Acquire general ideas on valuation.

Name of the Course	**Diploma in Civil Engineering**	Course duration	6 semester
Course Title	**Design of R.C.C and Steel Structure Practices**	Course Code	**CEPC503S**
Subject offered in Semester	Fifth	Number of Credits	1 (L:0, T: 0, P: 2)
Prerequisites	NIL	Course Category	**PC**
Question distribution		Marks distribution	As per standing norms of WBSCT&VE&SD

Course Objectives: Following are the objectives of this course:

- To learn the concept of limit state design of tension and compression steel members.
- To understand design of steel beams.
- To learn the concept of limit state design of RCC beams.
- To know the limit state design of RCC columns.

List of Practical to be performed [application of CAD is mandatory]:

Sl no 6 to 11 is compulsory, one from 13 &14 is desirable and at least one from the rest

1	Draw any five commonly used rolled steel sections and five built up sections.
2	Summarize the provisions of IS 800 required for the design of a. tension member b. compression members c. flexural members in report form.
3	Draw sketches for a. single lacing b. double lacing and c. battening of a given built up columns.
4	Draw cross section, strain diagram & stress diagram for a. singly reinforced section and b. doubly reinforced section.
5	Draw sketches of different types of column footings.
6	Prepare a plan and draw sectional elevation showing reinforcement detailing in the longer and shorter direction of a single span [four edges discontinuous] for a. one way slab and b. two way slab
7	Prepare a plan and draw sectional elevations showing reinforcement detailing in the longer and shorter direction of two or more span of a two-way slab (having two adjacent edges discontinuous) inclusive of reinforcement for negative moment at support and torsional reinforcement where required.

8	Draw a sectional elevation showing reinforcement detailing of a beam having three span of unequal /equal length with cross section details at midspan and support
9	Prepare a plan and draw a sectional elevation showing reinforcement detailing for a typical isolated square footing having one column
10	Prepare a plan and draw a sectional elevation showing reinforcement detailing of a typical waist slab and landing for single flight of a dog-legged stair
11	Prepare a plan and draw a sectional elevation showing reinforcement detailing of a typical lintel with chajja
12	Interpret the actual RCC Structural Drawings used on site with reference to reinforcement details of various structural elements and Prepare a checklist for reinforcement provided from actual drawings used on site for various structural elements.
13	Prepare a detailed report of site visit for reinforcement detailing of structural elements like beams, columns, staircase & footing.
14	Prepare a detailed report of site visit for study of different elements of a bridge and roof truss

Suggested learning resources:

1. Shah, V. L., and Gore, V., Limit State Design of Steel Structures, Structures Publications, Pune.
2. Dayarathnam, P., Design of Steel Structures, S. Chand and Company, Delhi.
3. Subramanian N., Design of Steel Structures, Oxford University Press.
4. Sairam, K.S., Design of Steel Structures, Pearson Publication, Chennai, Delhi.
5. Shah, V. L., and Karve, S.R., Limit State Theory and Design of Reinforced Concrete Structures, Structures Publications, Pune,2014.
6. Sinha N.C., and Roy S.K., Fundamentals of Reinforced Concrete, S. Chand & Co., New Delhi.
7. Krishna Raju, and N.Pranesh, R.N., Reinforced Concrete Design Principles and Practice, New Age International, Mumbai.
8. Pillai, S.U., and Menon, Devdas, Reinforced concrete Design, McGraw Hill Publications, New Delhi.
9. Varghese, P. C., Limit State Design of Reinforced Concrete, Prentice Hall India Learning Private Limited, Delhi.

Course outcomes: After completing this course, student will be able to perform:

- Design of steel tension and compression member.
- Design of steel beams including check for shear.
- Design of singly and doubly reinforced RCC beam.
- Design of shear reinforcement in RC beams.
- Design of RCC column as per IS 456.

Name of the Course	**Diploma in Civil Engineering**	Course duration	6 semester
Course Title	**Estimating, Costing and Valuation Practices**	Course Code	**CEPC504S**
Subject offered in Semester	Fifth	Number of Credits	1 (L:0, T: 0, P: 2)
Prerequisites	NIL	Course Category	**PC**
Question distribution		Marks distribution	

Objective:- Following are the objectives of this course:

- To learn the procedure for estimating and costing of Civil Engineering works.
- To perform rate analysis for different items associated with construction projects.

List of Practical to be performed:

Application of spreadsheet is advised

SL no 6 to 11, 13 and 17 is compulsory and at least two from the rest

1	Prepare the list of items to be executed with units for detailed estimate of a given structure from the given drawing.
2	Prepare a report on market rates for given material, labour wages, hire charges of tools & equipment required to construct the given structure as mentioned in at Serial number 1 above.
3	Study of items with specification given in the PWD Schedule of Rates. (for any ten item)
4	Recording in Measurement Book (MB) for any four items
5	Prepare bill of quantities of given item from actual measurements. (any four items).
6	Prepare approximate estimate for the given civil engineering works for buildings, road and culvert.

7	Prepare detailed estimate from the given set of drawings using "standard measurement and abstract format" for (G+1), 2 BHK RCC framed structure Residential Building using description of item from PWD Schedule of Rates along with face sheet and prepare quarry chart, lead statement.
8	Prepare detailed estimate from the given set of drawings of Door(panelled door) and Window(partly glazed) - frame and shutter [wooden]
9	Calculate the reinforcement quantities from the given set of drawings for a room size of 3 m X 4 m with bar bending schedule (footing, column, beam, lintel with chajja, slab)
10	Prepare rate analysis for the given five item of works.
11	Prepare detailed estimate of road of one kilometre length from the given drawing.
12	Prepare detailed estimate of small Septic tank from the given set of drawings.
13	Prepare detailed estimate of Tube well with Hand pump from the given set of drawing.
14	Assess the valuation of an old building from the given drawing by any one method.
15	Prepare detailed estimate for a two lane bituminous road and culvert (slab type)
16	methods of stack measurement at site for stone chips etc
17	Prepare detailed estimate of steel roof truss

Suggested learning resources:

1. Chakraborti,M., Estimating and costing, specification and valuation in civil engineering, Monojit Chakraborti, Kolkata.
2. Datta, B.N., Estimating and Costing in Civil engineering, UBS Publishers Distributors Pvt. Ltd. New Delhi.
3. PWD Schedule of Rates.
4. Ministry of Road Transport and Highways (MORT&H) Specifications and Analysis of Schedule of Rates.
5. Peurifoy,Robert L. Oberlender,Garold, Estimating construction cost (fifth edition), McGraw Hill Education, New Delhi.
6. Rangwala,S.C., Estimating and Costing, Charotar Publishing House PVT. LTD., Anand.

7. Birdie,G.S., Estimating and Costing, Dhanpat Rai Publishing Company(P) Ltd. New Delhi.
8. Patil,B.S., Civil Engineering Contracts and Estimates, Orient Longman, Mumbai.
9. Manual of Specifications and Standards for DBFOT projects, EPC works.

Course outcomes: After completing this course, student will be able to:

- Select modes of measurements for different items of works.
- Prepare approximate estimate of a civil engineering works.
- Prepare detailed estimate of a civil engineering works.
- Justify rate for given items of work using rate analysis techniques.

Name of the Course	**Diploma in Civil Engineering**	Course duration	6 semester
Course Title	**Water Resource Engineering Practices**	Course Code	**CEPC505S**
Subject offered in Semester	Fifth	Number of Credits	1 (L:0, T: 0, P: 2)
Prerequisites	NIL	Course Category	**PC**
Question distribution		Marks distribution	As per standing norms of WBSCT&VE&SD

Course Objectives: Following are the objectives of this course:

- To learn estimation of hydrological parameters.
- To understand water demand of crops and provisions to meet the same.
- To know planning of reservoirs and dams.
- To get acquainted with irrigation projects, canals and other diversion works.

List of Practical to be performed:

Sl no 1 to 6 mandatory, at least one from Sl no 7-11 and at least one from the rest

1.	Calculate average rainfall for the given area using isohyetal, Theissen polygon method [only simple cases]
2.	Compute the yield of the Catchment area demarcated in the previous problem
3.	Delineation of contributory area for the given outlet from the given toposheet
4.	Estimate water requirement of crops in different season from data provided by the subject teacher.
5.	Estimate capacity of the canal for the data provided by the subject teacher.
6.	Calculate reservoir capacity from the data provided by the subject teacher.
7.	Draw a neat sketch showing different control levels and storage for a reservoir.

8.	Draw a labeled sketch of the given different types of earthen dam section
9.	Draw the theoretical and practical profile of the given high type gravity dam section.
10.	Draw a labeled sketch of the given diversion head works and Cross Drainage works.
11.	Draw a labeled sketch of a canal section - a. in partly cutting and partly filling b. fully in cutting c. fully in filling.
12.	Prepare a presentation on the technical details of any one micro or minor irrigation scheme.
13.	Prepare a model of any irrigation structure using suitable material.
14.	Prepare summary of the technical details of any existing water resource project in West Bengal

Course outcomes: After completing this course, student will be able to:

- Estimate hydrological parameters.
- Estimate crop water requirements of a command area and capacity of canals.
- Understand Minor and Micro Irrigation Schemes.
- Select relevant Cross Drainage works for the specific site conditions.

Name of the Course	**Diploma in Civil Engineering**	Course duration	6 semester
Course Title	**Elective - II: Advanced Design of Structures**	Course Code	**CEPE506/I**
Subject offered in Semester	Fifth	Number of Credits	3(L:3, T: 0, P: 0)
Prerequisites	NIL	Course Category	**PE**
Question distribution	As per standing norms of WBSCT&VE&SD	Marks distribution	As per standing norms of WBSCT&VE&SD

Course Objectives: Following are the objectives of this course:

- To analyze T and L shaped beam sections.
- To understand the concept for design of one way and two way slabs.
- To identify short and long columns and their design provisions.
- To understand the concepts involved in the design of bolted and welded connections.
- To know the provisions of BIS code for design of built up sections.

Module /Group [as per directives from WBSCT&VE&SD in framing questions of end semester]	**Distribution of unit**
Module A/Group A	Unit I and II
Module B/Group B	Unit III
Module C /Group C	Unit IV and V

Course Content:

Unit 1: **Design of Concrete Slab (LSM)**

- Design and detailing of simply supported one-way slabs for flexure, check for deflection control, and shear.
- Design and detailing of one-way cantilever slabs and cantilevers chajjas for flexure, check for deflection control and check for development length and shear.

- Design and detailing of two-way simply supported slabs for flexure with corner free to lift.
- Design and detailing of dog-legged stair.
- Simple numerical problems on design of one-way simply supported slabs, cantilever slab, two-way simply supported slab, waist slab and landing slab of a dog-legged stair.

Unit 2: Design of Axially Loaded Concrete Column and Footing (LSM)

- Assumptions in limit state of collapse – compression
- Definition and classification of columns, effective length of column. Specification for minimum reinforcement; cover, maximum reinforcement, number of bars in rectangular, square and circular sections, diameter and spacing of lateral ties.
- Analysis and design of axially loaded short, square, rectangular and circular columns with lateral ties only; check for short column and check for minimum eccentricity may be applied.
- Types of footing, Design of isolated square footing of uniform thickness for flexure, and shear.
- Simple numerical problems on the design of axially loaded short columns and isolated square footing.

Unit-3 Design of Steel Tension Members and Compression Members by L.S.M

- Tension Members - introduction and different provision of relevant code, Design Strength of Tension members against yielding of gross section requirements: Deflection limits, Vibration, Durability and Fire resistance against rupture of critical section and due to block shear. Problems on determination of design strength of given members and designing tension members using rolled steel sections for given loads.
- Compression Members - Effective Length and Effective Sectional Area of Compression members, Design Stress and design strength – Buckling Class of cross sections– Imperfection factor– Stress reduction factor– Thickness of elements.
- Analysis and design of axially loaded column.
- Introduction to lacing and battening (No numerical problem on Lacing and Battening).

Unit- 4 Column Bases by L.S.M

- Slab base and Gusseted base – Code Provisions (IS:800-2007) – Minimum thickness and Effective Area of Base plate.
- Design of Slab base for axially loaded columns using bolts /welds.
- Introduction to Gusseted base(no numerical problems on gusseted Base).

Unit- 5 Steel Roof Truss

- Types of steel roof truss & its selection criteria.
- Calculation of panel point load for Dead load; Live load and wind load as per I.S. 875-1987.
- Analysis and Design of steel roof truss. Design of Angle purlin as per I. S. Arrangement of members at supports.

Suggested learning resources:

1. Shah, V. L., and Gore, V., Limit State Design of Steel Structures, Structures Publications, Pune.
2. Subramanian N., Design of Steel Structures, Oxford University Press.
3. Sairam, K.S., Design of Steel Structures, Pearson Publication, Chennai, Delhi.
4. Shah, V. L., and Karve, S.R., Limit State Theory and Design of Reinforced Concrete Structures, Structures Publications, Pune.
5. Sinha N.C., and Roy S.K., Fundamentals of Reinforced Concrete, S. Chand & Co.,
6. Krishna Raju, and N.Pranesh, R.N., Reinforced Concrete Design Principles and Practice, New Age International, Mumbai.
7. Pillai, S.U., and Menon, Devdas, Reinforced concrete Design, McGraw Hill
8. Varghese, P. C., Limit State Design of Reinforced Concrete, Prentice Hall India Learning Private Limited, Delhi.
9. Subramanian N., Design of Reinforced Concrete Structures, Oxford University Press.
10. Dr. N. R. Chandak, Design of Steel Structure, S.K.Kataria & Sons.

Course outcomes: After completing this course, student will be able to perform:

- Design of R.C.C one way, two way and waist slab
- Design of R.C.C axially loaded short column and isolated column footing
- Design of steel tension member, compression member and column base by limit state method
- Design of steel roof truss

Name of the Course	**Diploma in Civil Engineering**	Course duration	6 semester
Course Title	**Elective – II: Traffic Engineering**	Course Code	**CEPE506/II**
Subject offered in Semester	**Fifth**	Number of Credits	**3(L:3, T: 0, P: 0)**
Prerequisites	NIL	Course Category	**PE**
Question distribution	As per standing norms of WBSCT&VE&SD	Marks distribution	As per standing norms of WBSCT&VE&SD

Objective:- Following are the objectives of this course:

- To understand the issues involved in traffic flow.
- To know and understand the tools for traffic studies.
- To delineate various traffic control measures.
- To understand measures for preventing accidents.

Module /Group [as per directives from WBSCT&VE&SD in framing questions of end semester]	**Distribution of unit**
Module A/Group A	Unit I and II
Module B/Group B	Unit III and V
Module C /Group C	Unit IV

Contents:

Unit – I: Fundamentals of Traffic Engineering.

- Traffic engineering- Definition, objects, scope
- Relationship between speed, volume and density of traffic
- Road user's characteristics- physical, mental, emotional factors.
- Vehicular characteristics-width, length, height, weight, speed, efficiency of breaks.

- Road characteristics - gradient, curve of a road, design speed, friction between road and tyre surface.
- Reaction time - factors affecting reaction time. PIEV Theory.

Unit– II: Traffic Studies

- Traffic volume count data - representation and analysis of data.
- Necessity of Origin and Destination study and its methods.
- Speed studies - Spot speed studies, and its presentation.
- Need and method of parking study.

Unit– III: Road Signs and Traffic Markings

- Traffic control devices –definition, necessity, types.
- Road signs - definition, objects of road signs.
- Classification as per IRC: 67- Mandatory or Regulatory, Cautionary or warning, informatory signs, Location of cautionary or warning sign in urban and non-urban areas, Points to be considered while designing and erecting road signs.
- Traffic markings- definition, classification, carriage way, kerb, object marking and reflector markers.

Unit– IV Traffic Signals and Traffic Islands

- Traffic signals- Definition, Types, Traffic control signals, pedestrian signals.
- Types of traffic control signals - Fixed time, manually operated, traffic actuated signals andlocation of signals.
- Compute signal time by fix time cycle, Webster's and IRC method and sketch timing diagramfor each phase.
- Conflict points
- Traffic islands –Definition, advantages and disadvantages of providing islands.
- Types of traffic islands - rotary or central, channelizing or Refuge Island.
- Road intersections or junctions - Definition, Types of road intersection.
- Intersection at grade- Types, basic requirements of good intersection at grade.
- Grade separated intersection- advantages and disadvantages, types - flyovers-partial andfull Cloverleaf pattern, Diamond intersection, Trumpet type, underpass.

Unit– V Road Accident Studies and Arboriculture

- Road Accidents-Definition, types and causes for collision and non-collision accidents.
- Measures to prevent road accidents.

- Collision and condition diagram.
- Street lighting –definition, necessity, types-luminaire, foot candle, lumen, factors affectingtheir utilization and maintenance.
- Arboriculture- definition, objectives, factors affecting selection of type of trees.
- Maintenance of trees - protection and care of road side trees.

Suggested learning resources:

- Khanna S.K., Justo, C E G and Veeraragavan, A., Highway Engineering, Nem Chand and Brothers, Roorkee.
- Kadiyali L.R., Transportation Engineering, Khanna Book Publishing Co., Delhi
- Vazirani, V N , Chaondola, S P, Transportation Engineering Vol. I & II, Khanna Publishers. Delhi.
- Saxena, S C, Traffic planning and design, Dhanpat Rai & Sons Delhi.
- Kumar R S, Introduction to Traffic Engineering, University Press (India), Pvt. Ltd.
- Dr. Sharma, S. K., Principles, Practices and Design of Highway Engineering , S. Chand &Company Ltd., New Delhi

Course outcomes: After completing this course, student will be able to:

- Analyze road traffic characteristics.
- Undertake various types of road traffic studies.
- Use relevant road traffic signs, signal and markings.
- Identify the intersection depending on the traffic flow.
- Suggest preventive measures to avoid accidents by analyzing the traffic conditions at site.

Name of the Course	**Diploma in Civil Engineering**	Course duration	6 semester
Course Title	**Elective - III : Building Services and Maintenance**	Course Code	**CEPE507/I**
Subject offered in Semester	Fifth	Number of Credits	3(L:3, T: 0, P: 0)
Prerequisites	NIL	Course Category	**PE**
Question distribution	As per standing norms of WBSCT&VE&SD	Marks distribution	As per standing norms of WBSCT&VE&SD

Course Objectives:

Following are the objectives of this course:

- To know the procedure for classifying various types of buildingservices.
- To know the fire safety requirements for multi-storeyedbuilding.
- To devise suitable plumbing system for given type ofbuilding.
- To understand the procedure for rain water harvesting and solar waterheater.
- Toknowthesystemfordesigninglighting,ventilationandacousticsforanybuilding.

Module /Group [as per directives from WBSCT&VE&SD in framing questions of end semester]	**Distribution of unit**
Module A/Group A	Unit I and II
Module B/Group B	Unit III and V
Module C /Group C	Unit IV

Course Content:

Unit – I Overview of Building Services

- Introduction to building services, Classification of buildings as per National Building code, Necessity of building services, Functional requirements of building, Different types of buildingservice-si.e.HVAC(Heat,VentilationandAirConditioning),Escalatorsandlifts,firesafety, protectionandcontrol,plumbingservices,rainwaterharvesting,solarwaterheatingsystem, lighting, lightening proof arrangement in building, acoustics, sound insulation and electric installationetc.
- Role and responsibility of Building Service Engineers and Plumbers, licensing of a plumber, Introduction to BMS (Building Management Services), Role of BMS, concept of smart building.

Unit– II Modes of vertical communication

- Objectives and modes of vertical communication inbuilding.
- Lifts**:**Differenttypesofliftsanditsuses,ComponentpartsofLift-LiftWell,Travel,Pit,Hoist Way, Machine, Buffer, Door Locks, Suspended Rope, Lift Car, Landing Door, Call Indicators, EmergencyCall Push, machine roometc., Design provisions for basic size calculation of space enclosure to accommodate lift services, number of required lifts in multi-storeyed apartment, Safetymeasures.
- Escalators**:** Different Types of Escalators and its Uses, Components of escalators, Design provisions for basic size calculation of space enclosure to accommodate escalator services, Safetymeasures.
- Ramp: Necessity, design consideration, gradient calculation, layout and Special features required for physically handicapped andelderly,safety measures.

Unit– III Fire Safety

- Fire protection requirements for multi-storeyed building, causes of fire in building, Fire detecting and various extinguishing systems, Working principles of various fire protection systems.
- Safety against fire in residential and public buildings (cinema hall, theatrehall,

mall and other multi-storeyed buildings), National Building Code provision for fire safety, Fire resisting materials and their properties, Fire resistant construction, procedures for carrying out fire safety inspections of existing buildings, Provisions forevacuation.

Unit– IV Plumbing Services

- Importanceofplumbing,AHJ(AuthorityHavingJurisdiction)approval,PlumbingTerminology and fixtures: Terms used in plumbing, Different types of plumbing fixtures, shapes/ sizes, capacities, situation and usage, Traps,Interceptors.
- System of plumbing for building water supply: storage of water, hot and cold water supply system.
- System of plumbing for building drainage: Types of drainage system such as two pipe system, one pipe system, types of Vents and purpose of venting, Concept of grey water and reclaimedwater, maintenance of building drainage.
- Different types of pipes, common joints and valves, materials, and jointing methods, fittings, hanger, supports and valves usedin plumbing and their suitability.

Unit– V Lighting, Ventilation and Acoustics

- Concept of SWH (Solar water heating), component parts of SWH, various system of SWH (heat transfer, propulsion, passive direct system, active direct system, Do-it-yourself), installation andmaintenance.
- Concept of lighting, types of lighting (natural and artificial), factors influencing the brightness of room, factors affecting selection of artificial lighting, installation of light (direct, half-direct,indirect,half-indirectanddirect-indirect),typesoflightcontrol(manualswitch, one way and two way switch, remote switch, timer switch and photo-electric cell switch), typesof lamps(incandescent,tungstenhalogenandelectricdischarge),Lampselectionasperroom sizes.
- Concept of ventilation, necessity and Types ofventilation.
- Building Acoustic, Objectives, acoustic Control in a building, acoustic material (porous absorber and cavityresonator), factors to be considered in acoustic design

of cinema halls and auditoriums.

Suggested learning resources:

1. Patil, S. M., Building Services, Seema Publication,Mumbai.
2. MantriandSan-deep.,TheAtoZofPracticalBuildingConstructionanditsManagement,Satya Prakashan, NewDelhi.
3. Bag S P, Fire Services in India: History, Detection, Protection, Management, Mittal Publications, NewDelhi.
4. Deolalikar,S. G., Plumbing Design and Practice,McGraw-Hill,
5. AkhilKumar-Das.,PrinciplesofFireSafetyEngineering:UnderstandingFireandFireProtection, PHI Learning Pvt. Ltd, NewDelhi.
6. Shraman N L, Solar panel installation guide & user manual, The Memory Guru ofIndia.
7. Gupta M K, Practical handbook on building maintenance - Civil works, Nabhi-Publications.
8. BIS., National Building Code Part1, 4, 8, 9., Bureau of Indian Standard, NewDelhi
9. BIS., IS 12183(Part 1):1987 Code of practice for plumbing in multistoried buildings., Bureau of Indian Standard, NewDelhi
10. BIS., 2008 Uniform Plumbing Code – India (UPC- I)., Bureau of IndianStandard

Course outcomes:

After completing this course, student will be able to:

- Classify various types of building services as per functionalrequirements.
- Propose the fire safety requirements for multi-storeyedbuilding.
- Devise suitable water supply and sanitation system for given type ofbuilding.
- Evaluate the potential of rain water harvesting and solar water heater system for the given type ofbuilding.
- Justify the necessity of designing the system of lighting, ventilation and acoustics for the given type ofbuilding.

Name of the Course	**Diploma in Civil Engineering**	Course duration	6 semester
Course Title	**Elective - III : Repair and Maintenance of Structures**	Course Code	**(CEPE507/II)**
Subject offered in Semester	Fifth	Number of Credits	3 (L:3, T: 0, P:0)
Prerequisites	NIL	Course Category	**PE**
Question distribution	As per standing norms of WBSCT&VE&SD	Marks distribution	As per standing norms of WBSCT&VE&SD

Course Objectives: Following are the objectives of this course:

- To learn about types of maintenance techniques
- To understand causes of various types of damages.
- To know about relevant materials for repair.
- To learn methods of retrofitting for different structures.

Module /Group [as per directives from WBSCT&VE&SD in framing questions of end semester]	**Distribution of unit**
Module A/Group A	Unit I and II
Module B/Group B	Unit III and V
Module C /Group C	Unit IV

Course Content:

Unit – I Basics of maintenance

•Types of Maintenances - repair, retrofitting, re-strengthening, rehabilitation and restoration.

• Necessity, objectives and importance of maintenance.

•Approach of effective management for maintenance.

•Periodical maintenance: check list, maintenance manual containing building plan, reinforcement details, material sources, maintenance frequency, corrective maintenance procedures and sources. Pre- and post- monsoon maintenance.

Unit– II Causes and detection of damages

•Causes of damages due to distress, earthquake, wind, flood, dampness, corrosion, fire, deterioration, termites, pollution and foundation settlement.
•Various aspects of visual observations for detection of damages.
•Load test and non-destructive tests (brief description). NDT tests on damaged structure such as rebound hammer, ultrasonic pulse velocity, rebar locator, crack detection microscope, digital crack measuring gauge.
•Chemical test - Chloride test, sulphate attack, carbonation test, pH measurement, resistivity method, Half-cell potential meter (Introduction and demonstration only).

Unit– III Materials for maintenance and repairs

•Types of repair material, material selection.
•Essential parameters for maintenance and repair materials such - bond with substrate, durability.
•Waterproofing materials based on polymer modified cement slurry, UV resistant acrylic polymer, ferro-cement.
•Repairing materials for masonry: plastic/aluminum nipples, non-shrink cement, polyester putty or 1:3 cement sand mortar, galvanized steel wire fabrics and clamping rods, wire nails, ferro-cement plates.
•Repairing materials for RCC: epoxy resins, epoxy mortar, cement mortar impregnated with polypropylene, silicon, polymer concrete composites, sealants, fiber reinforcement concrete, emulsions and paints.

Unit– IV Maintenance and repair methods for masonry Construction

•Causes of cracks in walls - bulging of wall, shrinkage, bonding, shear, tensile, vegetation.
•Probable crack location: junction of main and cross wall, junction of slab and wall, cracks in masonry joints.
•Repair methods based on crack type - For minor & medium cracks (width 0.5 mm to 5mm): grouting and for major cracks (width more than 5mm): fixing mesh across cracks, RCC band, installing ferro-cement plates at corners, dowel bars, propping of load bearing.
•Remedial measures for dampness & efflorescence in wall.

Unit– V Maintenance and repair methods for RCC Construction

- Repair stages such as concrete removal and surface preparation, fixing suitable formwork, bonding/passive coat and repair application, various methods of surface preparation

- Repair options such as grouting, patch repairs, carbonated concrete, cleaning the corroded steel, concrete overlays, latex concrete, epoxy bonded mortar and concrete, polymer concrete, corrosion protection such as jacketing.
- Building cracks and its prevention, common methods for dormant crack repairs such as Epoxy injection, grooving and sealing, stitching, grouting, guniting, shotcreting
- Strengthening methods for live cracks such as addition of reinforcements, Jacketing, brackets, collars, supplementary members i.e. shoring, underpinning and propping of framed structure.

Suggested learning resources:

1. Gahlot, P. S., Sharma, S., Building Repair and Maintenance Management, CBS Publishers & Distributors Pvt. Ltd., New Delhi
2. Guha, P. K., Maintenance and Repairs of Buildings, New Central Book Agencies
3. Hutchin Son, B. D., Maintenance and Repairs of Buildings, Newnes-Butterworth
4. Relevant BIS codes
5. CPWD Maintenance Manual 2012

Course outcomes: After completing this course, student will be able to:

•Decide which type of maintenance is needed for a given damaged structure

•Assess causes of damages various types of structures.• Select the relevant material for repair of the given structure.

•Apply relevant method of retrofitting for re-strengthening of structures.

•Suggest relevant technique to restore the damages of the given structural elements

Name of the Course	**Diploma in Civil Engineering**	Course duration	6 semester
Course Title	**Internship-II after fourth Semester**	Course Code	**CEI508**
Subject offered in Semester	**Fifth**	Number of Credits	**1 (L:0, T: 0, P: 0)**
Prerequisites	NIL	Course Category	**I**
Question distribution		Marks distribution	As per standing norms of WBSCT&VE&SD

Course Objectives: Following are the objectives of this course:

- To participate in industrial environment and Civil Engineering activities.
- To study on supervision of field level works.
- To work part of working drawings.
- To be involved with online courses.

NOTE: Activity(s) from the following has to be performed and prepare a report on the activity together with presentation before the subject teacher.

Activities may be arranged by the West Bengal State Council of Technical and Vocational Education & Skill Development.

Board of Practical Studies, MSME or Department of Small Scale Industries or other engineering department of State Government may be involved. Initiative from the Department of Technical Education, Training and Skill Development is highly solicited.

Activities centering Private organization in the arena of civil engineering construction/ planning and design/ supervision/marketing etc may also be considered.

It may be arranged in-campus or off-campus; online or offline mode or blended mode.

Activities may be conducted continuously for stipulated period of time or may be arranged in a staggered fashion – in the later case Saturday and Sunday may be utilized for the Internship Program and accordingly class schedule will have be to arranged.

Activities may be performed by a group of students or may be done by individually under the guidance of subject teacher(s)

At least one activity has to be performed:

1	To participate and study site layout, excavation of sites, masonry works, concreting or any other field level construction activities at site.
2	To study and observe on Soil Testing and prepare Report required for multi-storey building construction or any other Civil Engineering Construction Project.
3	To visit Civil Engineering consultancy-cum-design office, study and get brief idea about Staad-Pro or any other design or civil engineering software.
4	To visit company related to workshop oriented or fabrication jobs and get brief ideas on these.
5	To be accustomed with Massive open online courses (Moocs) under Swayam Platform; register and study on course selected by Civil Engineering Faculty of the Institute for duration as fixed by WBSCTVESD. NO certificate to be produced during internal or external sessional examination. Only knowledge student can gather will be judged and evaluated accrodingly.
6	To undertake any other suitable training or Internship programme selected by Civil Engineering Faculty of the Institute for duration as fixed by WBSCTVESD

After completion of each internship, the student should prepare a comprehensive report to indicate what he/she has observed and learnt in the training period. The student may contact Industrial supervisor/Faculty member/TPO for assigning topics and problems and should prepare the final report on assigned topics. The training report should be signed by the Industrial supervisor/Internship Faculty member, TPO and HOD.

The internship report will be evaluated on the basis of following criteria (as applicable):

Sl no	Criteria for evaluation of Internship Report
1	Origianlity
2	Adequacy and purposeful writeup
3	Orgainizaton, format, drawing,sketches style language
4	Practical applications and relationship with basic

	theory
5	Concepts taught in the course outcomes
6	Attendance record, daily diary,quality of the internship report

Seminars must be arranged for the students based on his/her training report, before an internal committee constituted by the concerned department of the institute. The evaluation will be based on the following criteria:

Sl no	Criteria for evaluation of Internship Report
1	Quality of content presented
2	Proper planning for presentation
3	Effectiveness of presentation
4	Depth of knowledge and skills
5	Viva voce

Course outcomes: After completing this course, student will be able to:

- Understand the brief knowledge of industrial activities.
- Perform online courses under Moocs or other courses.
- Project the industrial activities with his/her different courses of Diploma in Civil Engineering.

Name of the Course	**Diploma in Civil Engineering**	Course duration	6 semester
Course Title	**Major Project I**	Course Code	**CEPR509S**
Subject offered in Semester	**Fifth**	Number of Credits	**2 (L:0, T: 0, P: 4)**
Prerequisites	NIL	Course Category	PR
Question distribution		Marks distribution	As per standing norms of WBSCT&VE&SD

Objective: - Following are the objectives of this course:

- To collect the information for a given project.
- To apply principles, theorems and bye-laws in the project planning and design.
- To interpret and analyze the data.
- To develop professional abilities such as persuasion, confidence, and perseverance and communication skill.
- To develop presentation skill.
- To enhance creative thinking.

The project report shall be in the following format:

(The project shall be undertaken by a group of 4 to 6 students)

- Topic and objectives
- Collection of data, required survey work,
- Management and construction procedure
- Resources scheduling and networking
- Design details
- Required drawing set
- Utility to society if any
- Conclusion

NOTE: Same Planning, Drawings and detailing of the problem given in the semester 4 will have be used in Major Project I and Major Project II in semester 5 and 6. All draw-

ing will be done using CAD.

Contents:-

- Title of the Project:-Planning and designing of (G+2) Residential Complex for Middle Income Group.
- The details of the Project are given below:-

(Same topic from the 4th semester shall be considered)

- Each building (RCC framed structure) shall comprise of two symmetrical flat per floor each containing two rooms, bath, WC, kitchen, front verandah with a provision of common staircase and mumty for utilization of roof space and overhead water tank (around 210 sq m. covered area for each building unit and total 100 Nos of flat in the Complex of around 10000 sq m. of total land area) Ground floor to be used for parking spaces.
- The following provisions are to be considered during the project planning:- a) Security room(Single room with WC, Load bearing wall structure), b) Central Park, c) Play Ground, d) Hume Pipe Culvert in between the complex and the 12m wide main road, e) Boundary Wall with main gate, f) Submersible Pump, g) Pump House (Load bearing wall structure), h) Surface Drainage System, i) Bituminous road over WBM inside the complex etc.

The project report shall include:-

1) Load calculation & design of all structural components, structural details – reinforcement of: floor slab (as a whole), critical beam, central column and corner column including footing (Drawing of trench plan), tie beam(tie beam layout plan, and reinforcement detailing), stair with landing etc.

2) Details structural drawing of

a. RCC footing, Column, Beam, Lintel and chajja, staircase – landing slab and waist slab, stringer beam etc., floor slab and roof slab

b. drawing of various types of doors and windows etc. used in construction

3) Prepare bar bending schedule and determine cutting length of all structural items of workshop

4) Prepare a design mix for the project

5) Specification of different items of works

Course outcomes: After completing this course, student will be able to:

- **Design various structural members of the building.**
- **Prepare the drawing and detailing of project by using various drafting softwares**.
- Prepare bar bending schedule of an items of works
- Write specifications for different items of works.
- **Solve the problem by working in a group.**

Name of the Course	**Diploma in Civil Engineering**	Course duration	6 semester
Course Title	**Safety Engineering & Management in the Construction Sector**	Course Code	**CEPC510**
Subject offered in Semester	**Fifth**	Number of Credits	2 (L:2, T: 0, P: 0)
Prerequisites	NIL	Course Category	**PC**
Question distribution	As per standing norms of WBSCT&VE&SD	Marks distribution	As per standing norms of WBSCT&VE&SD

Course Objectives: Following are the objectives of this course:

- to be able to comprehend the various safety guidelines applicable for different construction industries
- to be able to understand various hazard and contaminant discharges related to various construction industries
- to be able to plan and implement effective control measures to prevent health hazards, accident and unwanted contaminant discharge from different construction industries.

Module /Group [as per directives from WBSCT&VE&SD in framing questions of end semester]	**Distribution of unit**
Module A/Group A	Unit I and II
Module B/Group B	Unit III
Module C /Group C	Unit IV and V

Course Content:

Unit 1: Safety Philosophy:

Physical, Physiological and Psychological Factors of Safety. Safety Education and Training. Employees Participation in Safety. Economics of Safety. Behavioral Safety culture and motivation. Safety Laws: Provisions of Factories Act and Rules, Employees State Insurance Act.

Unit 2:Safety Management system & Guidelines:

Key elements of Safety Management system & Guidelines (ISO 14001, ISO 45001:2018/OHSAS 18001 etc.), ILO Legislation – Convention and Recommendation concerning Safety, Health and Environment – Safety, Health and Environment as Human Right Issue, Awareness programme, types of occupational health hazards in industries, physical, chemical, biological, mechanical and psychological hazards, common work related or occupational diseases, occupations involving risk of contracting these diseases, mode of causation of the diseases and its effect, diagnostic methods, methods of prevention, notifiable occupational diseases, compensation for occupational diseases, evaluation of injuries, medical services in an industrial establishment and its functions, occupational health audit and survey, occupational diseases relating to construction work, emergency medical treatment of injuries and rehabilitation at construction site. Personal protective equipment Introduction and requirements and assessment of PPE, type of PPE. Non respiratory personal protective devices: head protection, ear protection, face and eye protection, hand protection, feet protection, body protection. Supply, use, care and maintenance of personal protective equipment. Requirements under Factories Acts and Rules. Respiratory personal protective devices, Classification of hazards, classification of respiratory personal protective devices, selection of respirators, instructions and hints in the use of breathing apparatus, supply, use, care and maintenance of breathing apparatus, training in the use of breathing apparatus.

Unit 3: Safety in Construction and Demolition Operation:

a. Safety in Conventional Construction Operations: Underground works, above ground works, underwater portions, movements of construction machinery, special works, safety in use of explosives. Safety in stacking, storage and transport of construction materials: reinforcements, cement, sand, aggregates, chemicals, organic binders, gas cylinders. Safety in use of construction machinery and equipment: batching plant, mixers, earth moving equipment, cranes, pile driving equipment, excavators, drilling equipment, welding equipment, gas cutting equipment, grinding equipment, derricks, compressors, crushers, layers.

b. Safety in Special Construction Operations: Transmission towers, railways, power plants, transformer installations. Working at heights and prevention of falls of persons: high incidence of serious accidents in working at heights, types of operations, planning operations, safety features associated with construction, design and use of gangways, floors, ladders of different types, scaffolds of different types, other safety requirements while working at height, prevention of falls at floor level.

Potential tripping, slipping hazards. Precautions from falling of materials. Laws and Regulations – Relevant Provisions of Building and Other Construction Workers (Regulation of Employment and Conditions of Service) Act and Rules – National Building Codes.

Unit 4: Hygiene Concept, definition and importance of hygiene in construction industry:
Difference between domestic hygiene and industrial hygiene. Physical hazards – heat stress and its control, ventilation, noise, vibration, illumination, thermal radiation, X rays, ultra violet radiation, ionizing and non-ionizing radiations, sensitization to different air, water and waste water, soil contaminant related to construction industries, permissible exposure limits, effects of exposure, preventive and control measures.

Unit 5: Strategic management and planning for prevention and control of contaminant discharge from process operation in construction industries:
sensitization to different process operation in different construction industries, identification of sources of contamination and hazards, planning for control and prevention, implementation through proper measures, planning for control and prevention, safety audits, non compliance reports, turn around time for post audit compliance and implementation through proper measures.

Reference books:

- ISO 14001
- ISO 45001:2018/OHSAS 18001
- ILO Legislation
- Jha, N. Kumar, Patel, Dilip A, Singh Amarjit, Cosntruction Safety Management, Pearson India Education Seervices Pvt. Ltd, Noida, UP.
- Haldar, S.K., Occupational Health & Hygiene in Industry, CBS Publishers
- Das, Akhil Kumar, Principles of Industrial Safety Management by, PHI

Course outcomes: After completing this course, student will be able to:

- understand the basic tenets of safety engineering
- interpret the existing safety engineering guidelines and recommendations
- identify various safety requirements in conventional and special construction and associated demolition sectors.
- Understand hygiene and safety associated with construction industries
- analyze the problems related to safety hazard and contaminant discharge from construction industries and to suggest remedial measures.

West Bengal State Council of Technical & Vocational Education and Skill Development
(Technical Education Division)



Syllabus
of

Diploma in Civil Engineering [CE]

Part-III (6th Semester)

2023

CURRICULAR STRUCTURE FOR PART – III SECOND SEMESTER (SIXTH SEMESTER) OF THE FULL-TIME DIPLOMA COURSE IN CIVIL ENGINEERING

Sl. No.	Subject Code	Subject of Study	HOURS PER WEEK			CREDITS	Marks
		Theoretical Papers	Lecture	TUTORIAL	PRACTICAL		
1	CEPC601	Public Health Engineering	2	0	0	2	100
		LABORATORY/SESSIONAL PAPERS					
2	CEPC602S	Civil Engineering Lab-III: Module-VII: Public Health Engineering Lab	0	0	2	1	100
3	CEPC603S	Advanced Surveying Practices	0	0	2	1	100
		ELECTIVE, MANDATORY COURSES AND SEMINAR					
4	CEPE604	Elective IV: one subject out of Tendering and Accounts (CEPE604/I) / Advanced Construction Technology (CEPE604/II)	3	0	0	3	100
5		Entrepreneurship and Start-ups	3	0	0	3	100
6		Compulsory Open Elective: Open Elective I: Engineering Economics &Project Management	3	0	0	3	100
7		Open Elective II : one subject has to be taken from list of open elective subjects as provided by WBSCT&VE&SD*	3	0	0	3	100
8.	CEPR605S	Major Project-II	0	0	4	2	100
9.	CESE606S	Seminar and Viva -Voce	1	0	0	1	100
		TOTAL	15	0	8	19	900

NOTE: -

1. All subjects (theoretical as well as sessional/practical) are full paper with 100 marks in aggregate as per AICTE and WBSCT&VE&SD

2. Advanced Surveying Field Practices may also be conducted in 2-3 weeks field work continuously instead of having 2 practical classes per week for the entire semester.

Note: Civil students will be benefited if they choose Solid waste management/ Disaster Management/ Construction Management as the subject for Open Elective II from the list provided by WBSCT&VE&SD

Name of the Course	**Diploma in Civil Engineering**	Course duration	6 semester
Course Title	**Public Health Engineering**	Course Code	**CEPC601**
Subject offered in Semester	**Sixth**	Number of Credits	**2 (L:2, T: 0, P: 0)**
Prerequisites	NIL	Course Category	**PC**
Question distribution	As per standing norms of WBSCT&VE&SD	Marks distribution	As per standing norms of WBSCT&VE&SD

Course Objectives: Following are the objectives of this course:

- To learn the principles for identification of sources of surface and subsurface water
- To learn calculation of population and requirement of drinking water
- To understand the flow-diagram of water supply scheme highlighting different features
- To know evaluation of characteristics and treatment of sewage.

Module /Group [as per directives from WBSCT&VE&SD in framing questions of end semester]	**Distribution of unit**
Module A/Group A	Unit I and II
Module B/Group B	Unit III and V
Module C /Group C	Unit IV

Course Content

Unit – I Sources, Demand and Quality of water

- Water supply schemes - Objectives, components
- Sources of water: Surface and Subsurface sources of water,
- Intake Structures, Definition and types, Factors governing the location of an intake structure, Types of intakes.

- Demand of water: Domestic, Industrial, commercial & institutional, public use, losses & wastes, fire demand. Factors affecting rate of demand, Variations of water demands, Forecasting of population, Methods of forecasting of population, (Simple problems on forecasting of population), Design period, Estimation of quantity of water supply required for city or town.
- Quality of water**:** Need for analysis of water, Characteristics of water- Physical, Chemical and Biological, Meaning and importance of different parameters of water- Total solids, hardness, chlorides, dissolved Oxygen, pH, Fluoride, Arsenic, Nitrogen and its compounds, Bacteriological tests, Ecoli, Bcoli index, MPN, Sampling of water, Water quality standards as per IS:10500.

UNIT II Purification of water

- Purification of Water: Objectives of water treatment, Screening, Aeration- objects and methods of aeration, Plain sedimentation, Sedimentation with coagulation, principles of coagulation, types of coagulants, Jar Test, process of coagulation, types of sedimentation tanks, Clariflocculator.
- Filtration - mechanism of filtration, classification of filters: slow sand filter, rapid sand filter, pressure filter. Construction and working of slow sand filter and rapid sand filter, operational problems in filtration. Disinfection: Objects, methods of disinfection, Chlorination- Application of chlorine, forms of chlorination, types of chlorination practices, break-point chlorination, residual chlorine and its importance, Flow diagram of water treatment plants.
- Miscellaneous water Treatments: Removal of colour, taste and odour, Introduction to water softening and Defluoridation techniques.

UNIT III Conveyance and Distribution of water

- Conveyance: Types of Pipes used for conveyance of water, choice of pipe material, Types of joints & Types of valves- their use, location and function on a pipeline.
- Distribution of water: Methods of distribution of water- Gravity, pumping, and combined system, Service reservoirs- functions and types, Layouts of distribution of Water- Dead end system, grid iron system, circular system, radial system; their suitability, advantages and disadvantages.

UNIT IV Domestic sewage and System of Sewerages

- Building Sanitation: Necessity of sanitation, Necessity to treat domestic sewage, Definitions-Sewage, sullage, types of sewage. Definition of the terms related to Building Sanitation-Water-pipe, Rainwater-pipe, Soil-pipe, Sullage-pipe, Vent-pipe. Building Sanitary fittings-Water closet – Indian and European type, flushing cistern, wash basin, sinks, Urinals. Traps- types (P, Q, S, intercepting trap, gully trap, floor trap), qualities of good trap. Systems of plumbing-one pipe, two pipe, single stack, choice of system. Principles regarding design of building drainage, inspection and junction chambers, their necessity, location, size and shape.
- Systems of Sewerage and Sewer Appurtenances: Types of Sewers, Systems of sewerage, self-cleansing velocity and non-scouring velocity, Laying, Testing and maintenance of sewers, Manholes and Drop Manhole-component parts, location, spacing, construction details, Sewer Inlets, Street Inlets.

UNIT V Characteristics and treatment of Sewage

- Analysis of sewage: Characteristics of sewage - Major parameters, B.O.D., C.O.D. and its significance, Central Pollution Control Board Norms for discharge of treated sewage.
- Treatment of Sewage: Objects of sewage treatment and flow diagram of conventional sewage treatment plant - Screening, Types of screens, Grit removal, Skimming, Sedimentation of sewage, Aerobic and anaerobic process, Sludge digestion, trickling filters, Activated sludge process, Disposal of sewage, Oxidation pond, Oxidation ditch. Septic tank, Recycling and Reuse of domestic waste.

Suggested learning resources

- Sharma S.C, Environmental Engineering, Khanna Publishing House, NewDelhi
- Garg, S.K., Environmental Engineering Vol. I and Vol. II, KhannaPublishers
- Birdie, G. S. and Birdie, J. S.Water Supply and Sanitary Engineering, Dhanpat Rai
- Gupta, O.P., Elements of Environmental Pollution Control, Khanna Publishing House,Delhi
- Rao, C.S., Environmental Pollution Control Engineering, New Age International

- Punmia, B C, Environmental Engineering, vol. I and II, LaxmiPublishers
- Peavy H S, Rowe D R, and Tchobanoglous G, Environmental Engineering, McGraw Hill
- Basak N N, Environmental Engineering, McGraw HillPublishers.

Course outcomes: After completing this course, student will be able to:

- Know the procedure to identify the sources of surface and subsurface water
- Estimate the quantity of drinking water required for a population
- Draw labelled layout for water supply scheme.
- Select suitable water treatment technique.
- Evaluate the characteristics and suggest treatment of sewage.

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Name of the Course	**Diploma in Civil Engineering**	Course duration	6 semester
Course Title	**Civil Engineering Lab-III: Module-VII: Public Health Engineering Lab**	Course Code	**CEPC602S**
Subject offered in Semester	**Sixth**	Number of Credits	**1 (L:0, T: 0, P: 2)**
Prerequisites	NIL	Course Category	**PC**
Question distribution		Marks distribution	As per standing norms of WBSCT&VE&SD

Course Objectives: Following are the objectives of this course:

- To learn the tests for measuring quality of drinking water.
- To learn the tests for measuring quality of Domestic waste water.
- To learn determination of BOD and COD requirement in sewage.

List of Practical to be performed: (Items 1 & 2 compulsory and at least six experiments from the rest)

1	Draw sketches of various valves used in water supply pipe line
2	Draw a sketch of one pipe and two pipe system of plumbing
3	Determine pH value of given sample of water/sewage.
4	Determine the turbidity of the given sample of water.
5	Determine residual chlorine in a given sample of water.
6	Determine suspended, dissolved solids and total solids of given sample of water/sewage.
7	Determine the dissolved oxygen in a sample of water/sewage.
8	Determine Fluoride concentration in given water sample.
9	Determine Arsenic concentration (semi-quantative) in given water sample.
10	Determine the optimum dose of coagulant in a given raw water sample by jar test.
11	Determine B.O.D. & C.O.D. of given sample of sewage.
12	Prepare a report of a field visit to water treatment plant and/or sewage treatment plant if possible

Note: Item no. 12 may be included in internship.

Suggested learning resources:

- Sharma S.C, Environmental Engineering, Khanna Publishing House, New Delhi
- Basak N N, Environmental Engineering, McGraw Hill Publishers.
- Garg, S.K., Environmental Engineering Vol. I and Vol. II, Khanna Publishers

- Birdie, G. S. and Birdie, J. S. Water Supply and Sanitary Engineering, Dhanpat Rai
- Gupta, O.P., Elements of Environmental Pollution Control, Khanna Publishing House, Delhi
- Rao, C.S., Environmental Pollution Control Engineering, New Age International
- Punmia, B C, Environmental Engineering, vol. I and II, LaxmiPublishers
- Peavy H S, Rowe D R, and Tchobanoglous G, Environmental Engineering, McGraw Hill Publishers.
- BIS: 10500 - DRINKING WATER — SPECIFICATION, BIS, New Delhi.
- CPCB: GENERAL STANDARDS FOR DISCHARGE OF ENVIRONMENTAL POLLUTANTS, CPCB, New Delhi

Course outcomes: After completing this course, student will be able to:

- Perform various tests to assess quality of drinking water
- Perform various tests to assess quality of domestic sewage
- Understand the function of various components of water treatment and sewage treatment plants.

Name of the Course	**Diploma in Civil Engineering**	Course duration	6 semester
Course Title	**Advanced Surveying Practices**	Course Code	**CEPC603S**
Subject offered in Semester	**Sixth**	Number of Credits	**1 (L:0, T: 0, P: 2) ***
Prerequisites	NIL	Course Category	**PC**
Question distribution		Marks distribution	As per standing norms of WBSCT&VE&SD

*** Advanced Surveying Field Practices may also be conducted in 2 - 3 weeks field work continuously instead of having 2 practical class per week for the entire semester.**

Course Objectives: Following are the objectives of this course:

- To know methods of Theodolite surveying and their uses.
- To learn tacheometric surveying and curve setting.
- To understand the principles of Electronic Distance Measurement and Total station and their uses.
- To know the concept of GPS

List of Practical to be performed: [item number 3 and 5 are desirable]

1. Theodolite traverse Survey: A simple closed traverse of at least 5 sides for any suitable site. Preparation of Gale's traverse table. Plotting the traverse with details on A1 size imperial drawing sheet and calculation of area of the closed traverse. Interior details will have to be filled up by theodolite or by plane table which is found to be more suitable.

2. Simple circular curve setting: Setting out a simple circular curve by Rankine's method of Deflection angles (both one theodolite and two theodolite methods) for a given problem and plotting the curve showing the necessary supporting calculations in a tabular form mentioning suitable scale on A-1 size imperial drawing sheet.

3. Surveying with Total Station: Introduction, description of different parts of total station and reflector prism with stand, set up and orientation,

measurement of distances, measurement of horizontal and vertical angles, methods of measuring remote height and area, etc., Construction of a closed traverse of minimum 5 sides for any suitable site- collection and saving of field data in total station, downloading and transfer of raw data from total station to computer, processing of raw data with the help of any suitable software for preparation of drawing.

4. Use Theodolite as a Tacheometer to compute reduced levels and horizontal distances.

5. GPS Surveying: Working with hand held GPS instrument. Collection coordinates of different objects. Downloading raw data from GPS instrument and prepare a report sheet (excel or doc or pdf format).

Suggested learning resources:

- Kanetkar, T. P.; Kulkarni, S. V., Surveying and Levelling Part I and II, Pune Vidyarthi ruh prakashan, Pune.
- Basak, N. N.,Surveying and Levelling, McGraw Hill Education (India) Pvt. Ltd.,Noida.
- Duggal, S. K., Survey I and Survey II, Tata McGraw Hill Education Pvt. Ltd., Noida.
- Saikia, M D.; Das. B.M.; Das. M.M., Surveying PHI Learning Pvt. Ltd., New Delhi.
- Subramanian, R., Surveying and Levelling, Oxford University Press. New Delhi.
- Punmia, B.C,; Jain, Ashok Kumar; Jain, Arun Kumar, Surveying Vol. I and Surveying vol lI, Laxmi Publications Pvt. Ltd., New Delhi.
- Rao, P. Venugopala Akella, Vijayalakshmi, Textbook of Surveying, PHI Learning Pvt. Ltd., New Delhi.
- Venkatramaiah, C, Textbook of Surveying, Universities Press, Hyderabad.
- Anderson, James M and Mikhail, Edward M, Surveying theory and practice, Mc Graw Hill Education, Noida.
- De, Alak, Plane Surveying, S.Chand Publications, New Delhi

Course outcomes: After completing this course, student will be able to:

- Prepare plans using Theodolite surveys.
- Find distances and elevations using Tachometer.
- Make measurements using Total Station.
- Locate coordinates of survey stations using GPS

Name of the Course	**Diploma in Civil Engineering**	Course duration	6 semester
Course Title	**Elective IV: Tendering and Accounts**	Course Code	**CEPE604/I**
Subject offered in Semester	**Sixth**	Number of Credits	**3 (L: 3, T: 0, P:0)**
Prerequisites	NIL	Course Category	**PE**
Question distribution	As per standing norms of WBSCT&VE&SD	Marks distribution	As per standing norms of WBSCT&VE&SD

Objective:- Following are the objectives of this course:

- To understand terminologies in contract and tender document and their significance.
- To know different types of contracts and their uses.
- To learn preparation of typical Tender documents for civil engineering work.
- To get acquainted with rent fixation and valuation of civil structures.

Module /Group [as per directives from WBSCT&VE&SD in framing questions of end semester]	**Distribution of unit**
Module A/Group A	Unit I and II
Module B/Group B	Unit III
Module C /Group C	Unit IV and V

Contents:

Unit – I Procedure to execute the work

Administrative approval, Technical sanction, budget provision, expenditure sanction.

Methods for carrying out works- contract method, departmental method -rate list method, piece work method, day's work method, employing labours on daily wages basis.

Unit– II Contracts

• Definition of contract, objects of contract, requirements of contract, overview of Indian Contract Act.

• Types of engineering contract with advantages, disadvantages and their suitability - Lump sum contract, item rate contract, percentage rate contract, cost plus percentage, cost plus fixed fee, cost plus variable percentage and cost plus variable fee contract, labour contract, demolition contract, target contract, negotiated contract, Engineering Procurement Construction Contract (EPC), Annuity Contract.

• Introduction of FIDIC Conditions of contract.

• Classification of contractor on basis of financial limits, Requirement of documents for registration of contractor.

• Build Operate Transfer (BOT) Project, BOT Toll contract, BOT (Annuity) contract, Design, Build, Finance, Operate and Transfer (DBFOT) contract, Hybrid Annuity contract, Operate Maintain and Transfer (OMT) contract, Operation & Maintenance contract (Introduction only).

Unit– III Tender and Tender Documents

• Definition of tender, necessity of tender, types of tender- Local, Global, Limited.

• E -Tendering System – Online procedure of submission and opening of bids (Technical and Financial).

• Notice to invite tender (NIT)- Points to be included while drafting tender notice, Drafting of tender notice.

• Procedure of submitting filled tender Documents (Two envelope system), procedure of opening tender, comparative statement, scrutiny of tenders, award of contract, letter of award.

• Meaning of terms - Earnest Money Deposit (EMD), Performance Security Deposit, Validity period, corrigendum to tender notice and its necessity, Unbalanced bid.

• Tender documents – Index, tender notice, general instructions, special instructions, Schedule A, Schedule B, Schedule C etc.

• Terms related to tender documents – contract conditions- time limit, time extension, penalty, defective material and workmanship, termination of contract, suspension of work, subletting of contract, extra items, price variation clause(escalation), defect liability Period, liquidated Damages.

• Arbitration- Meaning, Qualification of an arbitrator, Appointment, Dispute and Settlement of disputes, Arbitration and Conciliation Act, Arbitration award.

Unit– IV Accounts

- Various account forms and their uses – Measurement Books, E- Measurement book (E-MB), Nominal Muster Roll(NMR), Imprest Cash, Indent, Invoice, Bill, Vouchers, Hand receipt Cash Book, Temporary Advance. Heads of Accounts.
- Mode of Payment to the contractor and its necessity -Interim Payment, Advance Payment Secured Advance, Petty advance, Mobilization advance, Running account bill, Final bill, Retention money, E - payment.

Unit– V Introduction to Valuation

• Definition and purpose of Valuation, role of valuer. Definition - Cost, Price and Value, Characteristics of Value, Factors Affecting Value.
• Types of Value - Book Value, Scrap Value, Salvage Value, Speculative Value, Distress Value, Market Value, monopoly Value, Sentimental Value. Factors affecting value.
• Depreciation, Obsolescence, Sinking Fund, Methods of Calculation of Depreciation – Straight Line Method, Sinking Fund Method, Constant Percentage Method.
• Fixation of rent, Lease – types of lease, lease hold property and free hold property. Mortgage– Mortgage deed, precautions to be taken while making mortgage.

Suggested learning resources:

- Datta, B. N., Estimating and Costing in Civil engineering, UBS Publishers Pvt. Ltd., New Delhi
- Raina, V. K., Construction Management and Contract Practices, Shroff Publishers & Distrib- uters Pvt. Ltd.
- Rangawala, S. C., Estimating and Costing, Charotar Publishing House PVT. LTD., Gujrat
- Birdie, G. S., Estimating and Costing, Dhanpat Rai Publishing Company(P) Ltd., New Delhi
- Patil, B. S., Civil Engineering Contracts and Estimates, Orient Longman, Mumbai
- Chakraborti, M., Estimating and Costing, Specification and Valuation in Civil Engineering, Monojit Chakraborti, Kolkata.

Course outcomes: After completing this course, student will be able to:

- Understand various types of contract and when they are used
- Suggest the relevant type of contract for the given civil engineering work.

- Prepare the typical Tender document for the given civil engineering work.
- Decide type of payment for the executed work.
- Justify the rent fixation and valuation of given civil structure.

Name of the Course	**Diploma in Civil Engineering**	Course duration	6 semester
Course Title	**Elective IV: Advanced Construction Technology**	Course Code	**CEPE604/II**
Subject offered in Semester	**Sixth**	Number of Credits	**3 (L: 3, T: 0, P:0)**
Prerequisites	NIL	Course Category	**PE**
Question distribution	As per standing norms of WBSCT&VE&SD	Marks distribution	As per standing norms of WBSCT&VE&SD

Course Objectives: Following are the objectives of this course:

- To gain knowledge on different materials in advanced construction
- To know different methods in concreting.
- To know the relevance of advanced construction methods for particular site condition.
- To identify the requisite hoisting and conveying machinery for the given situation.

Module /Group [as per directives from WBSCT&VE&SD in framing questions of end semester]	**Distribution of unit**
Module A/Group A	Unit I and II
Module B/Group B	Unit III
Module C /Group C	Unit IV and V

Course Content:

Unit – I Advanced Construction Materials

- Fibres: Use and properties of steel, polypropylene, carbon and glass fibres.
- Plastics: Use and properties of PVC, RPVC, HDPE, GFRP, CFRP.
- Miscellaneous Materials: Properties and uses of acoustics materials, wall claddings, plasterboards, micro-silica, waterproofing materials, adhesives, PMC (Polymer modified concrete)

- Use of waste products and industrial by products in bricks, blocks, concrete and mortar.

Unit– II Advanced Concreting Methods and equipment

- Ready Mix Concrete: Necessity and use of ready mix concrete. Products and equipment for ready mix concrete plant. Conveying of ready mix concrete, transit mixers.
- Vibrators for concrete consolidation: Internal, needle, surface, platform and form vibrators.
- Underwater Concreting: Procedure and equipment required for Tremie method, Drop bucket method. Properties, workability and water cement ratio of the concrete.
- Special concrete: procedure and uses of special concretes: High Strength Concrete, High Performance Concrete, Roller compacted concrete, Self-compacting concrete (SCC), Steel fibre reinforced concrete, Foam concrete, Guniting, shotcreting.

Unit– III Advanced Technology in Constructions

- Construction of bridges and flyovers: equipment and machineries required for foundation and super structure.
- Construction of multi-storeyed Building: equipment and machinery required for construction of multi-storeyed building such as use of lifts, belt conveyors, slip-form, pumping of concrete.
- Prefabricated construction: Methods of prefabrication, Plant fabrication and site fabrication, All prefabricated building elements such as wall panels, slab panels, beams, columns, door and window frames etc. equipment and machineries used for placing and Jointing of prefabricated elements.
- Strengthening of embankments by soil reinforcing techniques using geo-synthetics

Unit– IV Hoisting and Conveying Equipment

- Hoisting equipment: Principles and working of Derrick-Pole, Gin Pole, Crane, Power driven scotch derrick crane, Hand operated crane, Locomotive crane, Tower crane,

Lattice Girder, Winches, Elevators, ladders. Crawler cranes, Truck mounted cranes, Gantry cranes, Mast cranes.

- Conveying equipment: Working of belt conveyors, types of belts and conveying mechanism. Capacity and use of dumpers, tractors and trucks.

Unit– V Miscellaneous Machinery and Equipment

- Excavation Equipment: Use, working and output of following machinery – bull dozers, scrapers, graders, Clam Shell, trenching equipment, Tunnel boring machine, Wheel mounted belt loaders, power shovels, JCB, and drag lines.
- Compacting Equipment: Output of different types of rollers such as plain rollers, ship footed rollers, vibratory, pneumatic rollers rammers.
- Miscellaneous Equipment: Working and selection of equipment: Pile driving equipment, Pile hammers, Hot mix bitumen plant, bitumen paver, grouting equipment, guniting equipment, floor polishing and cutting machine selection of drilling pattern for blasting, Bentonite/mud slurry in drilling, Explosives for blasting, Dynamite, process of using explosives.

Suggested learning resources:

- Sharma S C and Deodhar S V, Construction Engineering and Management, Khanna Book Publishing, New Delhi
- Chudly, R., Construction Technology Vol. I to II, ELBS-Longman Group.
- Peurifoy, R. L., Construction Planning Equipment and Methods, McGraw Hill Co. Ltd. New York.
- Seetharaman, S., Construction Engineering and Management, Umesh Publication, New Delhi.
- Sengupta, B. and Guha., Construction Management and Planning, McGraw Hill Education, New Delhi.
- Smith, R. C., Materials of Construction, McGraw Hill Co. Ltd.
- Satyanarayana, R Saxena, S. C., Construction Planning and Equipment, Standard Publication, New Delhi.

- Rangawala,S. C., Construction of Structures and Management of works, Charotar Publication, Anand.
- Ghose,D. N., Materials of Construction, McGraw Hill Publishing Co, New Delhi.

Course outcomes: After competing this course, student will be able to:

- Use relevant materials in advanced construction of structures.
- Use relevant method of concreting and equipment according to type of construction.
- Apply advanced construction methods for given site condition.
- Select suitable hoisting and conveying equipment for a given situation.
- Select advanced equipment required for a particular site condition

Name of the Course	**Diploma in Civil Engineering**	Course duration	6 semester
Course Title	**Entrepreneurship and Start-ups**	Course Code	
Subject offered in Semester	**Sixth**	Number of Credits	**3 (L: 3, T: 0, P: 0)**
Prerequisites	NIL	Course Category	**HS**
Question distribution	As per standing norms of WBSCT&VE&SD	Marks distribution	As per standing norms of WBSCT&VE&SD

Detailed course content will be provided by the council separately

Name of the Course	**Diploma in Civil Engineering**	Course duration	6 semester
Course Title	**Compulsory Open Elective: Open Elective I: Engineering Economics & Project Management**	Course Code	
Subject offered in Semester	**Sixth**	Number of Credits	**3 (L: 3, T: 0, P: 0)**
Prerequisites	NIL	Course Category	**OE**
Question distribution	As per standing norms of WBSCT&VE&SD	Marks distribution	As per standing norms of WBSCT&VE&SD

Detailed course content will be provided by the council separately

Name of the Course	**Diploma in Civil Engineering**	Course duration	6 semester
Course Title	**Open Elective II : one subject has to be taken from list of open elective subjects as provided by WBSCT&VE&SD**	Course Code	
Subject offered in Semester	**Sixth**	Number of Credits	**3 (L: 3, T: 0, P:0)**
Prerequisites	NIL	Course Category	**OE**
Question distribution	As per standing norms of WBSCT&VE&SD	Marks distribution	As per standing norms of WBSCT&VE&SD

Detailed course content of all subjects under open elective II will be provided by the council separately

Name of the Course	**Diploma in Civil Engineering**	Course duration	6 semester
Course Title	**Major Project II**	Course Code	**CEPR605S**
Subject offered in Semester	**SIXTH**	Number of Credits	**2 (L:0, T: 0, P: 4)**
Prerequisites	NIL	Course Category	**PR**
Question distribution		Marks distribution	As per standing norms of WBSCT&VE&SD

Objective: - Following are the objectives of this course:

- To prepare detailed cost estimate of building and various ancillary items of the project.
- To develop professional abilities such as preparation of tender documents and scheduling of the project.
- To develop presentation skill.
- To enhance creative thinking and combination of a complete civil engineering project.

The project report shall be in the following format:

(The project shall be undertaken by a group of 4 to 6 students)

- Topic and objectives
- Collection of data, required survey work,
- Management and construction procedure
- Resources scheduling and networking
- Design details
- Required drawing set
- Utility to society if any
- Conclusion

NOTE: Same Planning, Drawings and detailings of the problem given in the semester 4 will have be used in Major Project I and Major Project II in semester 5 and 6. All drawing will be done by using CAD

Contents:-

- Title of the Project:-Planning and designing of (G+2) Residential Complex for Middle Income Group.
- The details of the Project are given below:

- Each building (RCC framed structure) shall comprise of two symmetrical flat per floor each containing two rooms, bath, WC, kitchen, front verandah with a provision of common staircase and mumty for utilization of roof space and overhead water tank (around 210 sq m. covered area for each building unit and total 100 Nos of flat in the Complex of around 10000 sq m. of total land area) Ground floor to be used for parking spaces.
- The following provisions are to be considered during the project planning:- a) Security room(Single room with WC, Load bearing wall structure), b) Central Park, c) Play Ground, d) Hume Pipe Culvert in between the complex and the 12m wide main road, e) Boundary Wall with main gate, f) Submersible Pump, g) Pump House (Load bearing wall structure), h) Surface Drainage System, i) Bituminous road over WBM inside the complex etc.

1) The project report shall include detailed Estimate and costing of:-

(i) (G+2) Building Unit.

(ii) Security Room.

(iii) Pump House.

(iv) Boundary Wall with main gate.

(v) Submersible Pump.

(vi) Hume Pipe Culvert.

(vii) Bituminous road over WBM.

(viii) Surface Drainage System.

2) The detailed report shall contain total Cost of the Project, Bar Chart, Project completion time using CPM/PERT & Preparation of tender documents for NIT (Notice inviting tender).
(Rate should be taken as per West Bengal PWD Schedule w.e.f 01.11.2017 with latest Corrigendum.)

3) PPT Presentation of the whole projects group wise

Course outcomes: After completing this course, student will be able to:

- Solve the problem by working in a group.
- Estimate the total cost of the project.
- Prepare tender documents and NIT.
- Know about project scheduling.

Name of the Course	**Diploma in Civil Engineering**	Course duration	6 semester
Course Title	Seminar and Viva-voce	Course Code	**CESE606S**
Subject offered in Semester	Sixth	Number of Credits	1 (L: 1, T: 0, P:0)
Prerequisites	NIL	Course Category	**SE**
Question distribution		Marks distribution	As per standing norms of WBSCT&VE&SD

NOTE: Seminar will cover any topic(s) related to the subjects taught from semester 1 to semester 6. Viva-voce will cover any subject covered from semester 1 to semester 6.

Course Title	Entrepreneurship and Start-ups
Course Code	HS 302
Number of Credits	3
Pre Requisites	None
Total Contact Hours	3(L: 2; T: 1)/week = 45 hrs
Course Category	HS

Course Learning Objectives

1. To raise awareness, knowledge and understanding of enterprise/ entrepreneurship.
2. To motivate and inspire students toward an entrepreneurial career.
3. To understand venture creation process and to develop generic entrepreneurial competences.
4. To introduce students to the basic steps required for planning, starting and running a business.
5. To familiarise students with the different exit strategies available to entrepreneurs.

Course Outcomes:

After completing the course students will able to:

CO 1	Identify qualities of entrepreneurs, develop awareness about entrepreneurial skill and mindset and express knowledge about the suitable forms of ownership for small business
CO 2	Comprehend the basics of Business idea, Business plan, Feasibility Study report, Project Report and Project Proposal
CO 3	Understand the concept of start-up business and recognise its challenges within legal framework and compliance issues related to business.
CO 4	Make a Growth Plan and pitch it to all stakeholders and compare the various sources of funds available for start-up businesses

Detailed Course Content

Unit	Name of the Topic	Hours
1.	**ENTREPRENEURSHIP – INTRODUCTION AND PROCESS** • Concept, Competencies, Functions and Risks of entrepreneurship • Entrepreneurial Values& Attitudes and Skills • Mindset of an employee/manager and an entrepreneur • Types of Ownership for Small Businesses o Sole proprietorship o Partnerships o Joint Stock company- public limited and private limited	10

	companies • Difference between entrepreneur and Intrapreneur	
2.	**PREPARATION FOR ENTREPRENEURIAL VENTURES** • Business Idea- Concept, Characteristics of a Promising Business Idea, Uniqueness of the product or service and its competitive advantage over peers. • Feasibility Study – Concept – Locational, Economic, Technical and Environmental Feasibility. Structure and Contents of a standard Feasibility Study Report • Business Plan – Concept, rationale for developing a Business Plan, Structure and Contents of a typical Business Plan • Project Report- Concept, its features and components • Basic components of Financial Statements- Revenue, Expenses (Revenue & capital exp), Gross Profit, Net Profit, Asset, Liability, Cash Flow, working capital, Inventory. Funding Methods-Equity or Debt. Students are just expected to know about the features and key inclusions under, Business Plan and Project Report. <u>They **may not** be asked to prepare a Business Plan/ Project Report/ Project Feasibility Report in the End of Semester Examination.</u>	20
3.	**ESTABLISHING SMALL ENTERPRISES** • Legal Requirements and Compliances needed for establishing a New Unit- o NOC from Local body o Registration of business in DIC o Statutory license or clearance o Tax compliances	03
4.	**START-UP VENTURES** • Concept & Features • Mobilisation of resources by start-ups: Financial, Human, Intellectual and Physical • Problems and challenges faced by start-ups. • Start-up Ventures in India – Contemporary Success Stories and Case Studies to be discussed in the class. Case studies have been included in the syllabus to motivate and inspire students toward an entrepreneurial career from the success stories. <u>No questions are to be set from the case studies.</u>	04

5.	**FINANCING START-UP VENTURES IN INDIA** • Communication of Ideas to potential investors – Investor Pitch • Equity Funding, Debt funding – by Angel Investors, Venture Capital Funds, Bank loans to start-ups • Govt Initiatives including incubation centre to boost start-up ventures • MSME Registration for Start-ups –its benefits	06
6.	**EXIT STRATEGIES FOR ENTREPRENEURS** • Merger and acquisition exit, Initial Public Offering (IPO), Liquidation, Bankruptcy – **Basic Concept only**	02

Examination Scheme

- **End Semester Examination: 60 marks**

  Suggested Question Paper Scheme for End Semester Examination

  Group A: 20marks

Question Type	Number of questions to be set	Number of questions to be answered
MCQ, Fill in the blanks, True or False (Carrying 1 mark each)	25	20

  Group B: 40marks

Question Type	Number of questions to be set	Number of questions to be answered
Subjective Type questions (Carrying 8 marks each)	10	5

- **Internal Assessment: 40 marks**
  - Class test : 20 marks
  - Assignment: 10 marks
  - Class attendance: 10 marks

Suggested Learning Resources

Sl. No.	Title of Book	Author	Publication
1.	Entrepreneurship Development	Sangeeta Sharma	Prentice Hall of IndiaLearning Private Ltd
2.	Entrepreneurship Development	S. Anil Kumar	New Age International
3.	Fundamentals of Entrepreneurship	Sangram Keshari Mohanty	Prentice Hall of India Learning Private Ltd
4.	Fundamentals of Entrepreneurship	Dr. G.K. Varshney	Sahitya Bhawan Publication
5.	Managing New Ventures: Concepts and Caseson Entrepreneurship	Anjan Raichaudhuri	Prentice Hall of India Learning Private Ltd
6.	How to Start a Business in India	Simon Daniel	Buuks, Chennai
7.	Entrepreneurship and Small Business Management	S.S. Khanka	S. Chand & Sons, New Delhi
8.	Entrepreneurship Development and Business Ethics	Abhik Kumar Mukherjee & Shaunak Roy	Oxford University Press
9.	Entrepreneurship Development and Business Ethics	Dr B Chandra & Dr B Biswas	Tee Dee Publications
10.	Entrepreneurship Development Small Business Entrepreneurship	Poornima Charantimath	Pearson Education India

Draft Syllabus of Engineering Economics & Project Management

Course Code:	OE301
Course Title:	Engineering Economics & Project Management
No. of Credits:	3 (L:3, T:0,P: 0)
Prerequisites:	NIL
Course Category:	Open Elective (Compulsory for all branches)

Course Objectives:

- To acquire knowledge of basic economics tofacilitate theprocess ofeconomic decision making.
- To acquire knowledge on basic financial management aspects.
- To develop the idea of project plan, fromdefining and confirming theproject goals and objectives, identifying tasks andhow goalswill be achieved.
- To develop an understanding of key project management skills and strategies.

Group-A

Unit-I (INTRODUCTION, THEORY OF DEMAND & SUPPLY) [9 hours]

1.1 Introduction to Engineering Economics, the relationship between Engineering and Economics

1.2 Resources, scarcity of resources, and efficient utilization of resources.

1.3 Opportunity cost, rationality costs, and benefits

1.4 Theory of Demand: the law of demand, different types of demand, determinants of demand, demand function, price elasticity of demand.

1.5 Theory of Supply: determinants of supply, supply function.

1.6 Market mechanism: Equilibrium, basic comparative static analysis (Numerical problems)

Unit-II (THEORY OF PRODUCTION & COSTS) [10 hours]

2.1 Concept of production (goods & services), Different factors of production (fixed and variable factors), Short-run Production function (Graphical illustration), and Long run production function (returns to scale).

2.2 Theory of Cost: Short-run and long-run cost curves with graphical illustration, basic concept on total cost, fixed cost, variable cost, marginal cost, average cost etc.

2.3 Economic concept of profit, profit maximization (numerical problems)

UNIT-III (DIFFERENT TYPES OF MARKET AND ROLE OF GOVERNMENT) [4 hours]

3.1 Perfect Competition: Features of Perfectly Competitive Market.

3.2 Imperfect Competition: Monopoly, Monopolistic Competition, and Oligopoly.

3.3 Role of government in Socialist, Capitalist and Mixed Economy structure with example.

Group-B

Unit-I (CONCEPT OF PROJECT) [4 hours]

1.1 Definition and classification of projects)

1.2 Importance of Project Management.

1.3 Project life Cycle [Conceptualization→Planning→Execution→Termination]

Unit-II (FEASIBILITY ANALYSIS OF A PROJECT) [10 hours]

2.1 Economic and Market analysis.

2.2 Financial analysis: Basic techniques in capital budgeting– Payback period method, NetPresentValue method, InternalRate of Returnmethod.

2.3 Environmental Impact study–adverse impact of the project on the environment.

2.4 Project risk and uncertainty: Technical, economical, socio-political, and environmental risks.

2.5 Evaluation of the financial health of a project–Understanding the basic concept of Fixed & Working Capital, Debt & Equity, Shares, Debentures etc., and different financial ratios like Liquidity Ratios, Activity Ratios, Debt-equity ratio & Profitability Ratio (Basic concept only).

N.B: Knowledge of financial statements is not required; for the estimation of ratios the values of the relevant variables will be provided.

Unit-III (PROJECT ADMINISTRATION) [8 hours]

3.1 Gantt Chart– a system of bar charts for scheduling and reporting the progress of a project (basic concept).

3.2 Concept of Project Evaluation and Review Technique (PERT) and Critical Path method (CPM): basic concept and application with real-life examples.

ExaminationScheme:

A. SemesterExaminationpatternof60 marks:

1. Objective type Question (MCQ, Fill in the blanks, and Very Short question-1 mark each): At least five questions from each unit. [Total marks: 20]

2. Subjective questions: Eight questions to be answered taking at least three from each group. (Two questions should be given from each unit). [Total marks: 40]

B. Assignment (10Marks)

Guidelinefor Assignment (10 Marks)

Students should be instructed to prepare a report on a project (preferably the Major Project they prepare in 6th Semester), using a popular project management software in IT/ Computer Laboratory, under the guidance of the Lecturer in Computer Science & Technology and Lecturer in Humanities.

C. Class Test: Two examinations 20 marks each. Take best of two.

D. Attendance: 10 Marks

Suggested reference books:

1 *Principles of Economics – Case and Fair, Pearson Education Publication*
2 *Principles of Economics – Mankiw, Cengage Learning*
3 *Project planning, analysis, selection, implementation and review – Prasannachandra–Tata McGraw Hill.*
4 *Project Management – Gopala Krishnan – Mcmillan India Ltd*

Name of the Course	**Diploma in Engineering**	Course duration	6 semester
Course Title	**Disaster Management**	Course Code	**OE**
Subject offered in Semester	**Sixth**	Number of Credits	3 (L:3, T: 0, P: 0)
Prerequisites	NIL	Course Category	**OE**
Question distribution	As per standing norms of WBSCT&VE&SD	Marks distribution	As per standing norms of WBSCT&VE&SD

Course Learning Objectives:

Following are the objectives of this course:

- To learn about various types of natural and man-made disasters.
- To know pre- and post-disaster management for some of the disasters.
- To know about various information and organisations in disaster management in India and Legal framework of disaster management.
- To get exposed to technological tools and their role in disaster management.

Module/ Group [as per directives from WBSCT&VE&SD in framing questions of end semester]	**Distribution of unit**
Module A/ Group A	Unit I and II
Module B/ Group B	Unit III and V
Module C/ Group C	Unit IV

Course Content:

Unit – I: Understanding Disaster
Understanding the Concepts and definitions of Disaster, Hazard, Vulnerability, Risk, Capacity – Disaster and Development, and disaster management.

Unit – II: Types, Trends, Causes, Consequences and Control of Disasters
Geological Disasters (earthquakes, landslides, tsunami, mining); Hydro-Meteorological Disasters (floods, cyclones, lightning, thunder-storms, hail storms, avalanches, droughts, cold and heat waves) Biological Disasters (epidemics, pest attacks, forest fire);
Technological Disasters (chemical, industrial, radiological, nuclear) and Manmade Disasters (building collapse, rural and urban fire, road and rail accidents, nuclear, radiological, chemicals and biological disasters, health disaster) Global Disaster Trends – Emerging Risks of Disasters – Climate Change and Urban Disasters.

Unit- III: Disaster Management Cycle and Framework
Disaster Management Cycle – Paradigm Shift in Disaster Management.
Pre-Disaster – Risk Assessment and Analysis, Risk Mapping, zonation and Microzonation, Prevention and Mitigation of Disasters, Early Warning System; Preparedness, Capacity Development; Awareness.

During Disaster – Evacuation – Disaster Communication – Search and Rescue – Emergency Operation Centre – Incident Command System – Relief and Rehabilitation –
Post-disaster – Damage and Needs Assessment, addressing Residual issues, Restoration of Critical Infrastructure – Early Recovery – Reconstruction and Redevelopment; IDNDR, Yokohama Strategy, Hyogo Framework of Action (HFA).

Unit– IV: Disaster Management in India and Legal framework of disaster management
Disaster Profile of India – Mega Disasters of India and Lessons Learnt.
Disaster Management Act 2005 – Institutional and Financial Mechanism
National Policy on Disaster Management, National Guidelines and Plans on Disaster Management; Role of Government (local, state and national), Non-Government and Inter Governmental Agencies
Refugee Camps and Settlements: Water Supply and Sanitation in Emergency: *Introduction*- Human rights, international humanitarian law and refugee conventions, water and sanitation, refugee camp planning.
Settlement planning- Environmental health risks in emergencies – needs and standards – public health approach to water supply and sanitation in emergencies – partners in the humanitarian response – working with disaster affected people – social diversity – local context Emergency settlements, site selection and planning – introduction – physical planning of emergency settlement – settlement location and physical layout: implications for water supply and sanitation.
Water supply – planning and implementation – water sources – treatment – pumping – tinkering – storage – distribution – collection and use – testing. Waste water – storm water – community involvement.
Waste Management- Phased response – organizational options – staffing needs – monitoring latrine programmers – technical options – options for problem sites- Health risk of solid waste from health centers – dead bodies disposal

Unit– V: Applications of Science and Technology for Disaster Management
Geo-informatics in Disaster Management (RS, GIS and GPS).
Disaster Communication System (Early Warning and Its Dissemination).
Land Use Planning and Development Regulations, Disaster Safe Designs and Constructions, Structural and Non Structural Mitigation of Disasters
S & T Institutions for Disaster Management in India

References:

1. Publications of National Disaster Management Authority (NDMA) on Various Templates and Guidelines for Disaster Management
2. Bhandani, R. K., An overview on natural & man-made disasters and their reduction, CSIR, New Delhi
3. Srivastava, H. N., and Gupta G. D., Management of Natural Disasters in developing countries, Daya Publishers, Delhi
4. Alexander, David, Natural Disasters, Kluwer Academic London
5. Ghosh, G. K., Disaster Management, A P H Publishing Corporation
6. Murthy, D. B. N., Disaster Management: Text & Case Studies, Deep & Deep Pvt. Ltd.
7. Singh Jagbir, Disaster Management-Future Challenges and Opportunities, IK International Publishing House Pvt. Ltd.

8. Gupta, Harsh K., Disaster Management, Universities Press (India) Pvt. Ltd.
9. Harvey, P.A., Baghri, S. and Reed, R.A. (2002) **Emergency Sanitation: Assessment and programme design**, WEDC, Loughborough University, UK.

Course outcomes:

After completing this course, student will be:

- Acquainted with basic information on various types of disasters
- Knowing the precautions and awareness regarding various disasters
- Decide first action to be taken under various disasters
- Familiarized with organization in India which are dealing with disasters and Legal framework of disaster management
- Able to select IT tools to help in disaster management